

HOMEWORKs & INCLASS ASSIGNMENT for CIRCUIT 1

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CLASS: ICT 4 Sinh - K67

W_3 : 8.5/10
 W_4 : 9/10
 W_{6+7} : 6, 25/10
 W_5 : 10/10

(dòng điện đi từ E → C
dòng điện là do lỗ trống đi
chuyển từ E qua B đến C)

WEEK 3

1st In-class Assignment: PN Junction and Applications
Instructor: Assoc. Prof. Pham Nguyen Thanh Loan

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I. Objectives

After this practice students will

- be familiar with the simulation tool Circuit Maker/Proteus/LTSPICE
- be able to analyze basic circuits such as rectifiers and amplifiers using electronics components.

II. Requirements

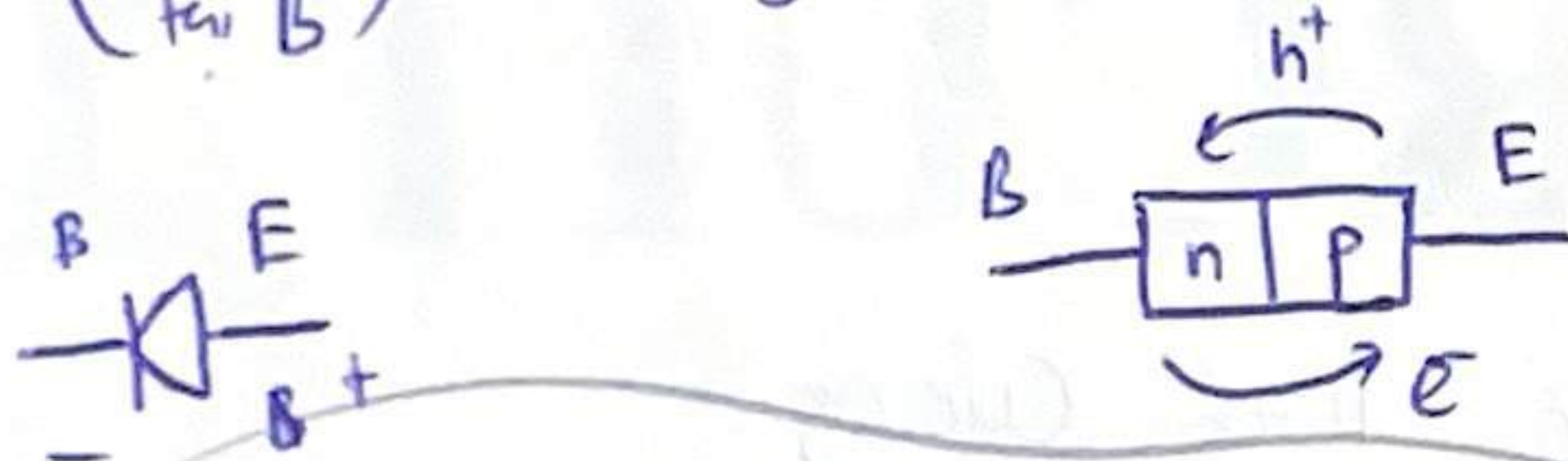
- Students are required to fulfill the work individually.
- Students are required to follow step by step all the simulations of design tools and to answer clearly and cleanly all the questions.
- Students are required to submit this document before the due date following the lecturer's instruction.

III. Lab work realization

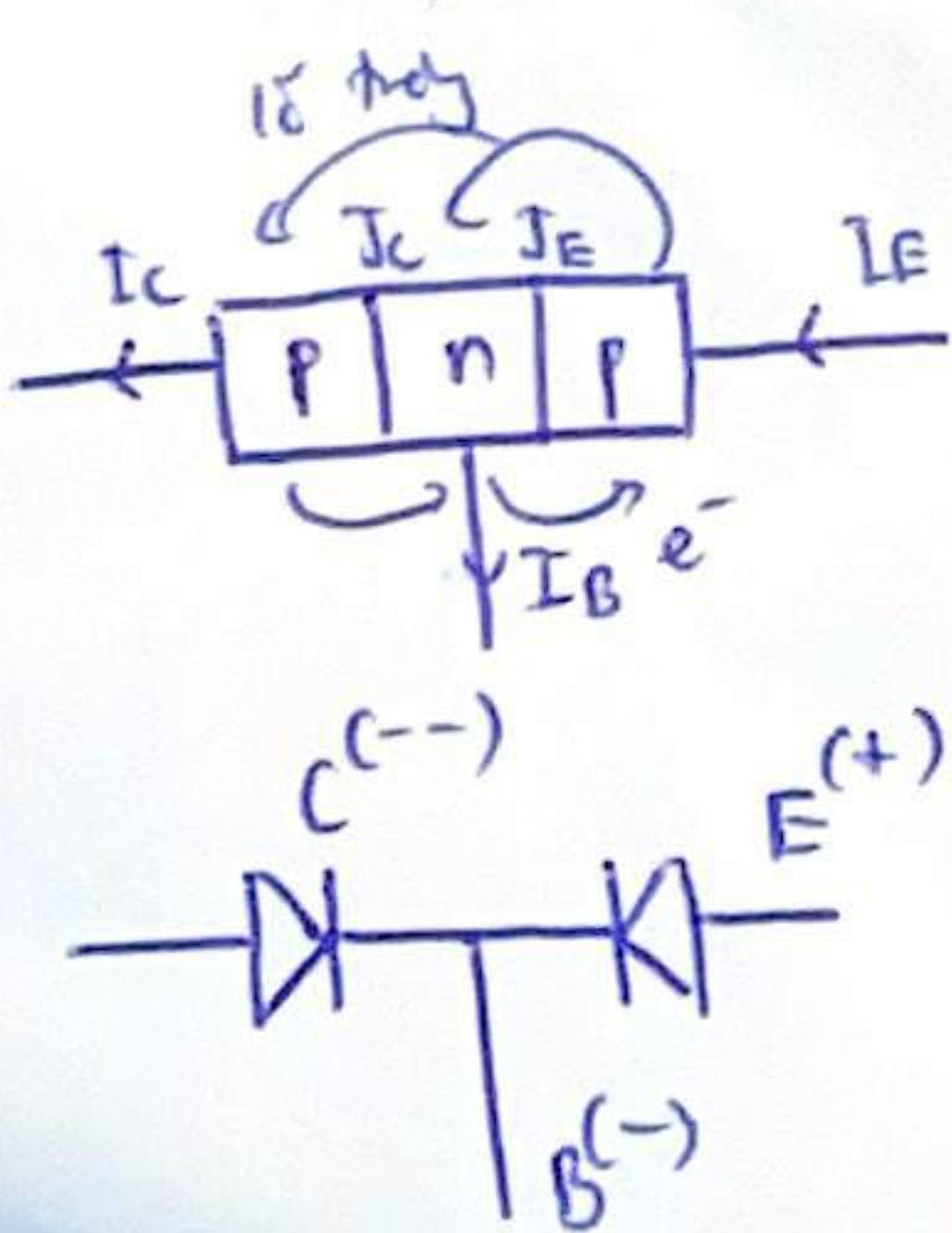
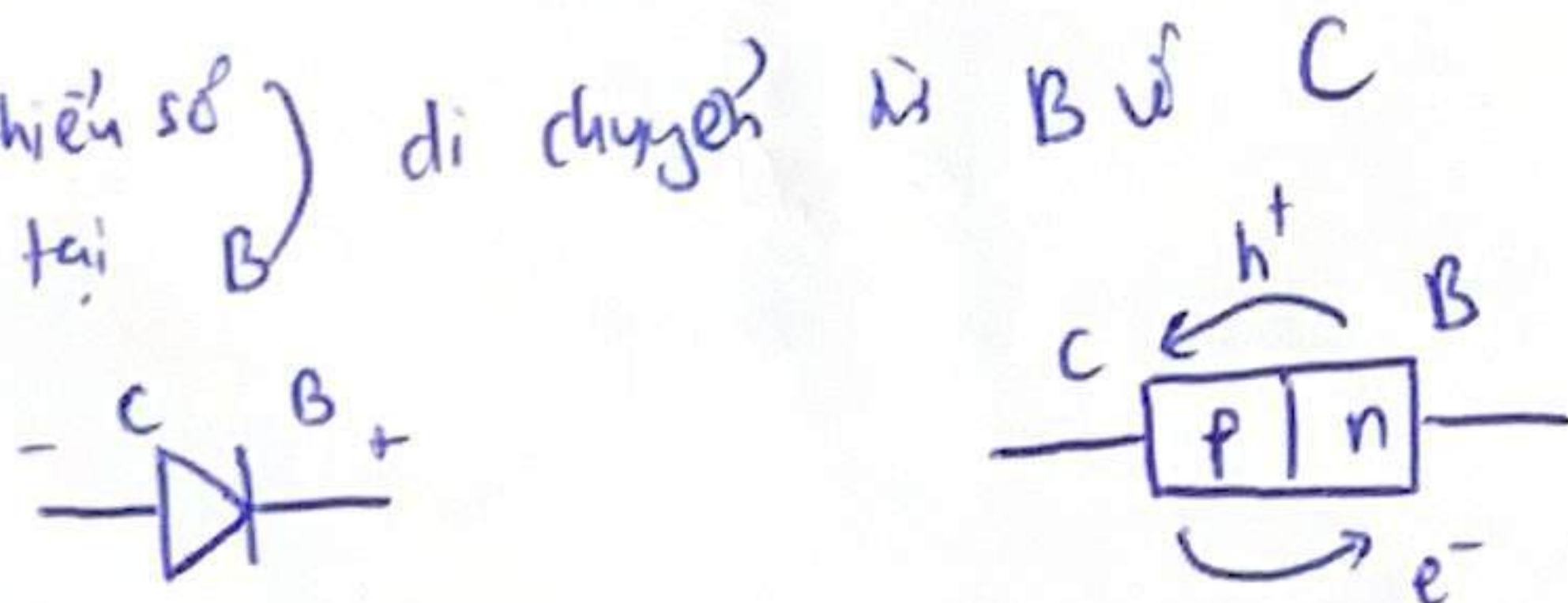
III.1 PN Junction

Q1: Explain the operation of PNP. How to bias for BE junction and BC junction?

* Ở J_E (tiếp giáp BE) cần phải phân cực thuận ($V_E > V_B$), thường $V_{EB} = 0,7V$ với Si
do lỗ trống (đa số tại E) di chuyển từ E sang B $\Rightarrow$ tạo dòng khuếch tán (tạo dòng)
và electron (đa số tại B) di chuyển từ B sang E



* Ở J_C (tiếp giáp BC) cần phải phân cực ngược ($V_B > V_C$) $\Rightarrow$ làm tăng hiệu điện thế
do electron (đa số tại B) không đi từ B về C mà electron di chuyển từ C sang B
và lỗ trống (thiểu số tại B) di chuyển từ B về C
 $\Downarrow$
tạo dòng I trở



$$\Rightarrow \begin{cases} I_E = I_C + I_B \\ V_C < V_B < V_E \end{cases}$$

* Tại miền p $\begin{cases} h^+ \text{ là đa số} \\ e^- \text{ là thiểu số} \end{cases}$ (E, C)

Tại miền n $\begin{cases} e^- \text{ là đa số} \\ h^+ \text{ là thiểu số} \end{cases}$ (B)

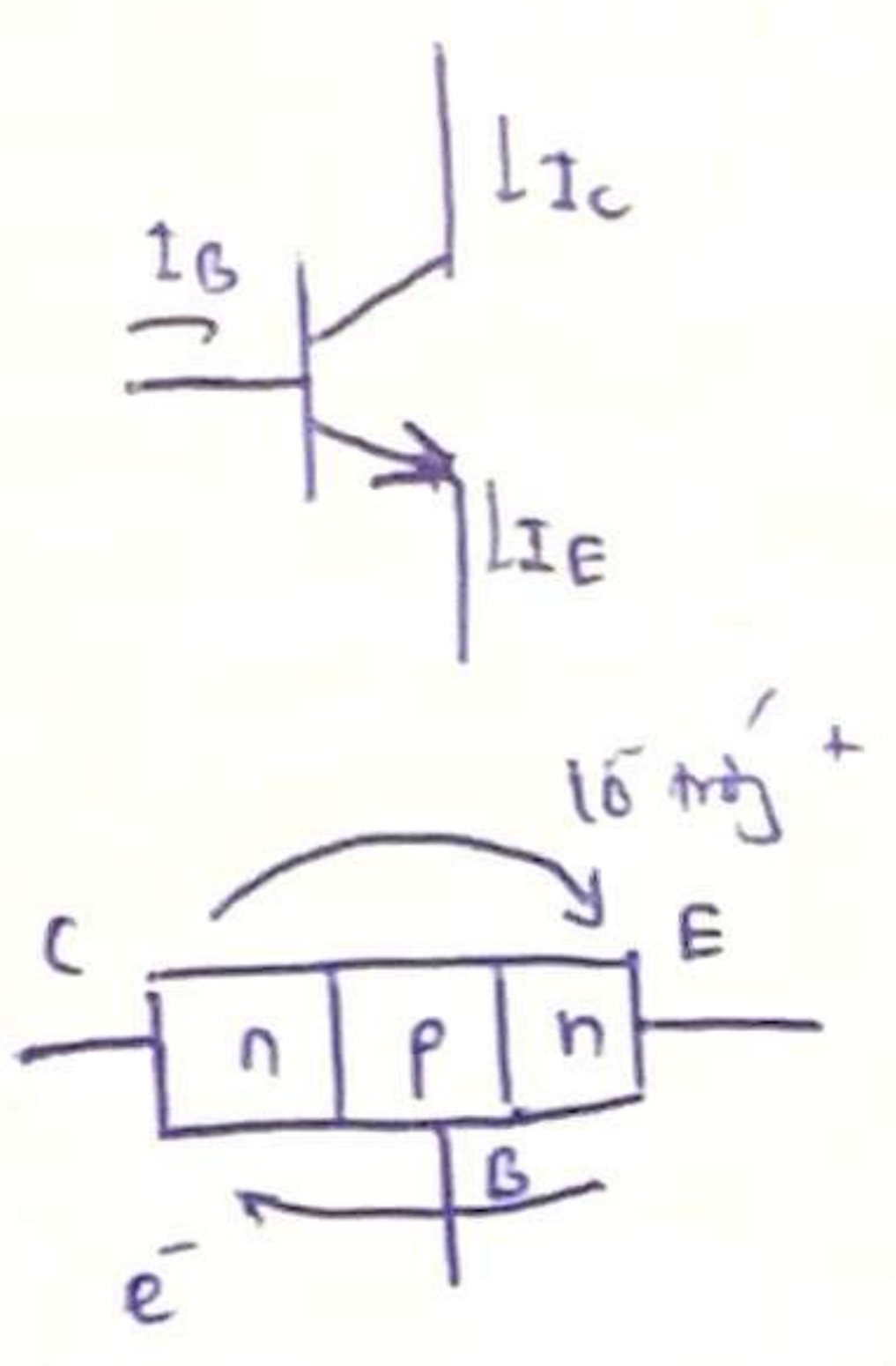
Base: máy nhử
Collector: thu phần tập ít hơn Emission

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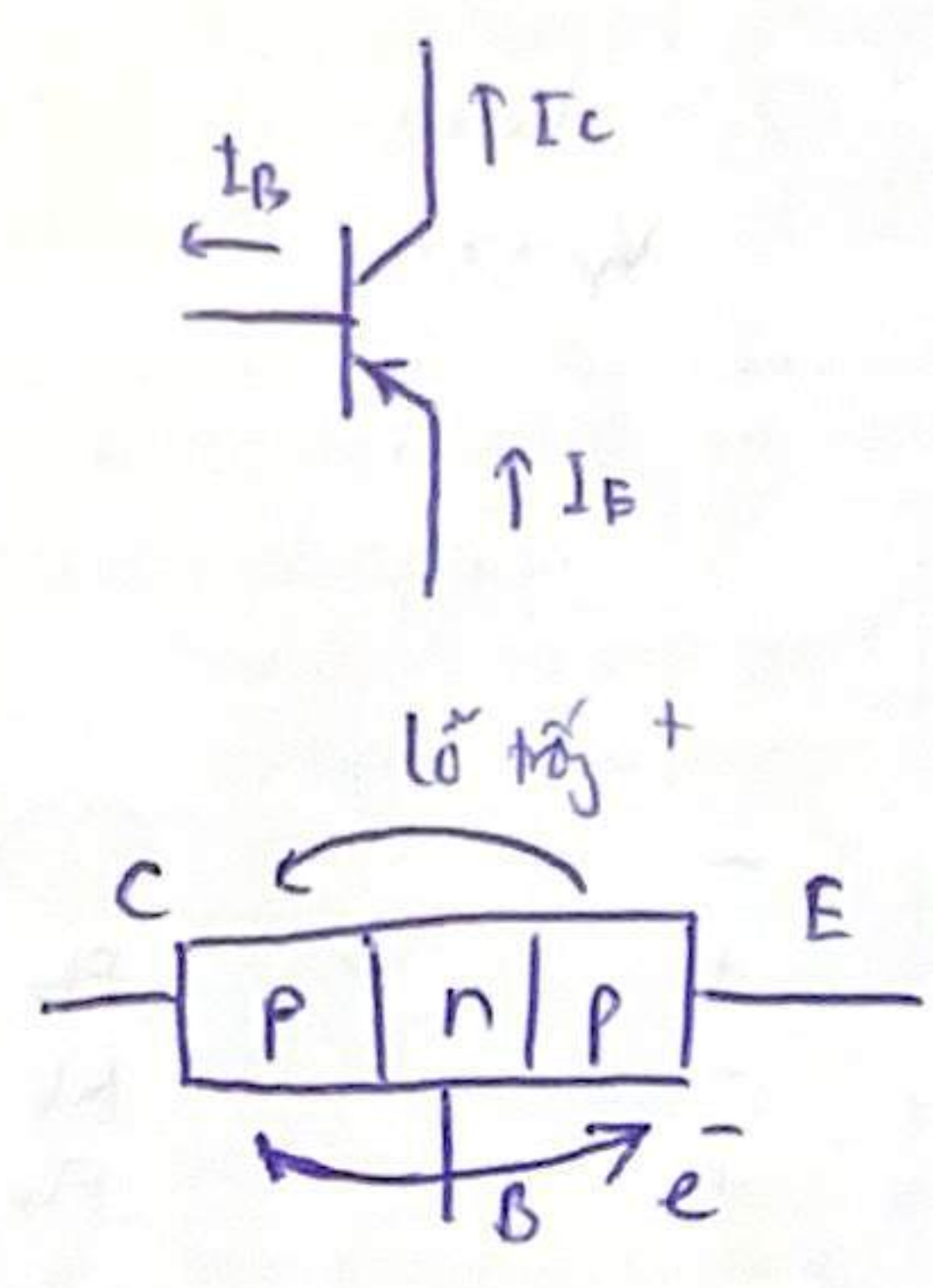
Q2: Explain the symbol's direction of PNP and NPN. Draw to illustrate your answer.

Chiều mũi tên là quy ước chiều chuyển động của lỗ trống mang điện tích dương (+) trong vùng tiếp giáp BE ở chế độ hoạt động.

BJT NPN

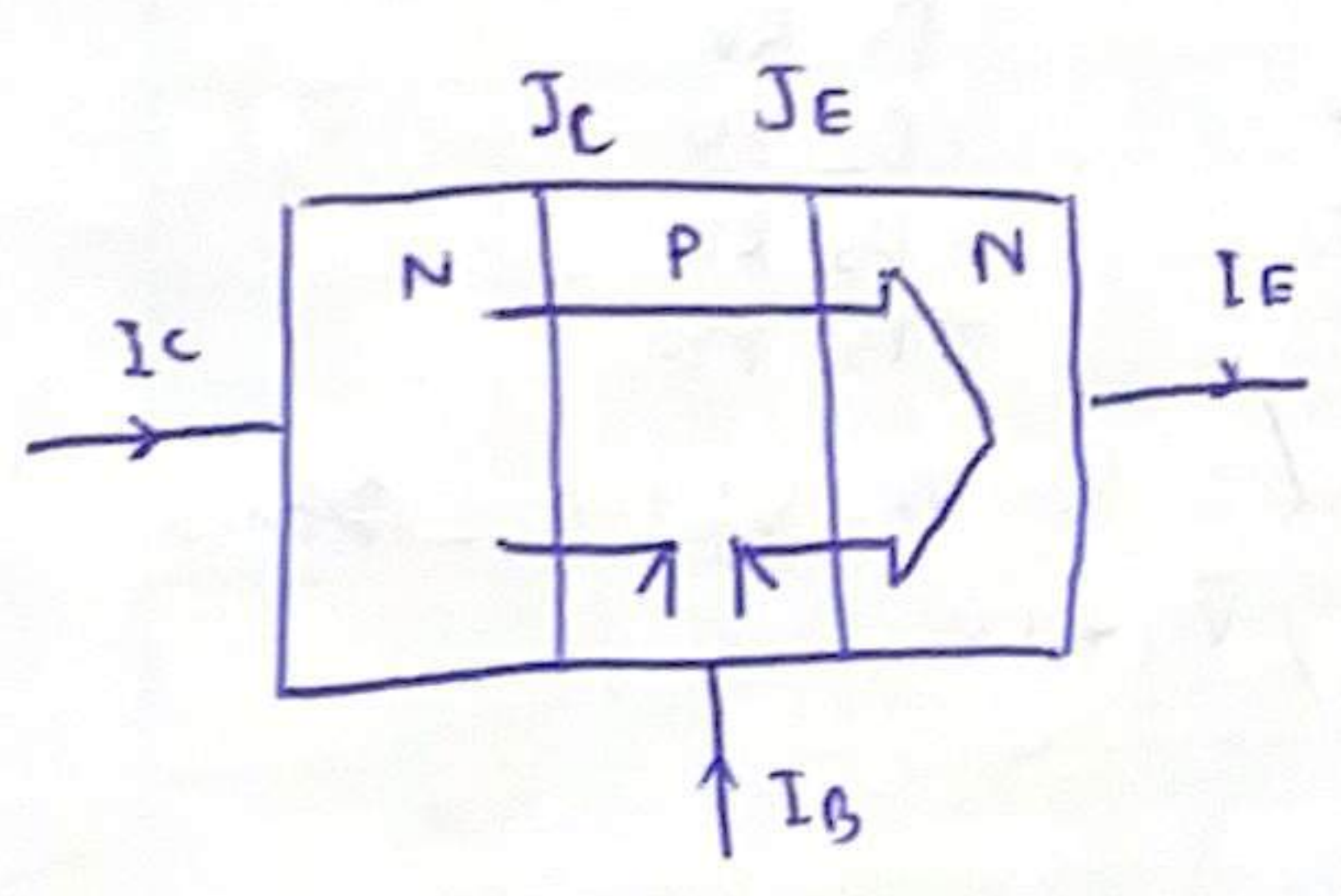


BJT PNP

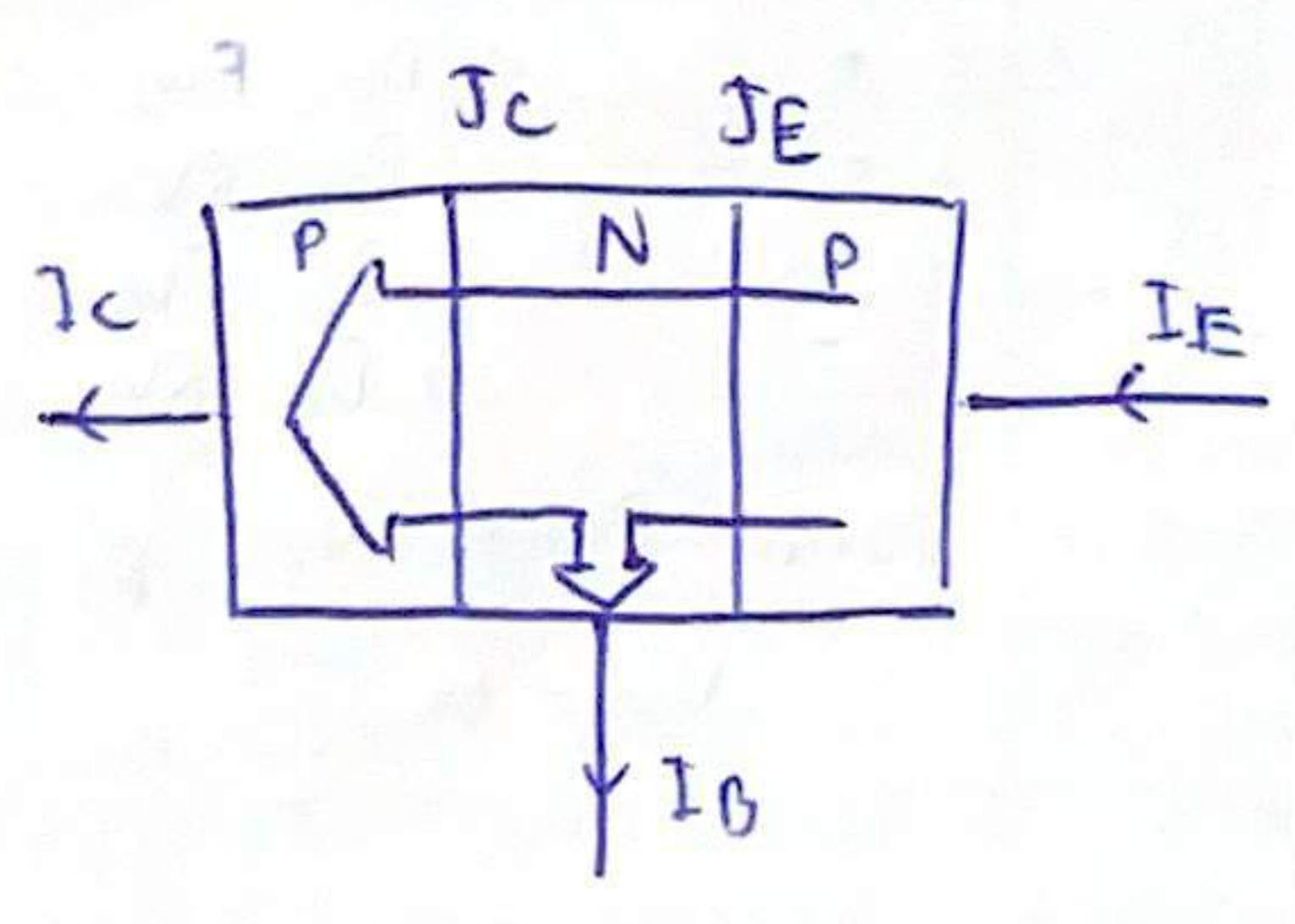


Trong BJT npn, lỗ trống chuyển động từ C về E và lượng rất nhỏ từ B về E

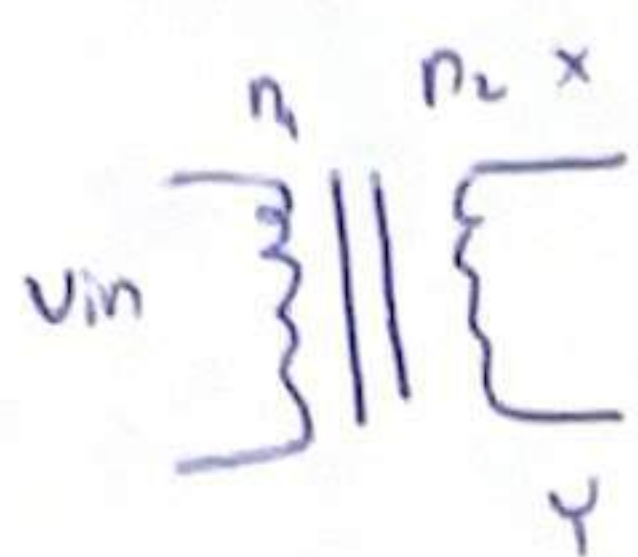
Trong BJT pnp, lỗ trống chuyển động từ E đến C và lượng rất nhỏ từ B



Khi có 1 dòng rất nhỏ đi vào Base, BJT cho phép dòng đi từ Collect đến Emitter



Khi có 1 dòng rất nhỏ đi ra từ Base, BJT cho phép dòng đi từ Emitter đến collector



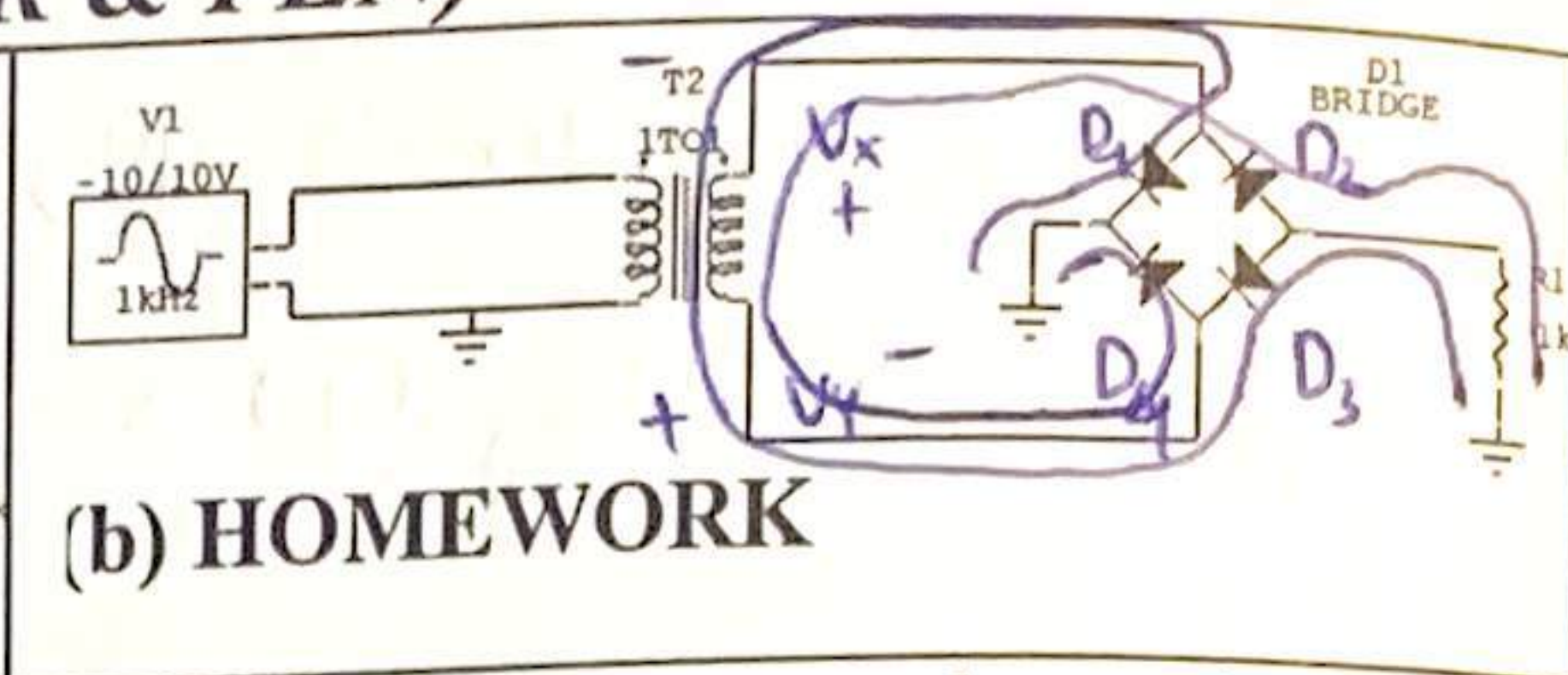
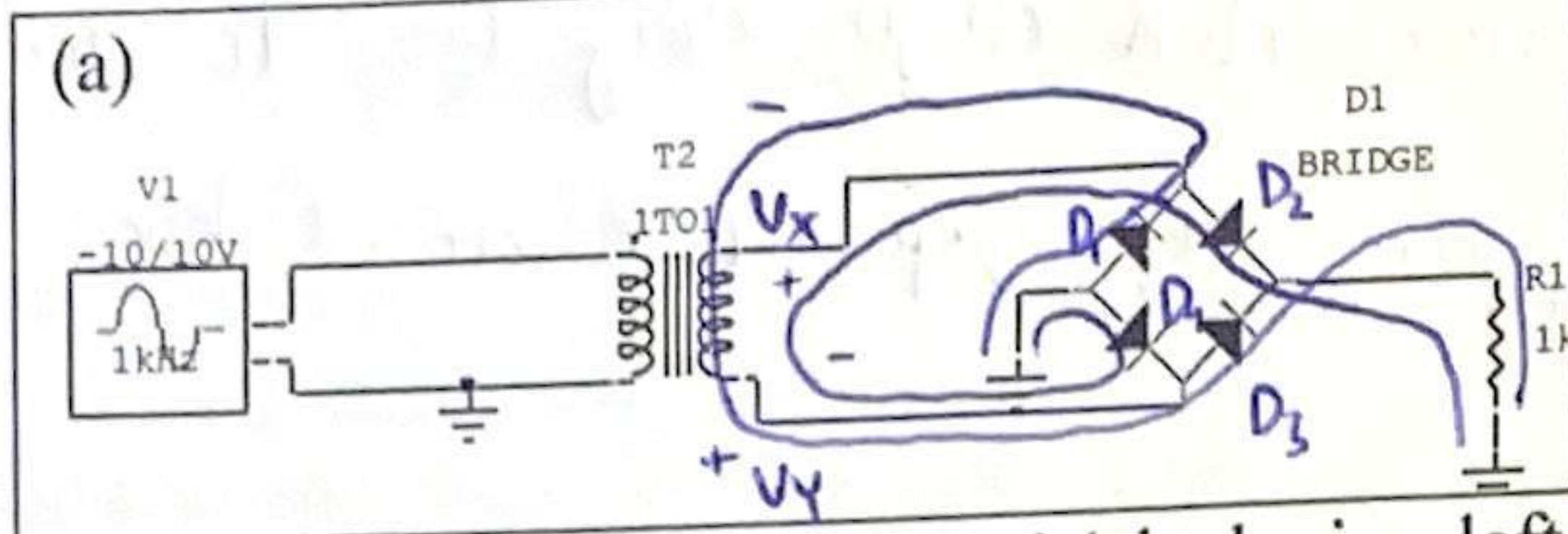
$$\frac{V_{in}}{V_{xy}} = \frac{n_1}{n_2}$$



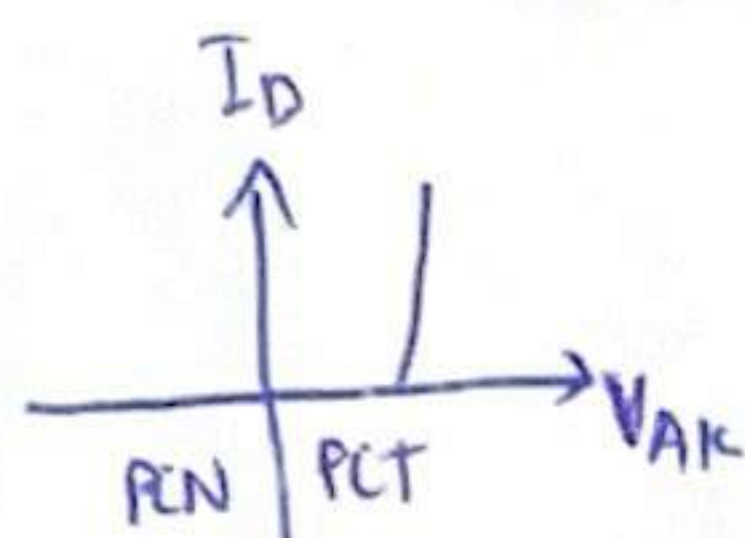
III.2 Wien bridge rectifier

a. Theoretical study (USE ONLY PAPER & PEN)

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(b) HOMEWORK



- Name the diodes from D1 to D4 (clockwise, left to right and from top to bottom). Consider the barrier voltages of 4 diodes are 0.7V. V_X & V_Y are voltage at 2 nodes (top bottom) right after the transformer. $V_{out} = 0$ ($I_D = 0$)
- Determine the equation of V_{AK} of each diode:
 - (a) $V_{AK}(D1) = -V_X$ $V_{AK}(D2) = V_X$ (b) $V_{AK}(D1) = V_X$ $V_{AK}(D2) = -V_X$
 - $V_{AK}(D3) = V_Y$ $V_{AK}(D4) = -V_Y$ $V_{AK}(D3) = -V_Y$ $V_{AK}(D4) = V_Y$

1 st half of period (Determine the sign of V_X , V_Y and V_{AK} of each diode)		
Nodes	Consider circuit (a)	Consider circuit (b)
V_X	Negative or Positive? +	-
V_Y	Negative or Positive? -	+
$V_{AK}(D1)$	= - $\rightarrow$ D1: RV or FW?	= + $\rightarrow$ D1: RV or FW?
$V_{AK}(D2)$	= + $\rightarrow$ D2: FW	= - $\rightarrow$ D2: RV
$V_{AK}(D3)$	= - $\rightarrow$ D3: RV	= + $\rightarrow$ D3: FW
$V_{AK}(D4)$	= + $\rightarrow$ D4: FW	= - $\rightarrow$ D4: RV
Current flow	(ground to ground): node... to node.... to... ground $\rightarrow$ D4 $\rightarrow V_Y \rightarrow V_X \rightarrow$ D2 $\rightarrow$ ground	(ground to ground) ground $\rightarrow$ D3 $\rightarrow V_Y \rightarrow V_X \rightarrow$ D1 $\rightarrow$ ground
Output voltage	$V_{out} = V_{in} - 1,4 = V_X - V_Y - 1,4$	$V_{out} = -V_{in} + 1,4 = -V_Y + V_X + 1,4$
$V_{KL}: V_{AD}(D_2) + V_{out} + V_{YX} + V_{AD}(D_4) \Rightarrow V_{out} = V_{YX} - V_{AD}(D_2) - V_{AD}(D_4)$		
2 nd half of period		
Diodes	Circuit (a) Diode status & Explanation	Circuit (b) Diode status & Explanation
V_X	-	+
V_Y	+	-
D1	= + $\Rightarrow$ D1 FW	= - $\Rightarrow$ D1 RV
D2	= - $\Rightarrow$ D2 RV	= + $\Rightarrow$ D2 FW
D3	= + $\Rightarrow$ D3 FW	= - $\Rightarrow$ D3 RV
D4	= - $\Rightarrow$ D4 RV	= + $\Rightarrow$ D4 FW
Current flow	(ground to ground) ground $\rightarrow$ D1 $\rightarrow V_X \rightarrow V_Y \rightarrow$ D3 $\rightarrow$ ground	(ground to ground) ground $\rightarrow$ D2 $\rightarrow V_X \rightarrow V_Y \rightarrow$ D4 $\rightarrow$ ground
Output voltage	$V_{out} = V_{in} - 1,4$	$V_{out} = -V_{in} + 1,4$

$$V_{out} - V_G = V_{AK}(D_1) + V_{AK}(D_2)$$

$$V_{out} = -V_X$$



$$\frac{V_X}{V_{in}} = \frac{n_2/2}{n_1}$$

$$\frac{V_Y}{V_{in}} = \frac{-n_2/2}{n_1}$$

$$V_{out} + V_{D4} + V_Y + V_X + V_{D2} = 0$$



b. Experimental study (USE DESIGN TOOLS: Circuit Maker/Proteus/LTSPICE)

- Build the above circuits in design tools
- Make run the simulation and then answer the following questions:

Parameters	Circuit (a)	Circuit (b)
Draw the input waveform and detail min and max value		
Draw the Waveform at the node X and detail min and max value		
Draw the Waveform at the node Y and detail min and max value		
Draw the output waveform and detail min and max value		
Value of V_{peak} of V_X	$ V_X _{max} = V_{peak} = 9.31$	$ V_X _{max} = V_{peak} = 9.31$
Value of V_{peak} (at output)	$ V_{out} _{max} = V_{peak} = 8.61$	$ V_{out} _{max} = V_{peak} = 8.61$
Theoretical V_B (or V_D)	$0.7V$	$-0.7V$
Practical V_D (observed from simulation)	$0.69V$	$-0.69V$

Comment on the difference between the theoretical results and simulation results for this Bridge RECTIFIER

Sự khác biệt giữa lý thuyết và thực nghiệm là vì thực tế có sự sụt áp do điện trở của linh kiện và dây dẫn (trong lý thuyết thường coi $r = 0\Omega$) $\Rightarrow$ có sụt áp ngược ở diode $\Rightarrow$ lý thuyết

VI. HOMEWORK (for a group)

A POPULAR POWER SUPPLY with IC 78XX

Group tasks: Analyze this circuit to theoretically and experimentally to choose a correct transformer ratio, to compare the theoretical and experimental results. Finally, student group have to realize a real circuit with PCB using Altium to design layout. The task is presented by the slides together with the real circuit which performs correctly after 2 weeks. The output voltage can be connected to a lamp (or LED) or an audio amplifier to test its performance.

WEEK 4 - *lên dạy quen dần*

2nd In-class Assignment: Analyze large and small signal of BJT amplifier circuit

Instructor: Assoc. Prof. Pham Nguyen Thanh Loan

Time: 40 minutes

I. Objectives

After this lab students will

- be familiar with the simulation tool Circuit Maker
- be able to distinguish the purpose of DC analysis and AC analysis
- be able to analyze a single amplifier circuit using BJT in different operation modes
- be able to design a single amplifier circuit to meet a certain requirement

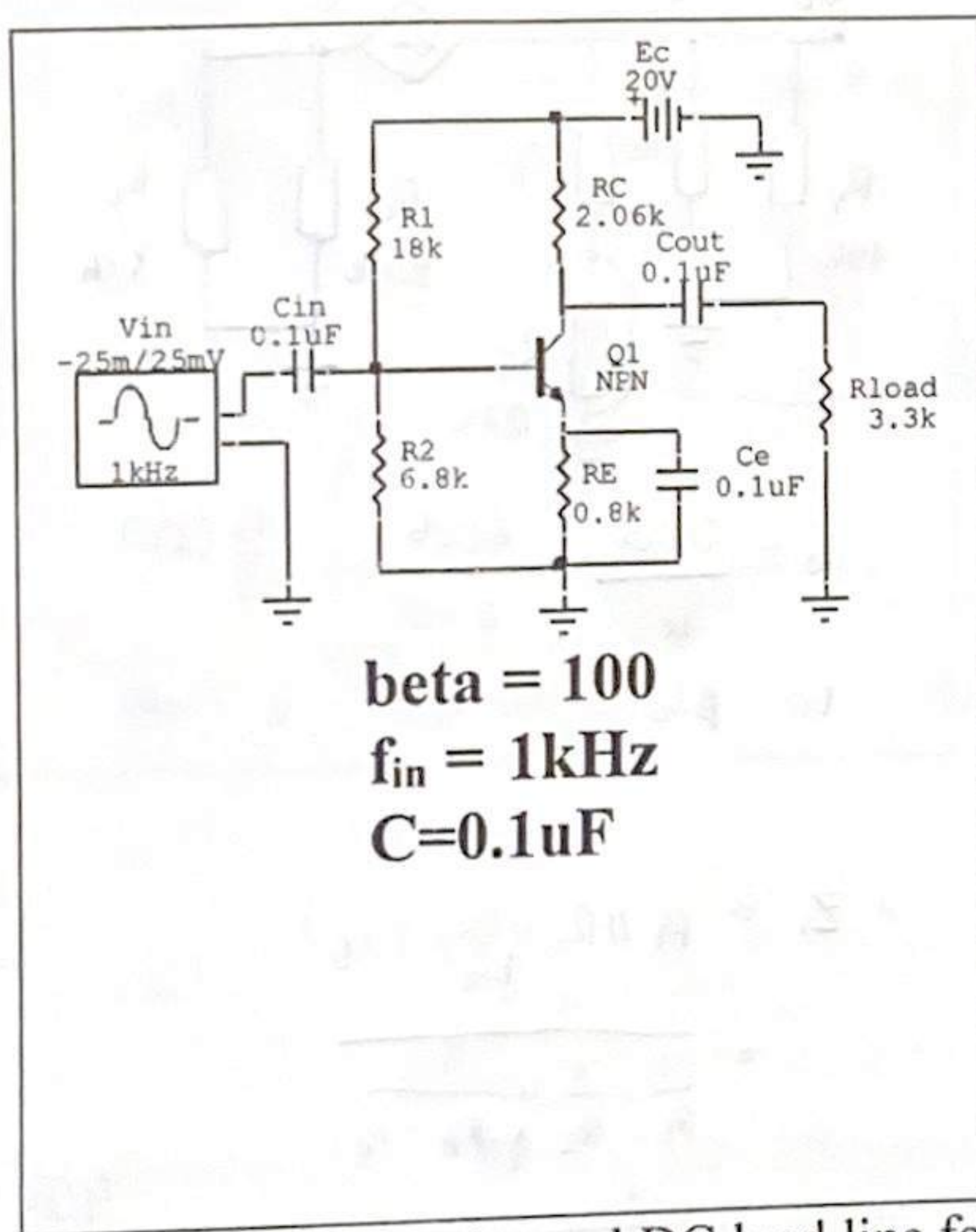
II. Requirements

- Students are required to fulfill the task in a team of 2 or 3 members.
- Students are required to follow step by step all the instruction for Circuit Maker and to answer **clearly and cleanly** all the questions.
- Students are required to submit this document on time.

III. Lab work realization: BJT amplifier circuit

III.1 Theoretical Study

(vào B
ra C)



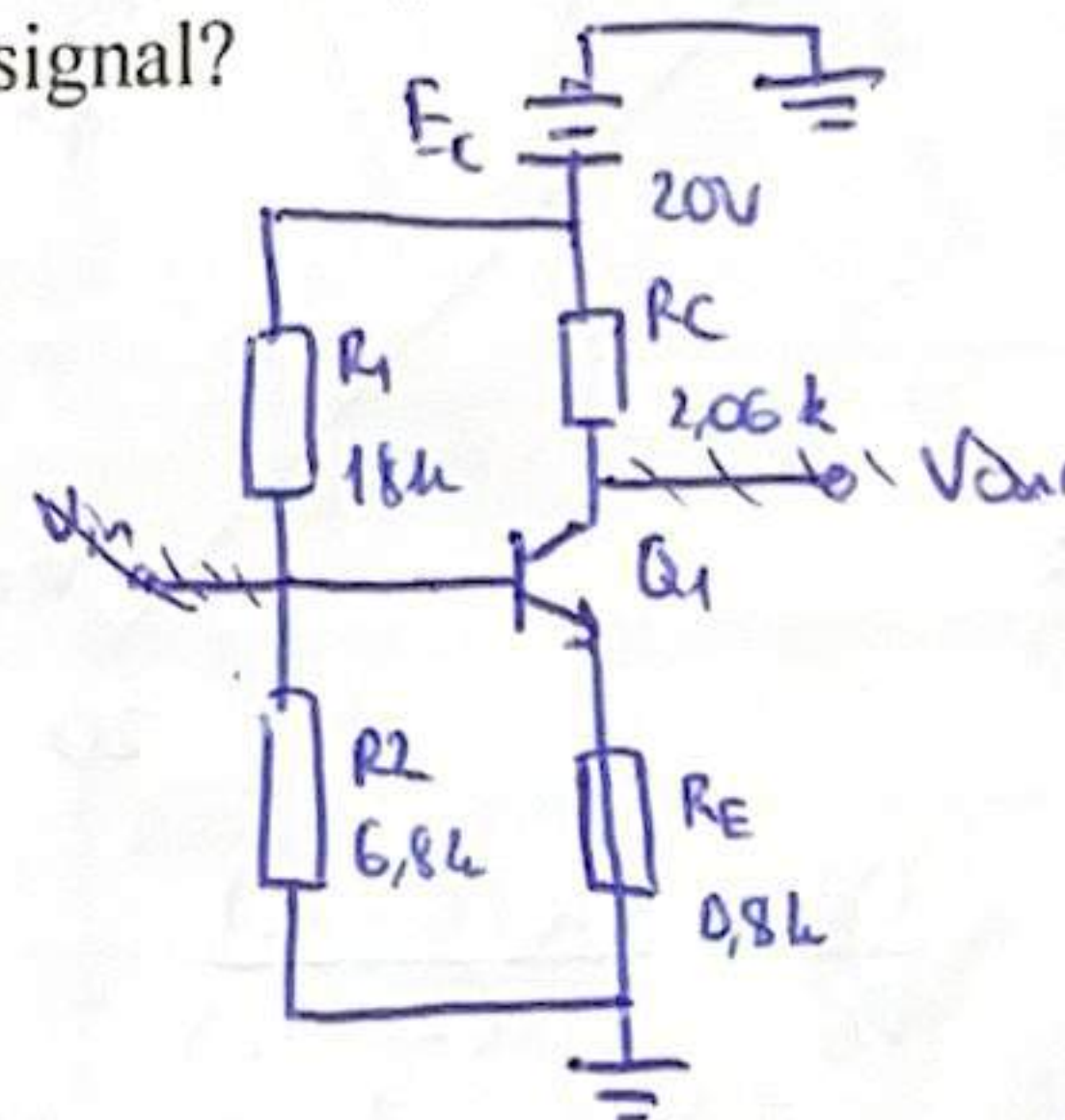
1. Determine the name of this amplifier?

Mạch Khuếch đại E chung (common emitter)

2. What kind of biasing?

Phân áp bằng chia áp (voltage divider biasing)

3. Draw the equivalent circuit for DC signal?



4. Determine the Q point and DC load line for this circuit.

(Instruction: Q point (I_{BQ} , I_{CQ} , V_{CEQ}); DC loadline: $V_{CE} = f(I_C)$)

Có PT đường tải tĩnh: $I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E}$

$$\Rightarrow V_{CE} = V_{CC} - I_C(R_C + R_E) = 20 - 2860 I_C$$

Điểm bão hòa (I_C max khi $V_{CE} = 0$) $\Rightarrow 0 = 20 - 2860 I_C$

$$\Rightarrow I_{C(sat)} = \frac{20}{2860} = 0,007 (A) \Rightarrow A(0; 0,007)$$

* Điểm cutoff ($I_C = 0$) $\Rightarrow V_{CE} = 20 - 2860 \cdot 0 = 20 \text{ (V)}$

$\Rightarrow V_{CE(\text{cutoff})} = 20 \text{ V} \Rightarrow B(20; 0)$

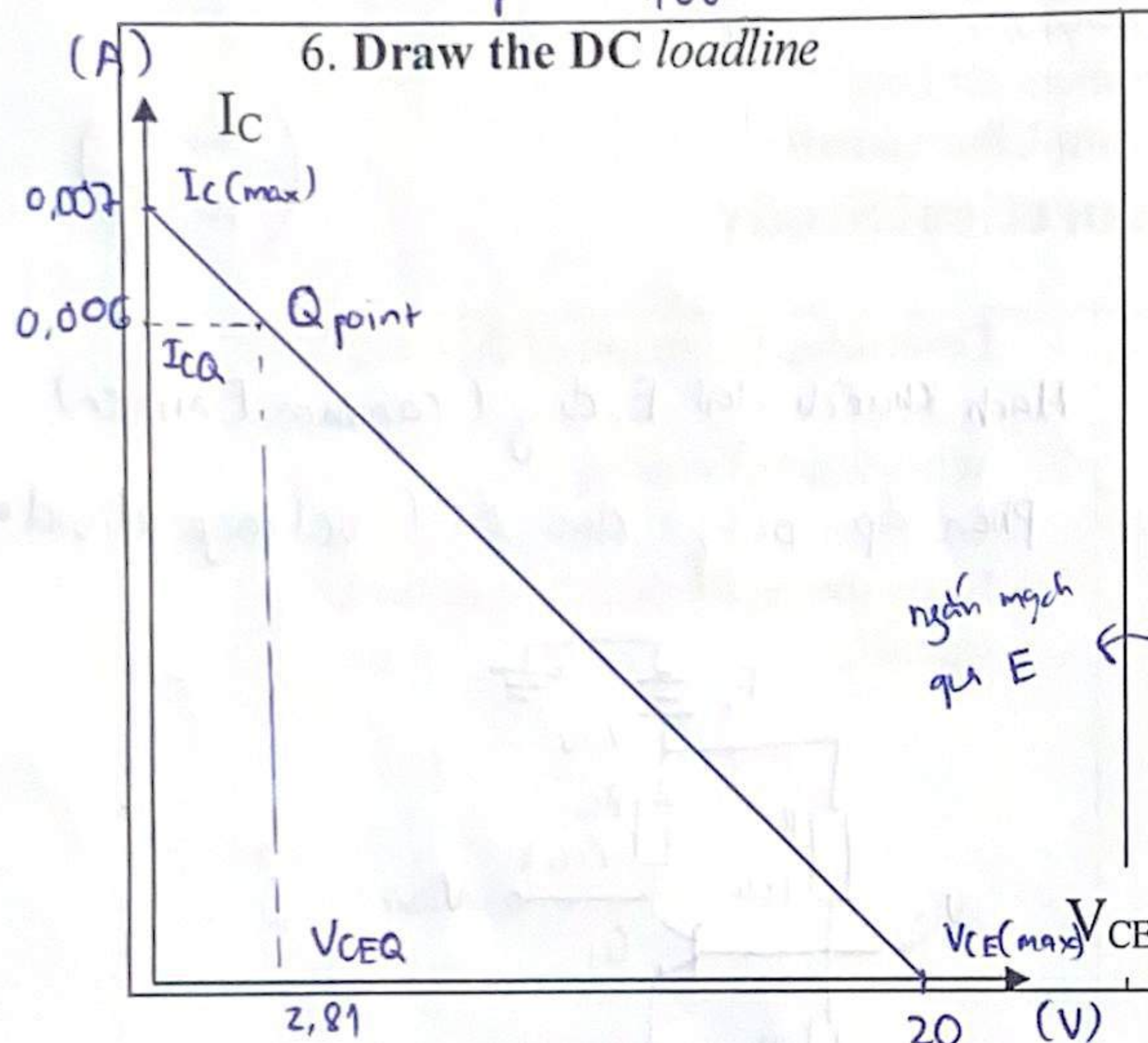
Ta có $V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{6,8}{18 + 6,8} \cdot 20 = 5,484 \text{ (V)}$

mà $V_B = V_E + V_{BE} \Rightarrow V_E = V_B - V_{BE}$

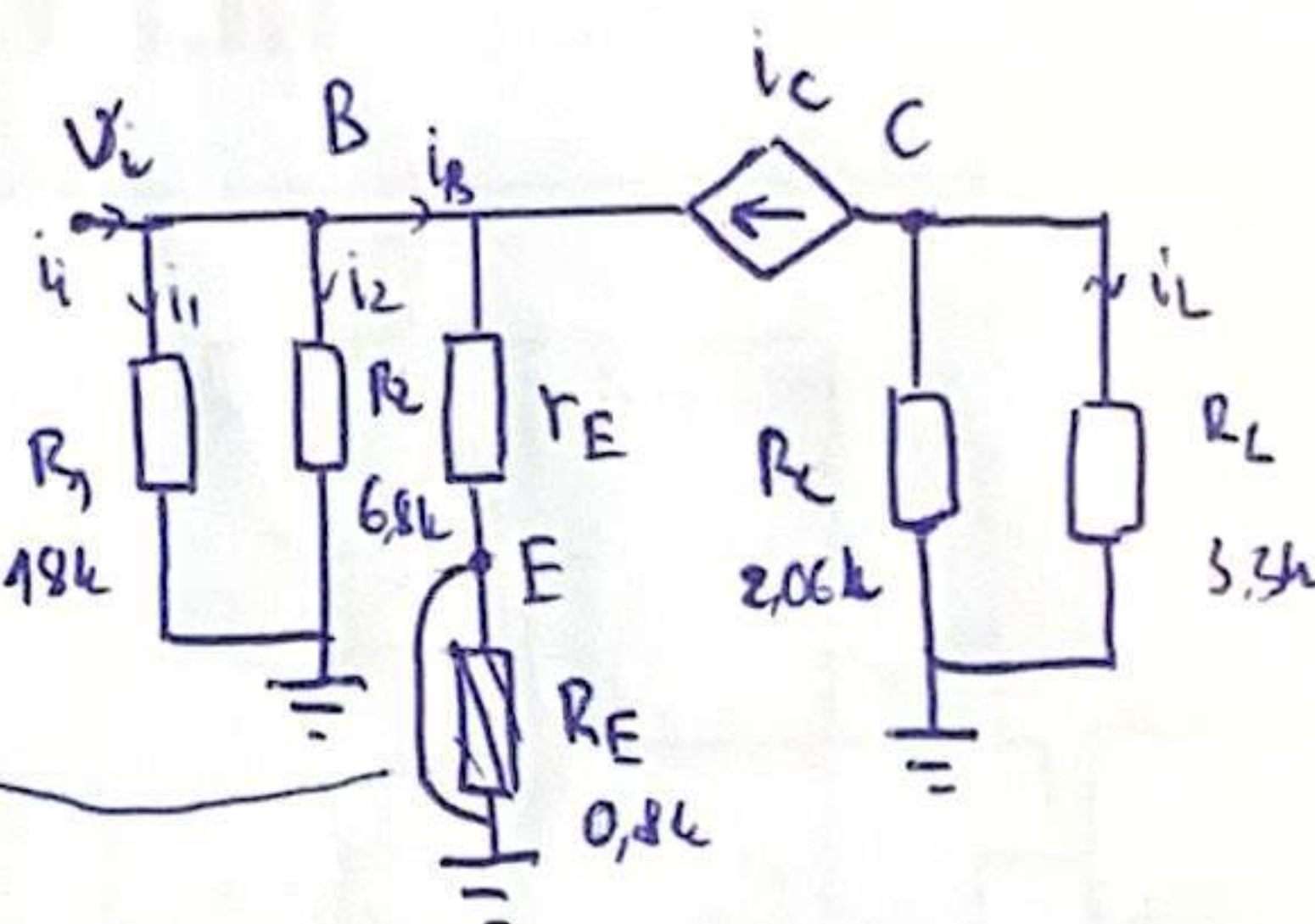
Lấy $V_{BE} = 0,7 \text{ (V)}$ (s) $\Rightarrow V_E = 5,484 - 0,7 = 4,784 \text{ (V)}$

$\Rightarrow I_{CQ} = \frac{V_E}{R_E} = \frac{4,784}{0,8k} \approx 6 \text{ (mA)}$

$\Rightarrow I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{6}{100} = 0,06 \text{ (mA)}$ $V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = 20 - 6(206 + 0,8) = 2,81 \text{ V}$



7. Draw the equivalent circuit for small signal (AC equivalent circuit)



$r_e = \frac{0,026}{I_E} = \frac{0,026}{6 \cdot 10^{-3}} = \frac{13}{3} \text{ (}\Omega\text{)}$

$i_c = \beta I_B = 100 \cdot 0,06 = 6 \text{ (mA)}$

8. Determine the A_i , A_v , Z_i , Z_{out}

* $A_v = \frac{V_o}{V_i} = \frac{V_{R_L}}{V_B} = \frac{-i_c(R_C \parallel R_L)}{i_b(R_E \parallel r_e)}$

$= -\beta \frac{R_C \parallel R_L}{\beta + 1} \approx -\frac{R_C \parallel R_L}{r_e} = \frac{2,06 \cdot 3,3}{\frac{13}{3}} = -292,7$

* $A_i = \frac{I_o}{I_i} = \frac{I_{R_L}}{I_i} = \frac{V_c}{R_L}$

$= \frac{i_c(R_C \parallel R_L) / R_L}{i_b + i_e(R_E + r_e)} = \frac{R_C}{R_C + R_L} \cdot \frac{1}{1 + \frac{r_e + R_E}{\beta}}$

$= \frac{1}{1 + \frac{r_e + R_E}{\beta}}$

* $Z_i = R_1 \parallel R_2 \parallel (\beta(r_e + R_E))$

$= \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{\beta(r_e + R_E)}}$

$= \frac{1}{\frac{1}{18k} + \frac{1}{6,8k} + \frac{1}{100 \cdot \frac{13}{3} + 0,8k}} = 402 \text{ }\Omega$

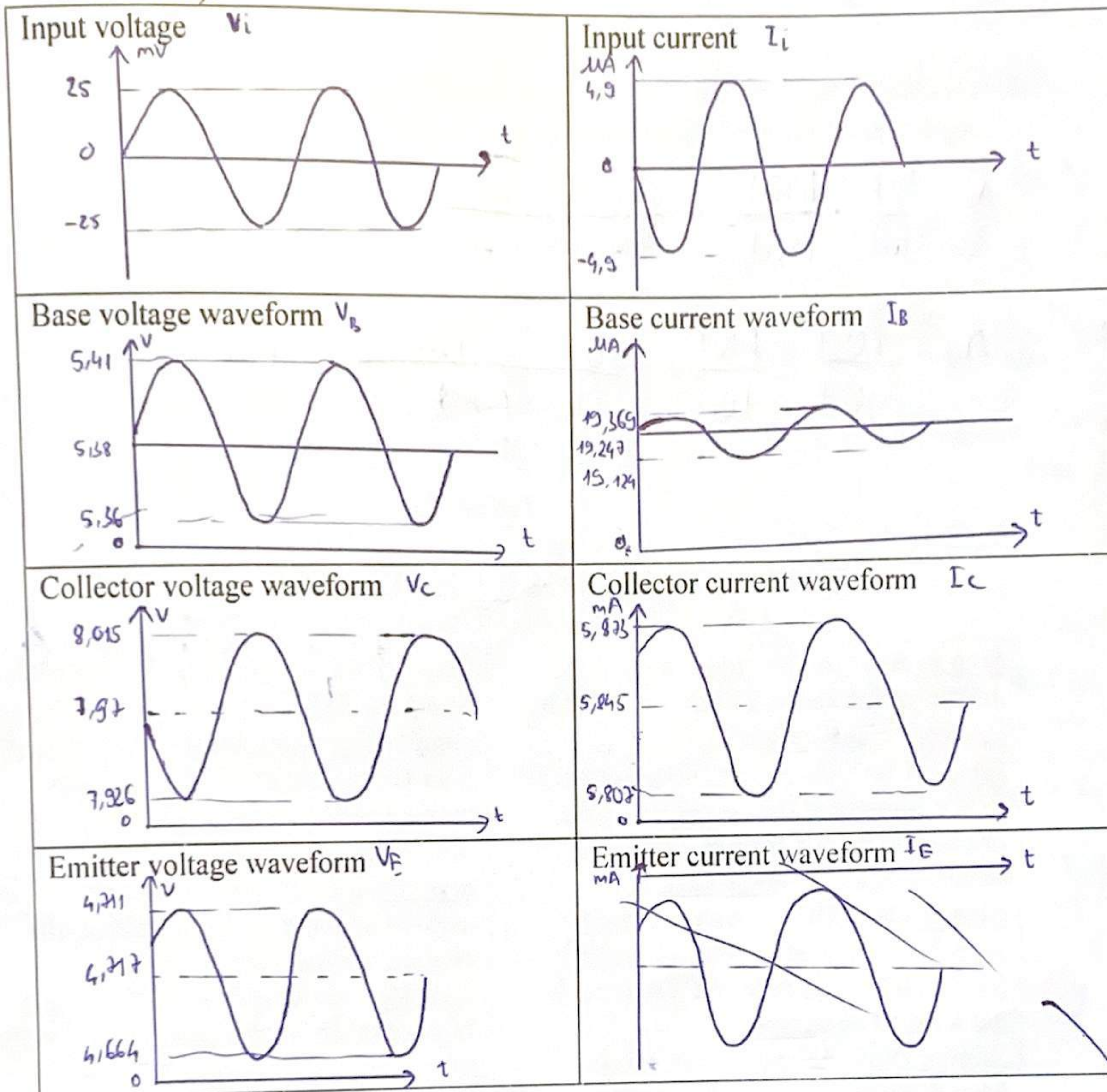
* $Z_{out} = R_C \parallel R_L = \frac{R_C \cdot R_L}{R_C + R_L}$

$= \frac{2,06 \cdot 3,3}{2,06 + 3,3} = 1,268k \text{ (}\Omega\text{)}$

$I_1 + I_2 = \frac{i_e(r_e + R_E)}{R_1 \parallel R_2} = 35,33$

III.2 Experimental study

1. Draw the circuit above in Circuit Maker
2. Draw the waveforms of voltage and current of the input and at the node Base, Collector and Emitter in the corresponding cases below (Click the button RUN in Circuit maker at each node and just copy the waveform from Circuit Maker to fill the table below):

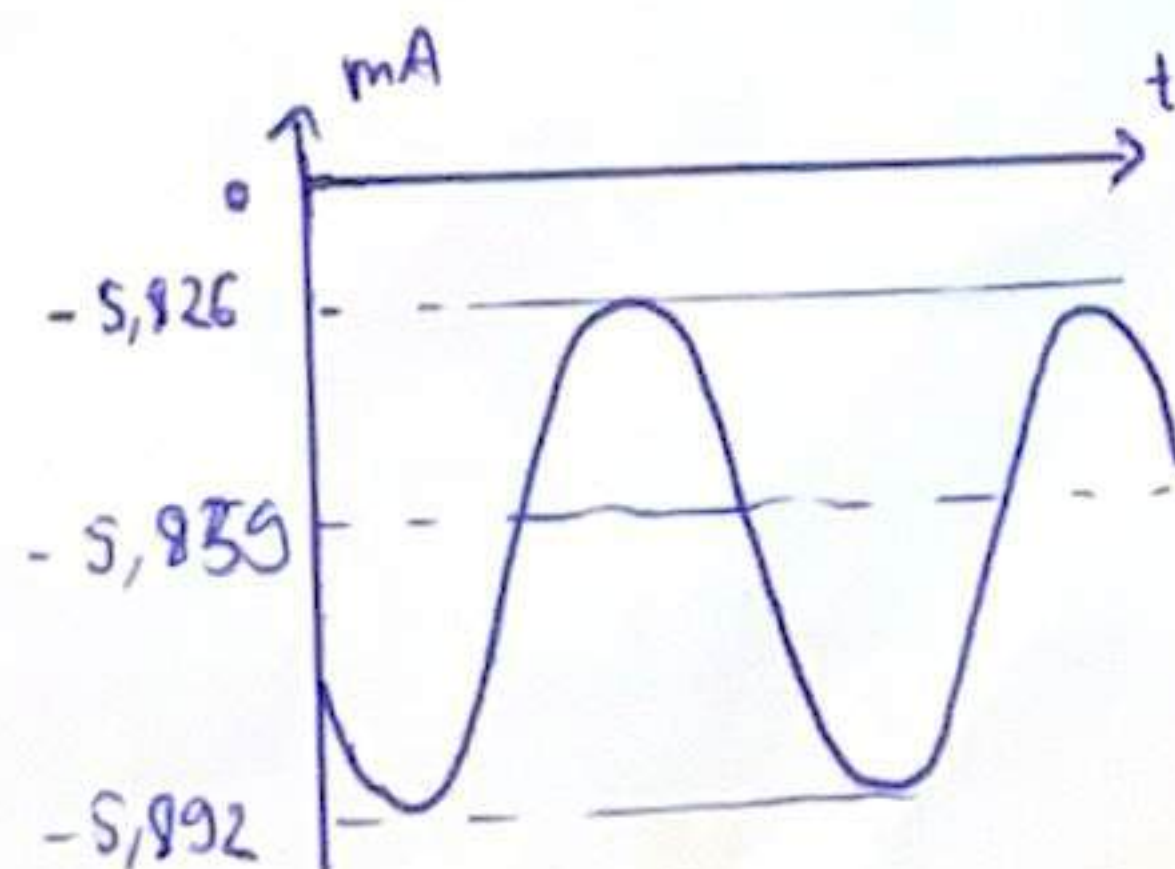


3. From the waveforms above, determine the Q point and DC load line for this circuit. (**Instruction:** Q point (I_{BQ} , I_{CQ} , V_{CEQ}); DC loadline: $V_{CE} = f(I_C)$, DC value is the offset of your waveform). Comments ?/k

$$I_{BQ} = 19.247 \text{ (uA)} ; I_{CQ} = 5.8235 \text{ (mA)}$$

$$V_{CE} = V_C - V_E = 7.97 - 4.1887 = 3.7813 \text{ (V)}$$

$$\Rightarrow Q(19.247 \cdot 10^{-6} ; 5.8235 \cdot 10^{-3} ; 3.7813)$$



⇒ Q point thì nó phụ thuộc vào Q point theo lý thuyết

4. Determine A_i , A_v from Circuit Maker Results (**Instruction:** from the waveform of input and output voltage/current, the gain is the ratio of its amplitude)

$$A_i = \frac{|i_o|}{|i_i|} = \frac{|i_c|}{|i_b|} = \frac{5,873 - (-5,807)}{4,2 - (-4,2)} = 1,19$$

$$A_v = \frac{|U_o|}{|U_i|} = \frac{|U_c|}{|U_b|} = \frac{8,1015 - (-2,926)}{25 - (-25)} = 0,31$$

THE END ./

PREPARE FOR YOUR GROUP PROJECT: BUILD YOUR OWN AUDIO AMPLIFIER

Step 0: Analyze the requirement and determine the specifications provided by lecturers (requirements)

Step 1: Choose topology: single amplifier EC or compound circuit (EC+darlington or cascade amplifier or EC + Power amplifier (class A, B or AB)

Step 2: Draw DC equivalent circuit

Step 3: Determine Q point so that it locates in the middle of DC loadline → get a set of equations

Step 4: Draw AC equivalent circuit

Step 5: Determine gain A_v , A_i and impedances Z_{in} , Z_{out} → get a set of equations

Step 6: Analyze the frequency response - > get a set of equations

Step 7: Map all equations from step 3 to 6 to the requirements → define/choose the suitable component values (BJT with β , Resistors, Power supply E_c , capacitor C etc.)

Step 8: Use design tools (Circuit Maker/ Proteus/LTSPICE to verify your choice of components from step 7 by comparing your simulation and calculation results with the requirements and make comments.

Step 9: Optimize your circuit to meet the requirement and get a **new set of all component values!**

Step 10: Layout with ALTIUM, make PCB and then weld components to PCB and testing

Step 11: Compare the testing results with the requirement and then Optimize your circuit if it doesn't meet the requirement!

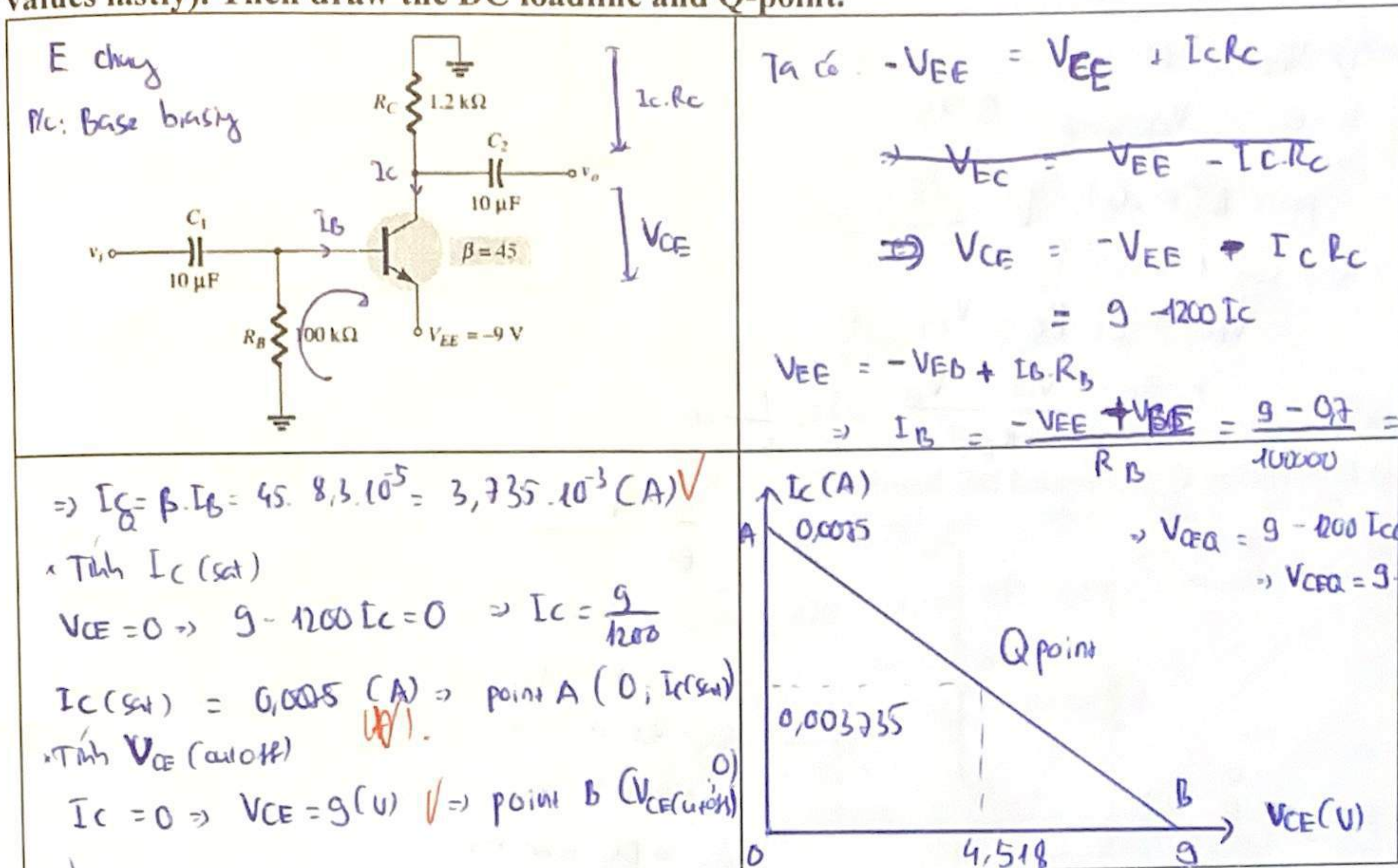
Step 12: Report your work: oral and writing presentation.

ENJOY YOUR DESIGN TASK!

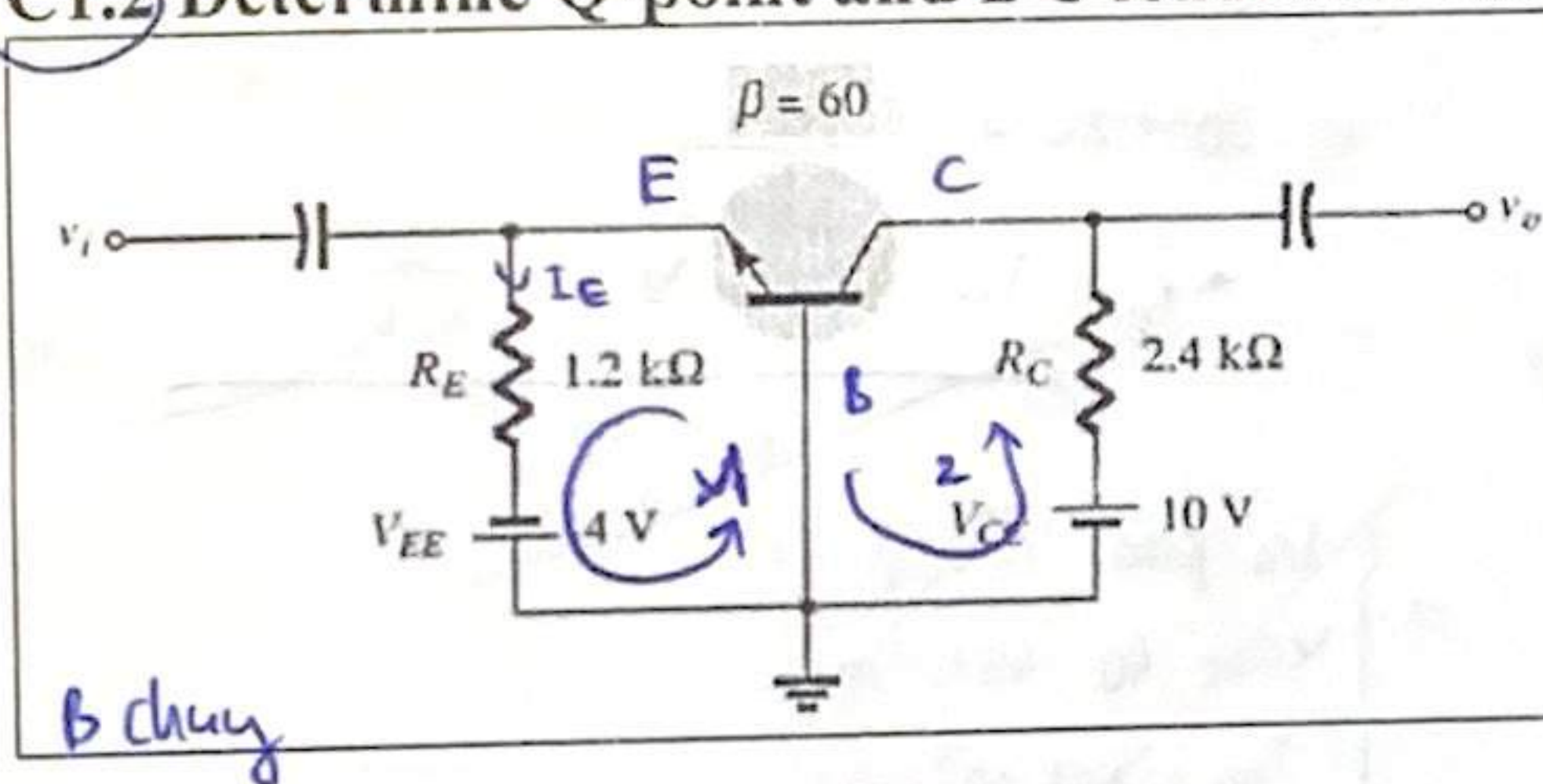
WEEK 4

HOMEWORK of Chapter 1. BJT

C1.1 Determine Q-point and DC loadline? (Always show the equations first and replace the values lastly). Then draw the DC loadline and Q-point.



C1.2 Determine Q-point and DC loadline? Then draw the DC loadline and Q-point.



ta có 1+2: $V_{CC} + V_{EE} = I_E R_E + I_C R_C + V_{CE}$
($I_E \approx I_C$)

$\Rightarrow V_{CE} = V_{CC} + V_{EE} - I_C (R_E + R_C) = I_E R_E$

$\left(\frac{I_E}{I_C} = \frac{1}{\alpha} = \frac{\beta+1}{\beta} \right) \Rightarrow V_{CE} = V_{CC} + V_{EE} - I_C R_C - \frac{\beta+1}{\beta} I_C R_E$

$\Rightarrow V_{CE} = 14 - 3620 I_C$

• Tính $I_{C(sat)}$

$$V_{CE} = 0 \Rightarrow 14 - 3620 I_C = 0 \Rightarrow I_C = \frac{14}{3620} = 3,87 \text{ (mA)}$$

→ point A (0, 0,00387)

• Tính $V_{CE(cutoff)}$

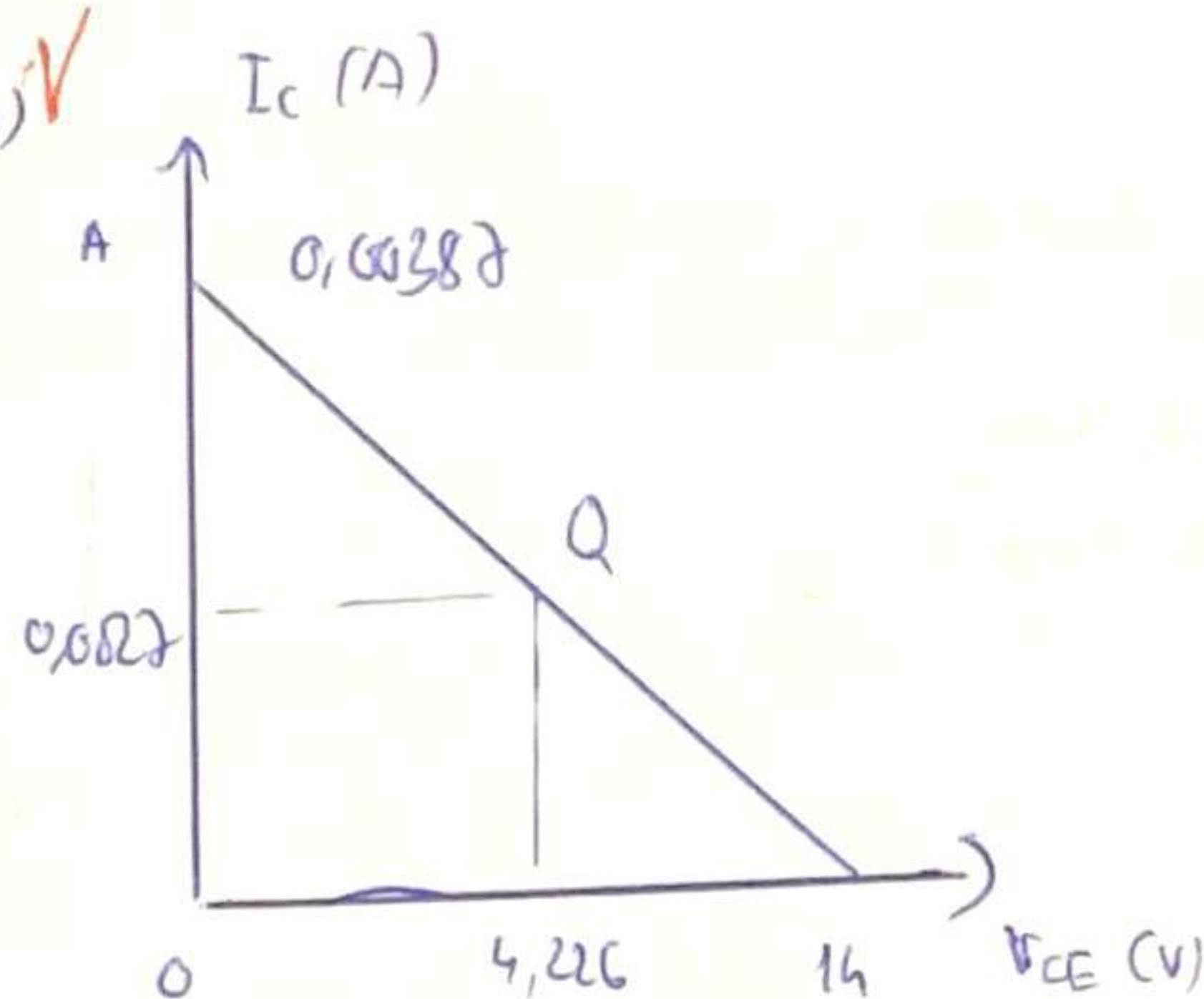
$$I_C = 0 \Rightarrow V_{CE(cutoff)} = 14$$

→ point B (14, 0)

• Xét loop 1 k có

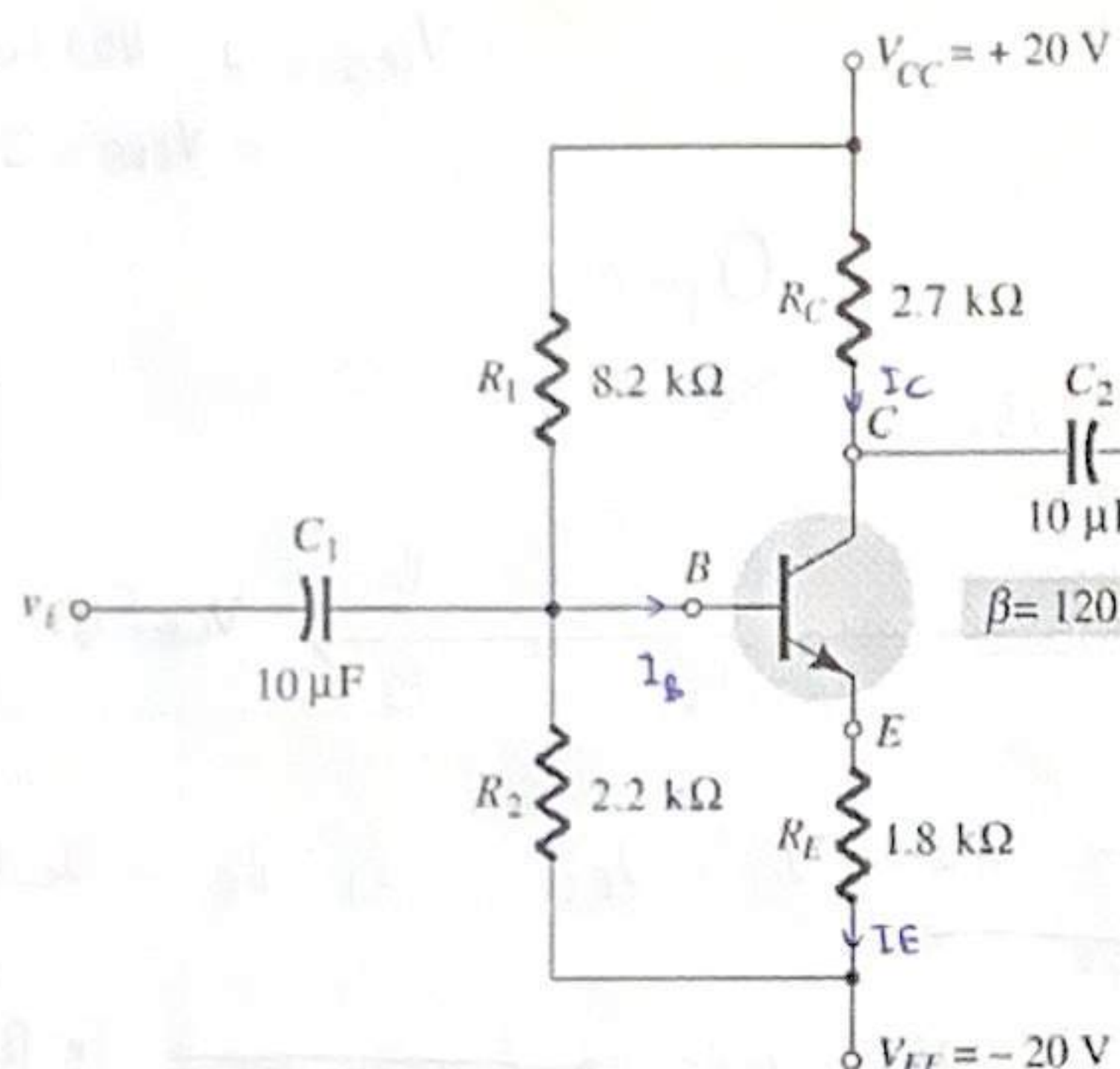
$$V_{BE} + I_E R_E - V_{EE} = 0$$

$$\Rightarrow I_E = \frac{V_{EE} - V_{BE}}{R_E} \Rightarrow I_{CQ} = \frac{\beta}{\beta + 1} I_E$$



C1.3

a) Determine Q-point and DC loadline?



thay vào direct V_{CEQ} ✓
 $\frac{60}{1200} - 0,7 = 0,0027 \text{ (A)} \Rightarrow 4,226 \text{ (V)}$

$$V_{CC} - V_{EE} = V_{RC} + V_{CE} + V_{RE} \Rightarrow V_{CE} = V_{CC} - I_C R_C - I_E R_E - V_{EE}$$

$$\Rightarrow V_{CE} = 40 - 4515 I_C - 12 I_E$$

$$V_{CE} = 0 \Rightarrow I_{C(sat)} = \frac{40}{4515} = 8,86 \text{ (mA)}$$

$$I_C = 0 \Rightarrow V_{CE(cutoff)} = 40 \text{ (V)}$$

→ DC load line

$$\text{• Tính } Q: V_{CC} - V_{EE} = V_{BE} + V_{CE} + V_{RE}$$

$$\Rightarrow 20 - 20 = 20 \cdot \frac{8,2}{2,2+8,2} + 0,7 + 12 I_E$$

$$I_B = \frac{V_{CC} - V_{EE} - V_{BE}}{R_1 + R_2} = \frac{20 - 20 - 0,7}{8,2 + 2,2} = -108 \cdot 10^{-4} \text{ (A)}$$

b) Determine AC loadline with R-load = 1kΩ?

$$r_e = \frac{26}{I_E} = \frac{26 \cdot 10^3}{0,0043} = 6046 \Omega$$

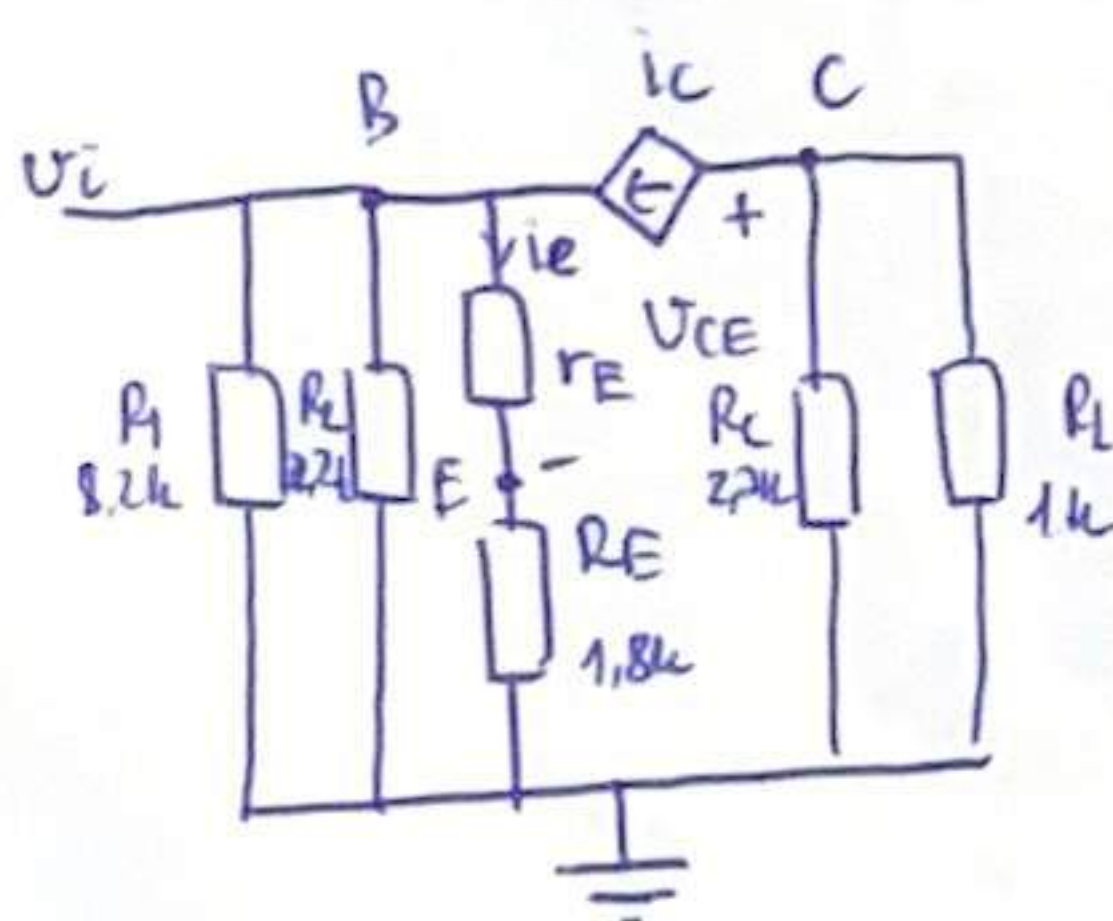
$$\begin{cases} i_C = i_c + I_{CQ} \\ V_{CE} = V_{CEQ} + V_{CEQ} \end{cases}$$

$$KVL: -i_C (R_C \parallel R_L) + i_E R_E = V_{CE}$$

$$\Rightarrow i_C = \frac{-V_{CE}}{(R_C \parallel R_L) + R_E}$$

$$i_C + I_{CQ} = \frac{-1 \cdot V_{CE}}{(R_C \parallel R_L) + R_E} + \frac{V_{CEQ}}{(R_C \parallel R_L) + R_E}$$

$$I_{CQ} = \frac{V_{CC} - V_{CEQ}}{R_C \parallel R_E}$$



$$I_C = 0 \Rightarrow V_{CE} = V_{CEQ} + I_{CQ} \cdot R$$

$$V_{CE} = 20,5885 + 0,0043 \left(\frac{2,7 \cdot 1}{2,7 + 1} + 1,8 \right) k$$

$$V_{CE} = 31,46 \text{ (V)}$$

$$V_{CE} = 0 \Rightarrow i_C = I_{CQ} + \frac{V_{CEQ}}{R} = 0,0043 + \frac{20,5885}{R} = 0,0124 \text{ (A)}$$

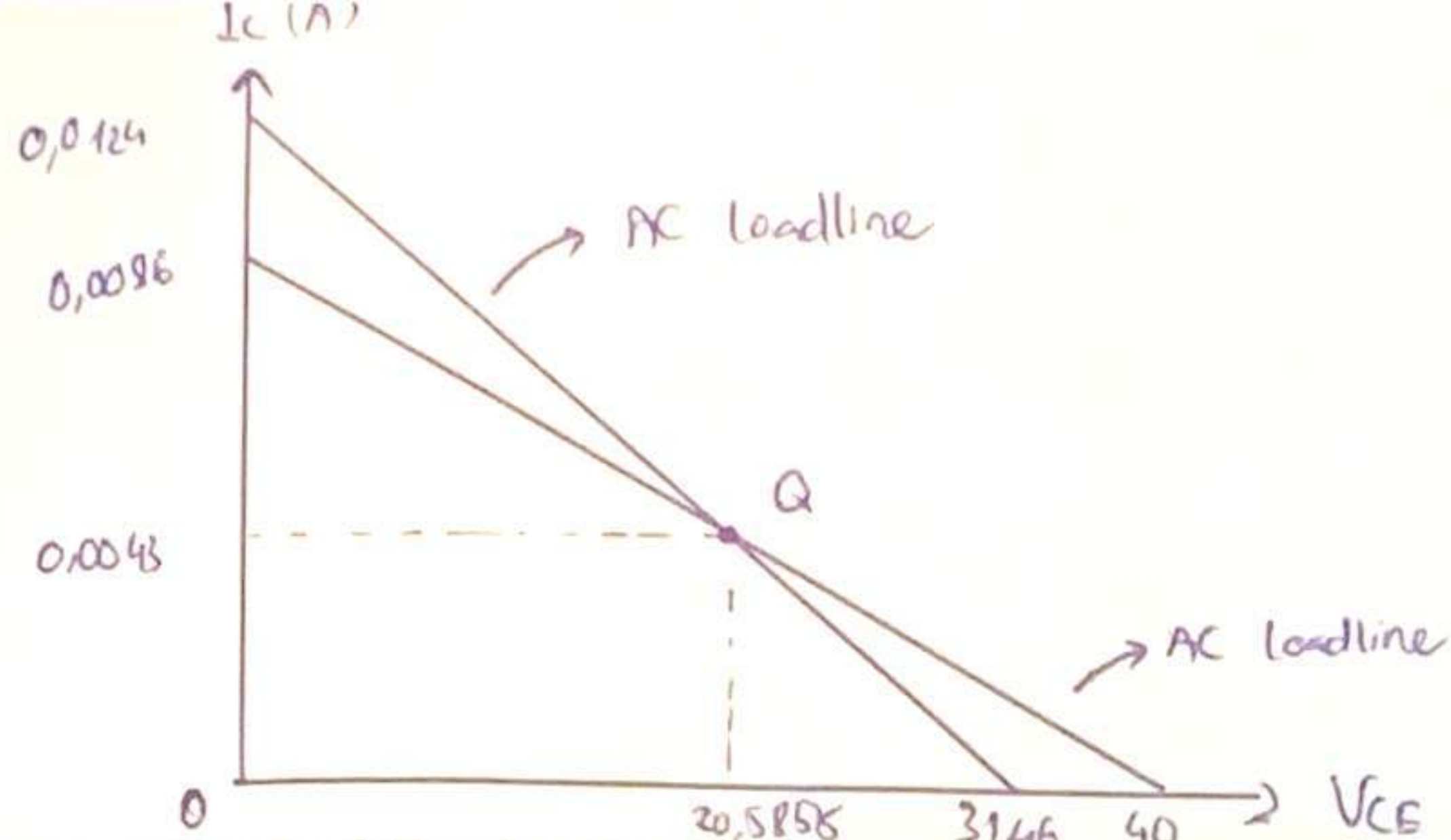
→ AC loadline

$$\begin{cases} I_{CQ} = \beta I_{BQ} = 0,013 \text{ (A)} \\ V_{CEQ} = 40 - 4515 I_{CQ} = \\ I_{BQ} = 108 \cdot 10^{-4} \text{ (A)} \end{cases}$$

$$V_{B} - V_{EE} = V_{BE} + V_{RE} \Rightarrow \frac{R_2 (V_{CC} - V_{EE})}{R_1 + R_2} = 0,7 + I_B (\beta + 1) R_E$$

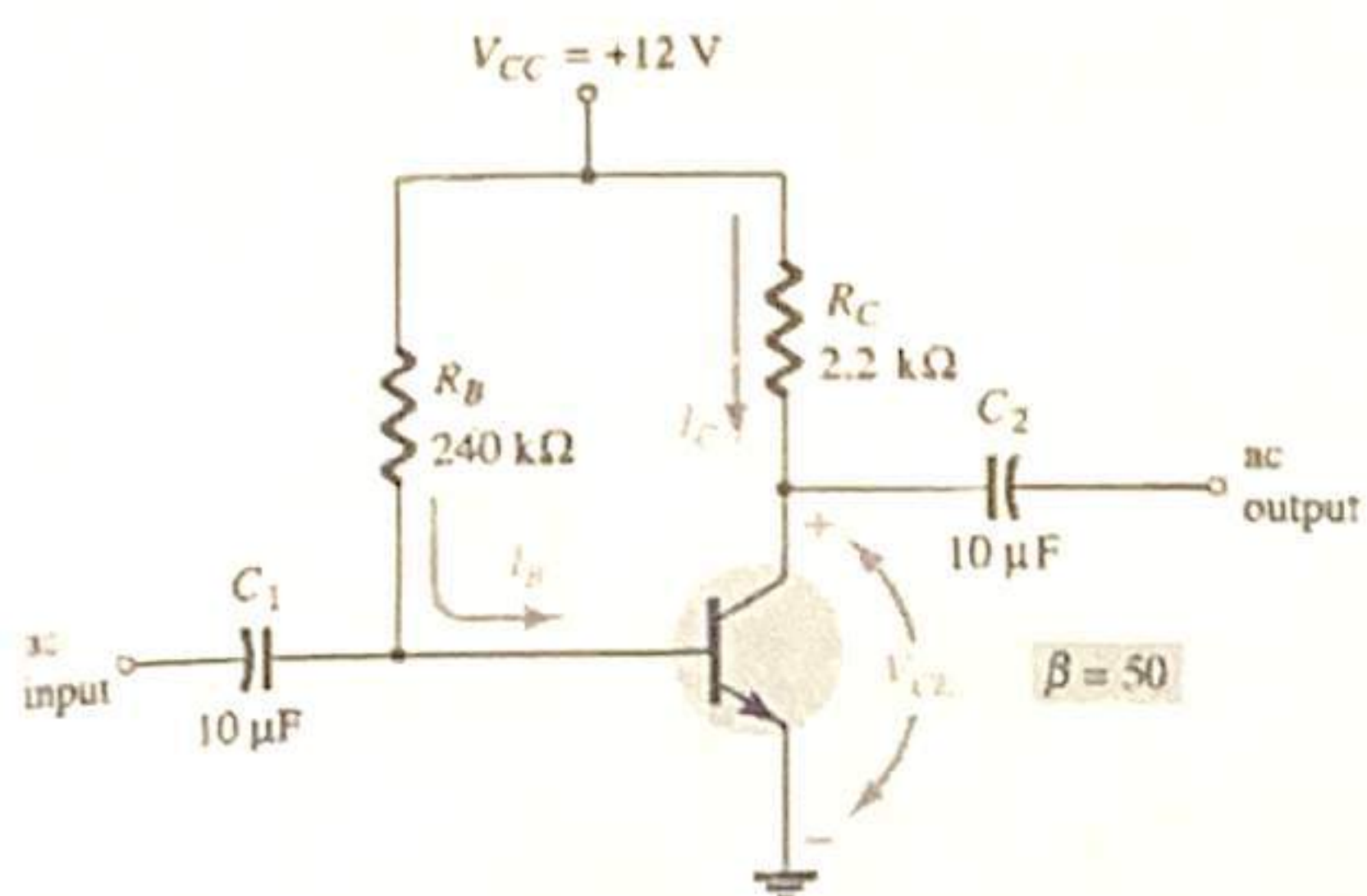
$$\Rightarrow I_B = \frac{R_2 (V_{CC} - V_{EE})}{R_1 + R_2} - \frac{0,7}{(\beta + 1) R_E} = \frac{3,56 \cdot 10^{-5}}{1,02 \cdot 10^4} \text{ (A)}$$

$$\begin{cases} I_{BQ} = 108 \cdot 10^{-4} \text{ (A)} \\ I_{CQ} = 0,0043 \text{ (A)} \\ V_{CEQ} = 20,5885 \text{ (V)} \end{cases}$$



C1.4: Determine the following for the fixed-bias configuration of Fig:

- I_{BQ} and I_{CQ}
- V_{CEQ}
- V_B and V_C
- V_{BE}



a. Ta có: $V_{CC} = V_{R_B} + V_{BE} + V_{CE} \Rightarrow V_{CC} = I_{BQ} \cdot R_B + V_{BE}$

$\Rightarrow I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = 4.71 \cdot 10^{-5}$ (A) ✓

$\Rightarrow I_{CQ} = \beta I_{BQ} = 2.355 \cdot 10^{-3}$ (A) ✓

b. Ta có: $V_{CC} = V_{R_C} + V_{CE} \Rightarrow V_{CC} = I_{CQ} \cdot R_C + V_{CE}$

$\Rightarrow$ p1 đs tại hnh: $V_{CE} = V_{CC} - I_{CQ} \cdot R_C$

$\Rightarrow V_{CEQ} = 12 - 2.355 \cdot 10^{-3} \cdot 2.2 \text{ k} = 6.819$ (V) ✓

c. $V_B = V_{CC} - I_{BQ} \cdot R_B = V_{BE} = 0.7$ (V) ✓

$V_C = V_{CC} - I_{CQ} \cdot R_C = V_{CE} = 6.819$ (V) ✓

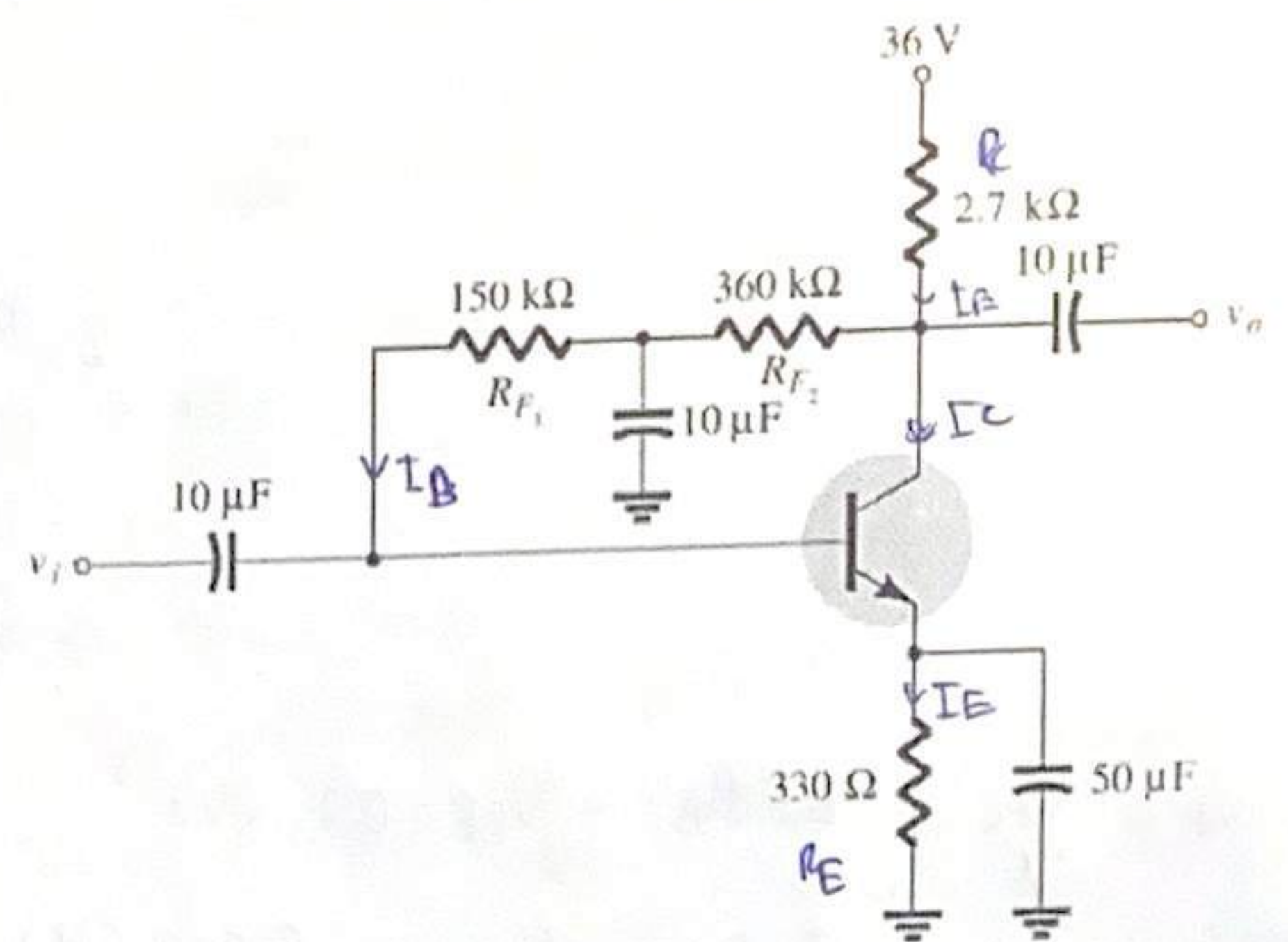
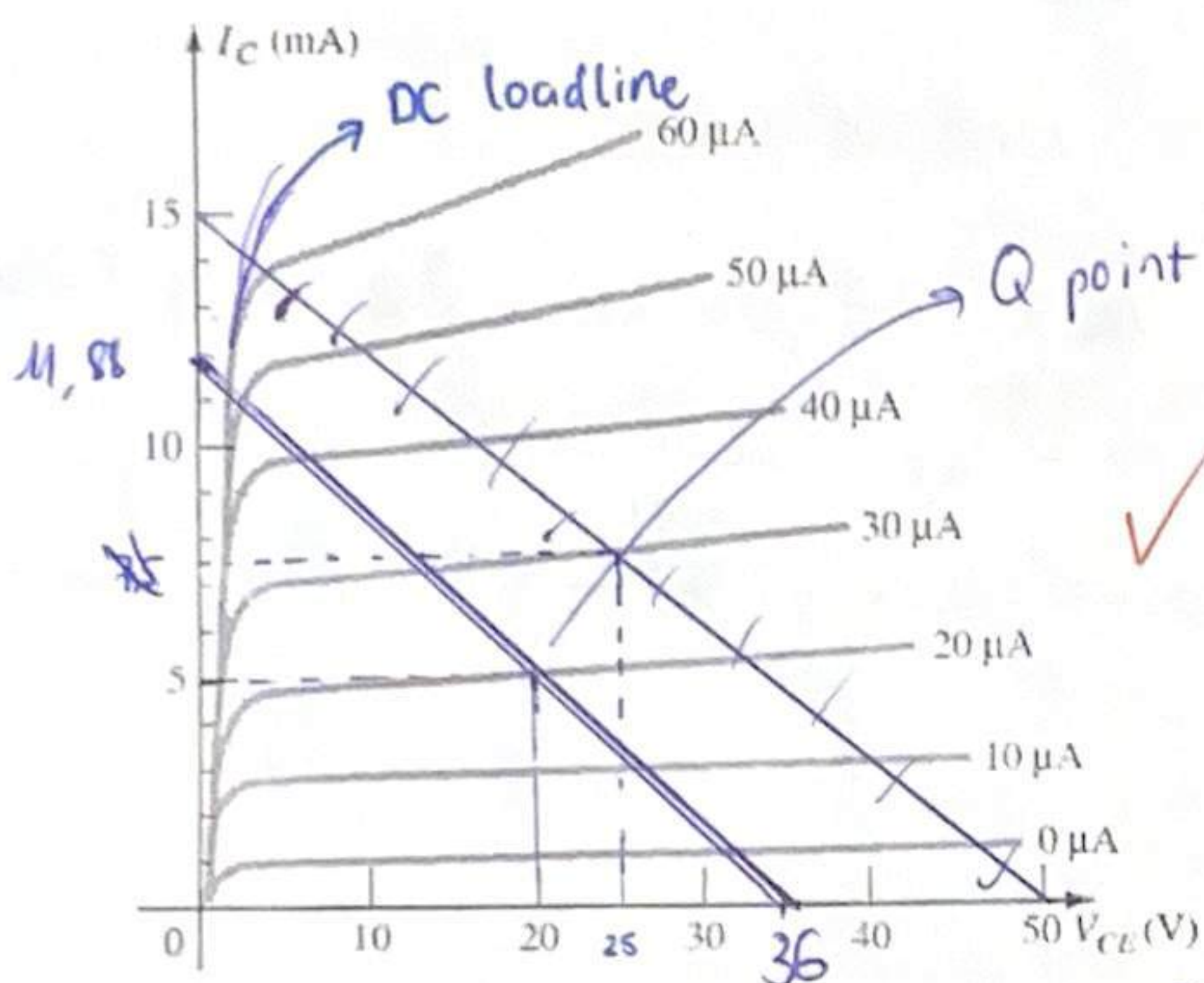
d. $V_{BE} = V_{BE} + V_{EC} = V_{BE} - V_{CE} = V_B - V_C = 0.7 - 6.819 = -6.119$ (V) ✓

C1.5: Given the network of Fig. 1.3b and the BJT characteristics of Fig. 1.3a.

- Draw the load line for the network on the characteristics.
- Determine the dc beta in the center region of the characteristics. Define the chosen point as the Q-point.
- Using the dc beta calculated in part b, find the dc value of I_B .
- Find $I_{C(Q)}$ and $V_{CE(Q)}$.

BJT characteristics.

Network for Example



a. DC loadline: đường nối giữa $(0; I_{C(sat)})$ và $(V_{CE(sat)}; 0)$

a. Phương trình đường tải tĩnh: $V_{CC} = V_{RC} + V_{CE} + V_E$
 $\Rightarrow V_{CC} = I_E(R_C + R_E) + V_{CE}$
 $\Rightarrow V_{CE} = V_{CC} - I_C(R_C + R_E)$

$\Rightarrow I_C = 0 \Rightarrow V_{CE(sat)} = V_{CC} = 36$ (V) ✓

$V_{CE} = 0 \Rightarrow I_{C(sat)} = \frac{V_{CC}}{R_C + R_E} = 11,8$ (mA) ✓

b. Chọn $I_B \approx 20 \mu A$

$\Rightarrow$ dựa vào đồ thị $\Rightarrow I_C \approx 5$ mA ✓

$\Rightarrow \beta_{dc} = \frac{I_{CQ}}{I_{BQ}} \approx \frac{5 \cdot 10^{-3}}{20 \cdot 10^{-6}} = 250$ ✓

b. $I_{CQ} = \frac{1}{2} I_{C(sat)} = 7,5$ mA
 dòng sang $\Rightarrow I_{BQ} = 30 \mu A$
 $\Rightarrow \beta = \frac{I_{CQ}}{I_{BQ}} = \frac{7,5 \cdot 10^{-3}}{30 \cdot 10^{-6}} = 250$
 (vì $\beta = 250 \gg 1 \Rightarrow \alpha \approx 1 \Rightarrow I_C \approx I_E$)

c. Dùng KVL ta có:

$$V_{CC} - V_{RC} - V_{F1} - V_{F2} - V_{BE} - V_{RE} = 0$$

$$\Rightarrow V_{CC} = I_E R_C + I_B (R_{F1} + R_{F2}) + V_{BE} + I_E R_E$$

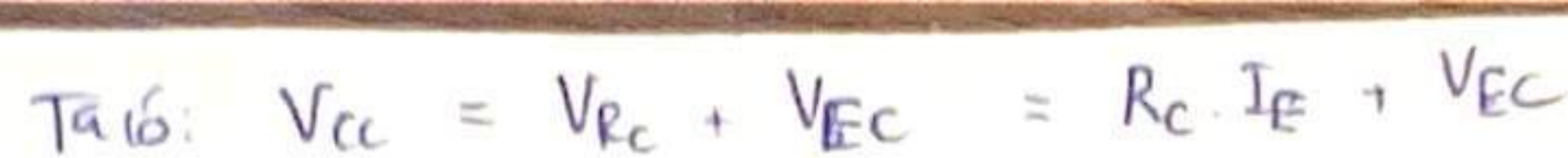
$$V_{CC} = [(\beta + 1)(R_C + R_E) + (R_{F1} + R_{F2})] I_B + V_{BE}$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{(\beta + 1)(R_C + R_E) + (R_{F1} + R_{F2})} = 27,78 \text{ (}\mu\text{A)} \quad \checkmark$$

d. Dùng công thức có ở câu a:

$$\begin{cases} I_{CQ} = \beta I_B = 6,945 \text{ (mA)} \quad \checkmark \\ I_{EQ} = (\beta + 1) I_B = 6,973 \text{ (mA)} \quad \checkmark \\ V_{CEQ} = V_{CC} - I_{CQ} (R_C + R_E) = 14,957 \text{ (V)} \quad \checkmark \end{cases}$$





$$\Rightarrow I_{C(sat)}|_{V_{EC}=0} = \frac{V_{CC}}{R_C} = 0,00425 \cdot 10^{-3} (A)$$

$$V_{EC}(C_{H\text{alt}})_{F_c=0} = V_{CC} = 20V$$

$$V_{EC0} = 11,21 \text{ (V)}$$



$$V_{EC} = V_{RC} + V_{RB} + V_{BE} = (\beta + 1)I_B R_C + I_B R_B + V_{BE}$$

$$\Rightarrow I_B = \frac{V_{IC} - V_{BE}}{R_B + (\beta + 1)R_C} = 15,47 \cdot 10^{-6} \text{ A}$$

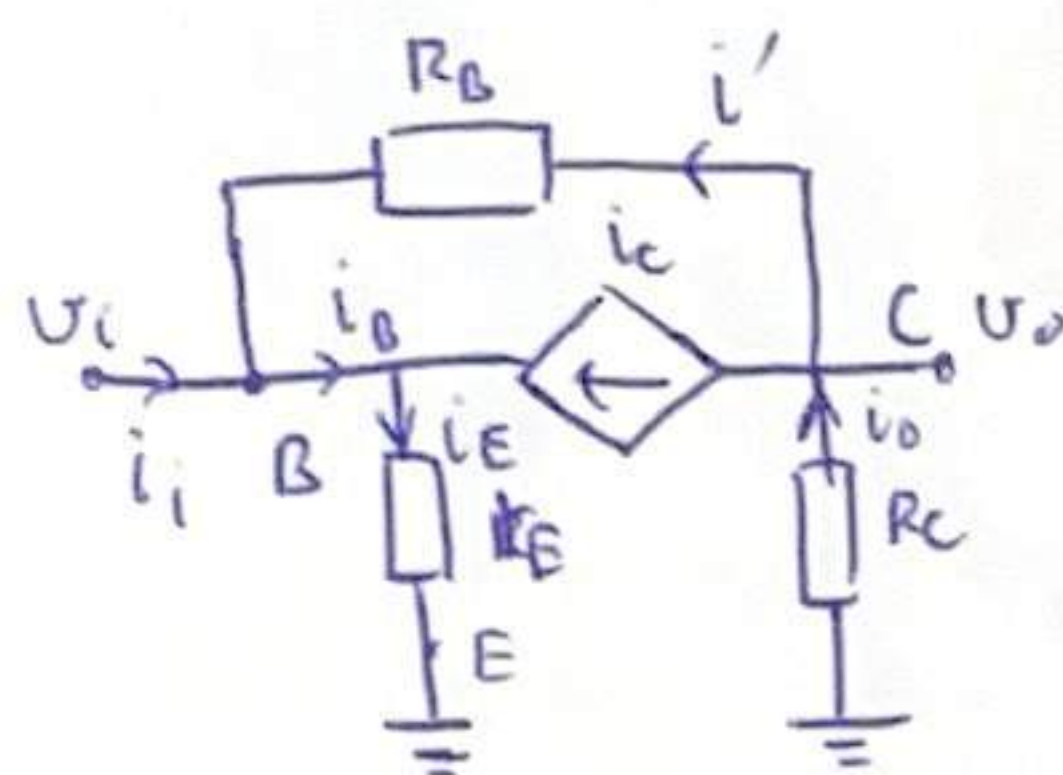
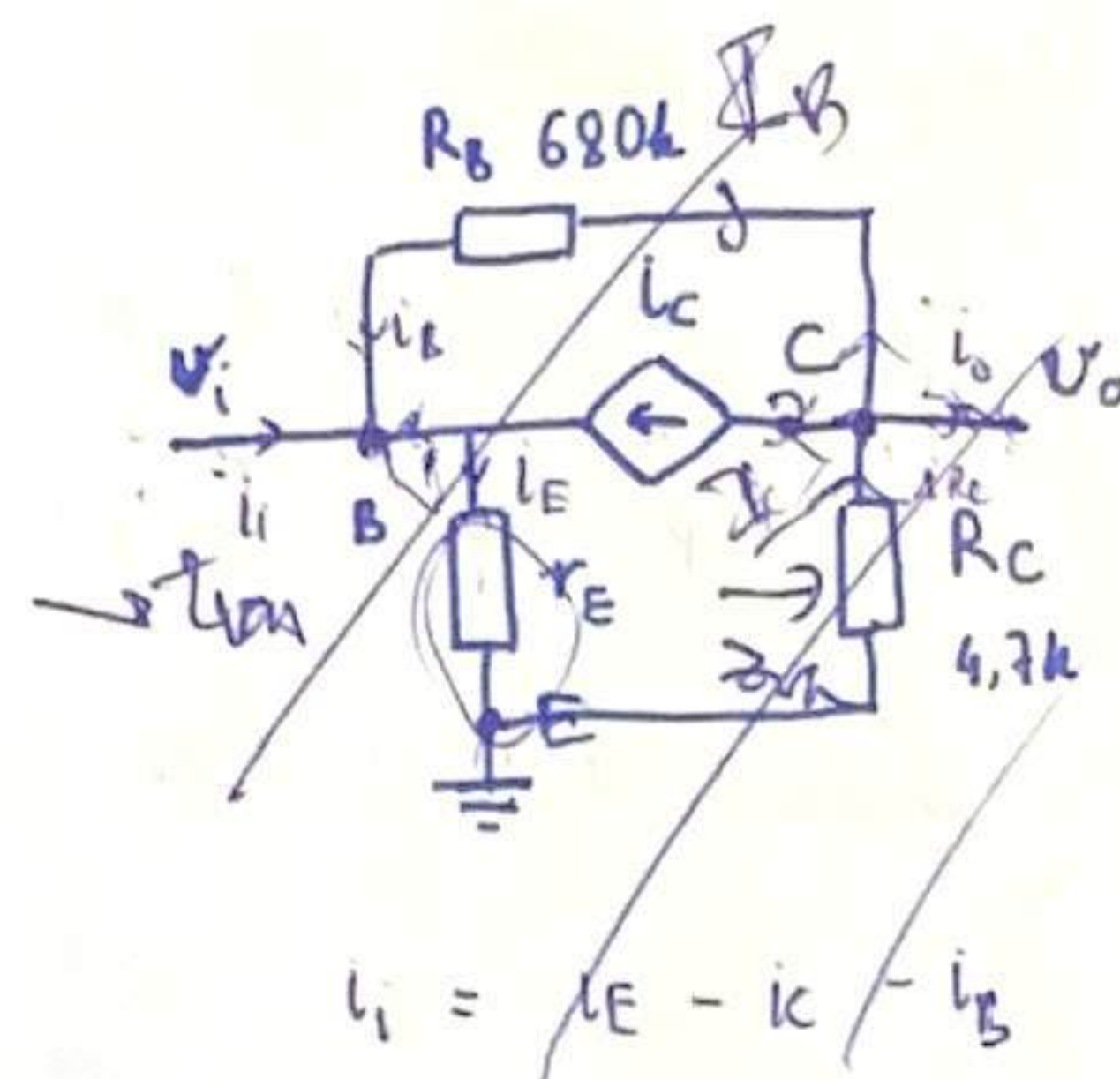
$$\Rightarrow I_E = (\beta + 1) I_B = 1,87 \text{ (mA)} = I_C$$

$$\Rightarrow r_e = \frac{26 \text{ mV}}{I_E} = 13,9 (\Omega)$$

WEEK 5

HOMWORK of Chapter 1. BJT (cont.)

C1.6 Determine voltage gain, current gain, input and output impedance of this circuit?



KCL: tại nút B: $i_0 = i_c + i'$
 — C: $i_B = i_1 + i'$

$$V_{CB} = V_C - V_B = -i_o R_C - i_E r_E = i' R_B$$

$$\Leftrightarrow -i_c R_c - i'_c R_c - i_E R_E = i'_c R_b$$

$$\Rightarrow i'(R_B + R_C) = -i_C R_C - i_E R_E$$

$$\Rightarrow i' = \frac{-i_C R_C - i_E R_E}{R_B \parallel R_E}$$

$$Z_i = \frac{v_i}{i_i} \Big|_{i_o=0} = \frac{i_E R_E}{i_B - i'} = \frac{i_E R_E}{i_B + \frac{i_C R_C + i_E R_E}{R_E + R_C}} = \frac{R_E}{\frac{1}{\beta} + \frac{R_C + R_E}{R_E + R_B}} = 913.4 (\Omega)$$

$$Z_o = \frac{V_o}{i_o} \Big|_{V_i=0} = R_C \parallel R_B = \frac{4.7 \cdot 10^3 \cdot 680 \cdot 10^3}{4.7 \cdot 10^3 + 680 \cdot 10^3} = 4.67 \text{ (k}\Omega\text{)} \quad \checkmark$$

$$A_i = \frac{i_o}{i_i} = \frac{i_c + i'}{i_b - i'} = \frac{i_c - \frac{i_c R_c + i_E R_E}{R_B + R_C}}{i_b + \frac{i_c R_c + i_E R_E}{R_B + R_C}} = \frac{i_c (R_B + R_C) - i_c R_c + i_E R_E}{i_b (R_B + R_C) + i_c R_c + i_E R_E} = \frac{\beta (R_B + (\beta + 1) r_E)}{R_B + (\beta + (\beta + 1) r_E)} \approx 65.26 \quad \checkmark$$

$$A_D = \frac{V_o}{V_i} = \frac{-i_e R_c}{i_e r_E} = \frac{\left(-i_c + \frac{i_c R_c + i_E r_E}{R_B + R_c} \right) R_c}{i_E r_E} = \frac{(-i_c (R_B + R_c) + i_c R_c + i_E r_E) R_c}{i_E r_E (R_B + R_c)}$$

$$= \frac{(-\beta R_B + (\beta+1)r_E) R_C}{(\beta+1)r_E(R_B + R_C)} = -333 \quad \checkmark$$

DC. $V_{BE} \approx \frac{R_2}{R_1 + R_2} (V_{CC} - V_{BE}) + I_B (\beta + 1) R_E = V_{BE}$

$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_1 + R_2} \cdot \frac{R_2}{(\beta + 1) R_E} = 3,56 \cdot 10^{-5} (A) \Rightarrow \begin{cases} V_{CEQ} = 20,585 (V) \\ I_{CQ} = I_{EQ} = 0,0043 (A) \end{cases} \quad (gdy \text{ bđ } 1.3)$

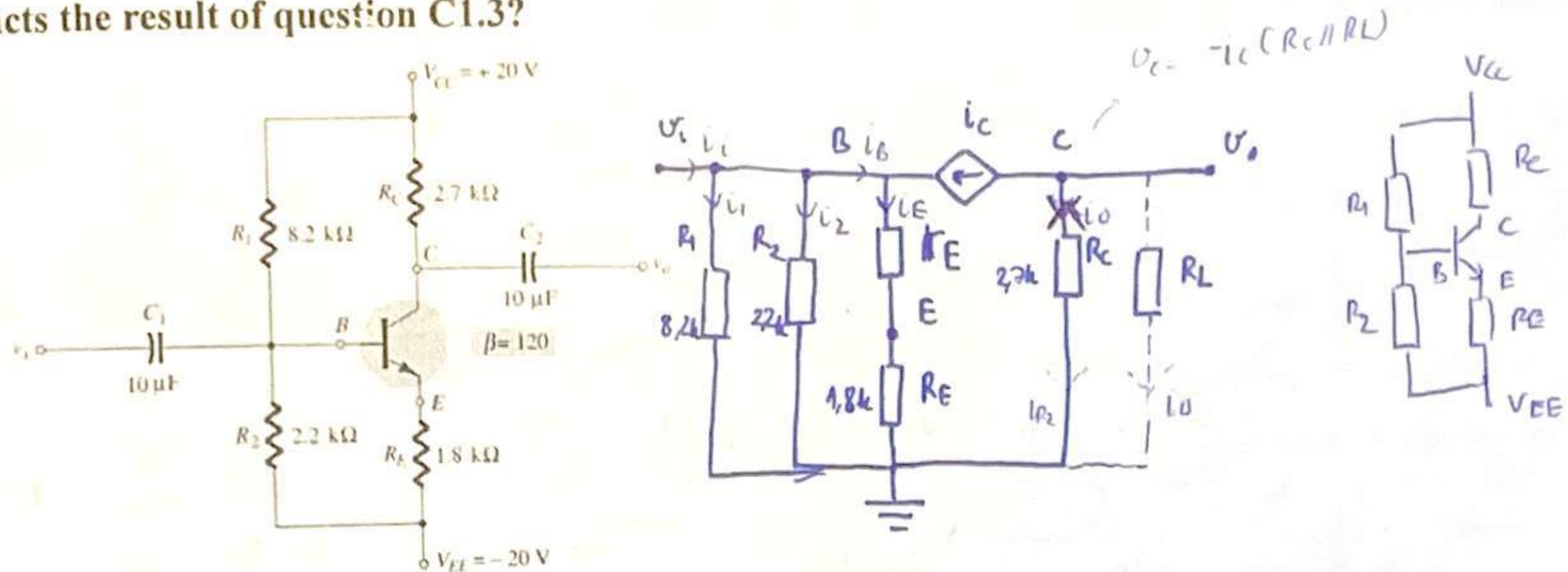
$\Rightarrow \text{trng AC} \begin{cases} r_e = \frac{26 \cdot 10^{-3}}{0,0043} = 6,05 (\Omega) \\ I_C = \beta I_B \end{cases}$

(26)

C1.7 a) Determine voltage gain, current gain, input and output impedance for the circuit without load?

b) Determine voltage gain, current gain, input and output impedance for the circuit with load equals 4 kΩ and for 4 Ω?

c) Comment on the results of question (b) and propose a solution? Verify if this solution impacts the result of question C1.3?



a. Kđ cđ kđ

$Z_i = \frac{U_i}{I_i} \Big|_{I_o=0} = \frac{I_E (R_E + r_E)}{I_B + I_1 + I_2} = \frac{I_E (R_E + r_E)}{I_B + \frac{I_E (R_E + r_E)}{\beta + 1}} = \frac{\beta (R_E + r_E)}{1 + \beta (R_E + r_E)} \quad \left((I_1 + I_2) R_1 \parallel R_2 = I_E (r_E + R_E) = V_{BE} \right)$

$= \frac{120 (6,05 + 1800)}{1 + \frac{120 (6,05 + 1800)}{8200 + 2200}} = 1720,8 (\Omega) \checkmark$

$Z_o = R_C = 2700 \Omega$

$A_v = \frac{U_o}{U_i} = \frac{-I_C R_C}{I_E (r_E + R_E)} = \frac{-R_C}{r_E + R_E} = -1,495 \checkmark$

$A_i = \frac{I_o}{I_i} = \frac{I_C}{I_B + I_1 + I_2} = \frac{I_C}{I_B + \frac{I_E (R_E + r_E)}{\beta + 1}} = \frac{\beta}{1 + \beta (R_E + r_E)} \approx 0,95 \checkmark$

b. Có tải $\Rightarrow$ kết quả

$Z_o = R_C \parallel R_L = \frac{U_o}{I_o} \Big|_{I_i=0} = \frac{R_C \cdot R_L}{R_C + R_L} \checkmark$

$Z_i = \frac{U_i}{I_i} \Big|_{I_o=0} = \dots = \frac{\beta (R_E + r_E)}{1 + \beta (R_E + r_E)} = 1720,8 (\Omega) \quad (\text{trng } kđ \text{ cđ } a) \checkmark$

$A_i = \frac{I_o}{I_i} = \frac{-I_C}{I_B + I_1 + I_2} = \frac{-I_C (R_C \parallel R_L) / R_L}{I_B + \frac{\beta (R_E + r_E)}{1 + \beta (R_E + r_E)}} = \frac{+ \frac{R_C}{R_C + R_L}}{\frac{1 + \beta (R_E + r_E)}{\beta (R_C \parallel R_L)}} \checkmark$

$A_v = \frac{U_o}{U_i} = \frac{U_{R_L}}{I_E (R_E + r_E)} = \frac{U_C}{I_E (R_E + r_E)} = \frac{-I_C (R_L \parallel R_C)}{I_E (R_E + r_E)} = \frac{-R_L \parallel R_C}{R_E + r_E} \checkmark$

$\Rightarrow$ thay số vào với $R_E = 4k\Omega$ và $R_L = 4\Omega$

c. Dựa vào công thức tính A_i, A_v, Z_i, Z_o mà tính a, b, và có kết luận:

• $A_v \leftarrow R_L \text{ nhỏ} \rightarrow R_c // R_L \downarrow \rightarrow A_v \downarrow$
 $R_L \text{ rất lớn} (R_L \rightarrow \infty) \rightarrow A_v \approx A_v (k_o \text{ tải})$

• $A_i \leftarrow R_L \text{ lớn} \rightarrow R_c // R_L \downarrow \rightarrow A_i \downarrow$
 $R_L \text{ rất lớn} (R_L \rightarrow \infty) \rightarrow A_i \approx A_i (k_o \text{ tải})$

• $A_i \leftarrow R_L \text{ nhỏ} \rightarrow \frac{R_c}{R_c + R_L} \rightarrow 1 \rightarrow A_i \approx A_i (k_o \text{ tải})$

$R_L \text{ lớn} \rightarrow A_i \downarrow$

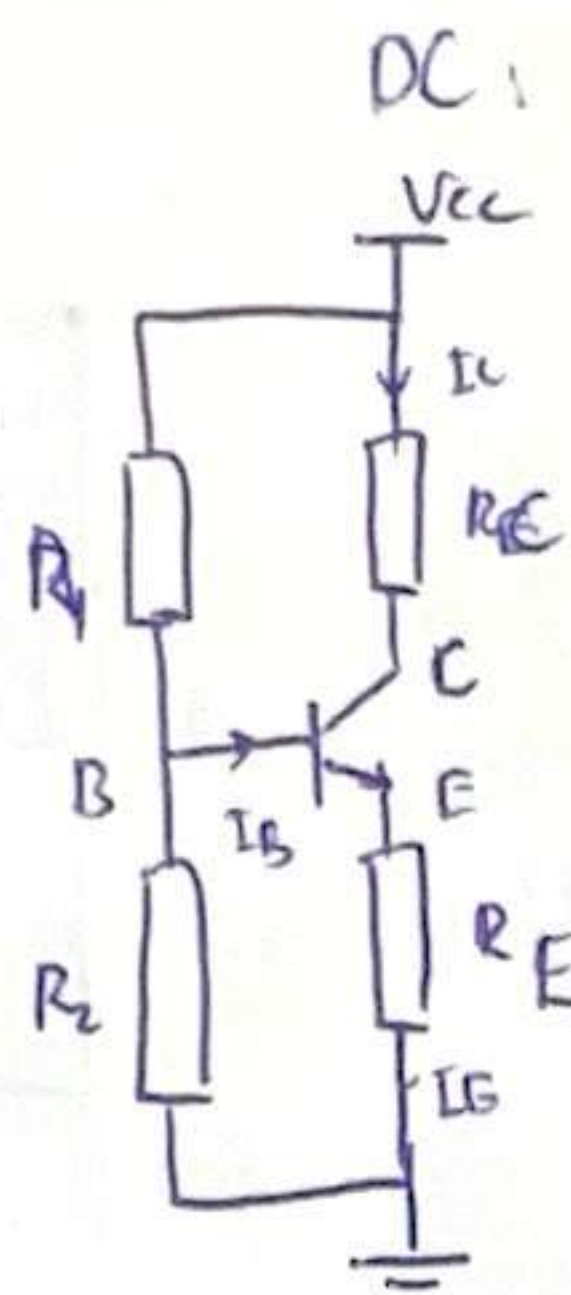
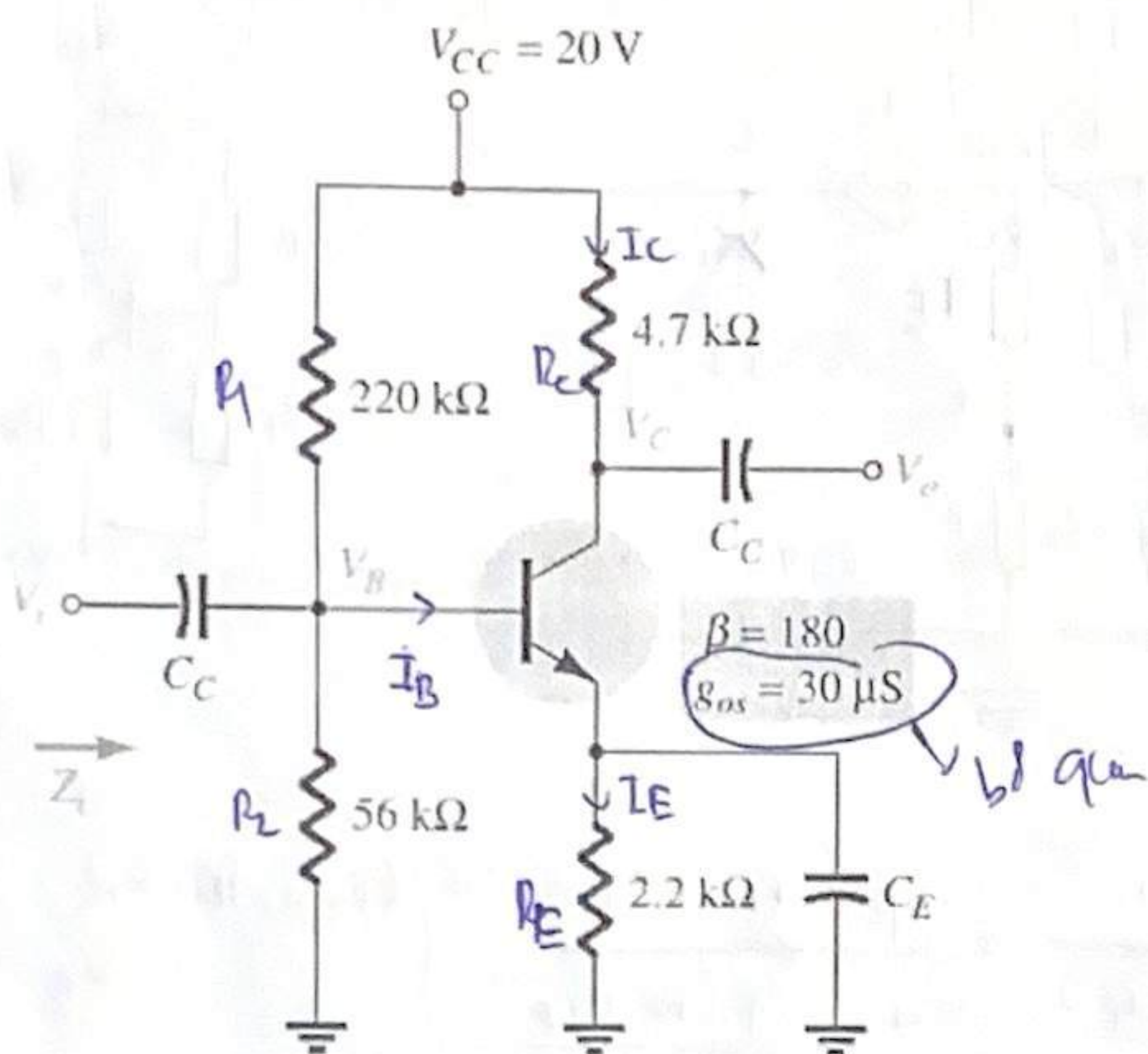
• Z_i không phụ thuộc $R_L \rightarrow$ không đổi

• $Z_o \leftarrow R_L \text{ nhỏ} \rightarrow Z_o \text{ cũng nhỏ}$
 $R_L \text{ càng lớn} (R_L \rightarrow \infty) \rightarrow Z_o \rightarrow R_c$ ($Z_o = R_c // R_L$)

• Khi ghép tải, dùng thêm CC sau EC để R_L không ảnh hưởng đến hệ thống (vì CC có Z_i cao, Z_o thấp, $A_v \approx 1$)
 Nếu thay đổi giá trị R_c, R_E thì V_{CE} sẽ thay đổi $\rightarrow$ ảnh hưởng đến Q-point $\rightarrow$ không đồng thời R_c, R_E

C1.8 For the network of Fig. 1.1 :

- Determine r_e .
- Calculate V_B and V_C .
- Determine Z_i and $A_v = V_o/V_i$.



KVL:

$$V_B = V_{BE} + V_{RE}$$

$$\frac{R_2}{R_1 + R_2} V_{CC} = V_{BE} + (\beta + 1) R_E I_B$$

$$\Rightarrow I_B = \frac{V_{CC} \frac{R_2}{R_1 + R_2} - V_{BE}}{(\beta + 1) R_E} = 8.43 \mu A$$

$$\Rightarrow I_C = \beta I_B = 1.52 \text{ mA}$$

$$I_E = (\beta + 1) I_B = 1.53 \text{ mA}$$

Fig 1.1

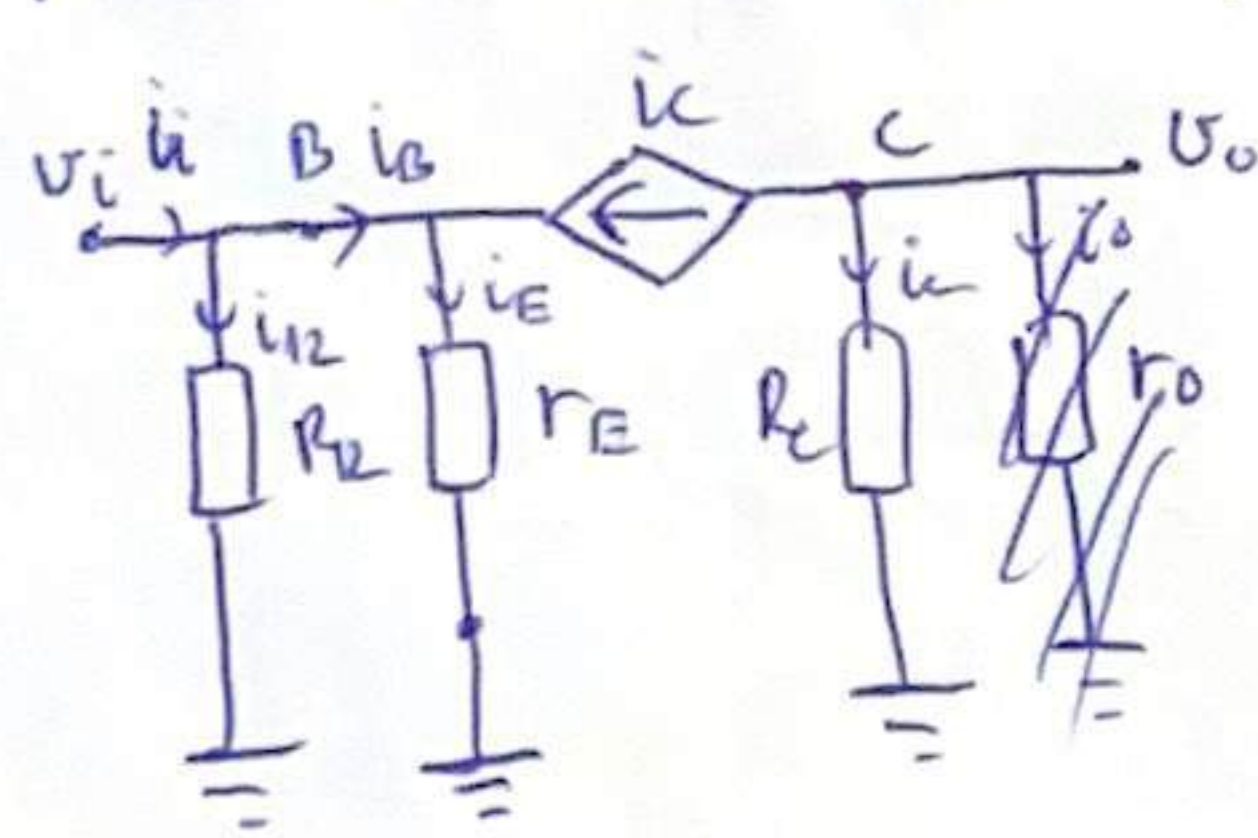
a. Ta có: $r_e = \frac{V_T}{I_E} = \frac{26}{1.53} = 17 \Omega$

b. $V_B = \frac{R_2}{R_1 + R_2} V_{CC} = 4.058 \text{ V}$

$$V_C = V_{CC} - V_{RC} = V_{CC} - I_C R_C = 12.856 \text{ V}$$

c. $r_o = \frac{1}{g_{os}} = 33.33 k\Omega$

Ta có mô hình tương đương T model.



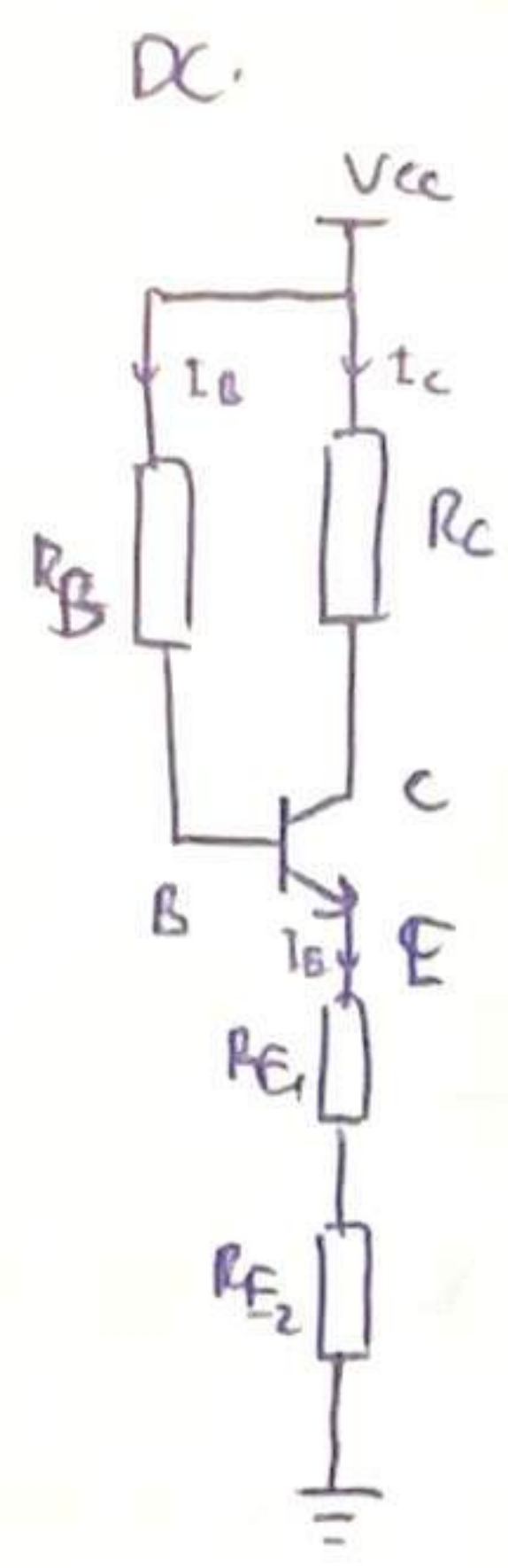
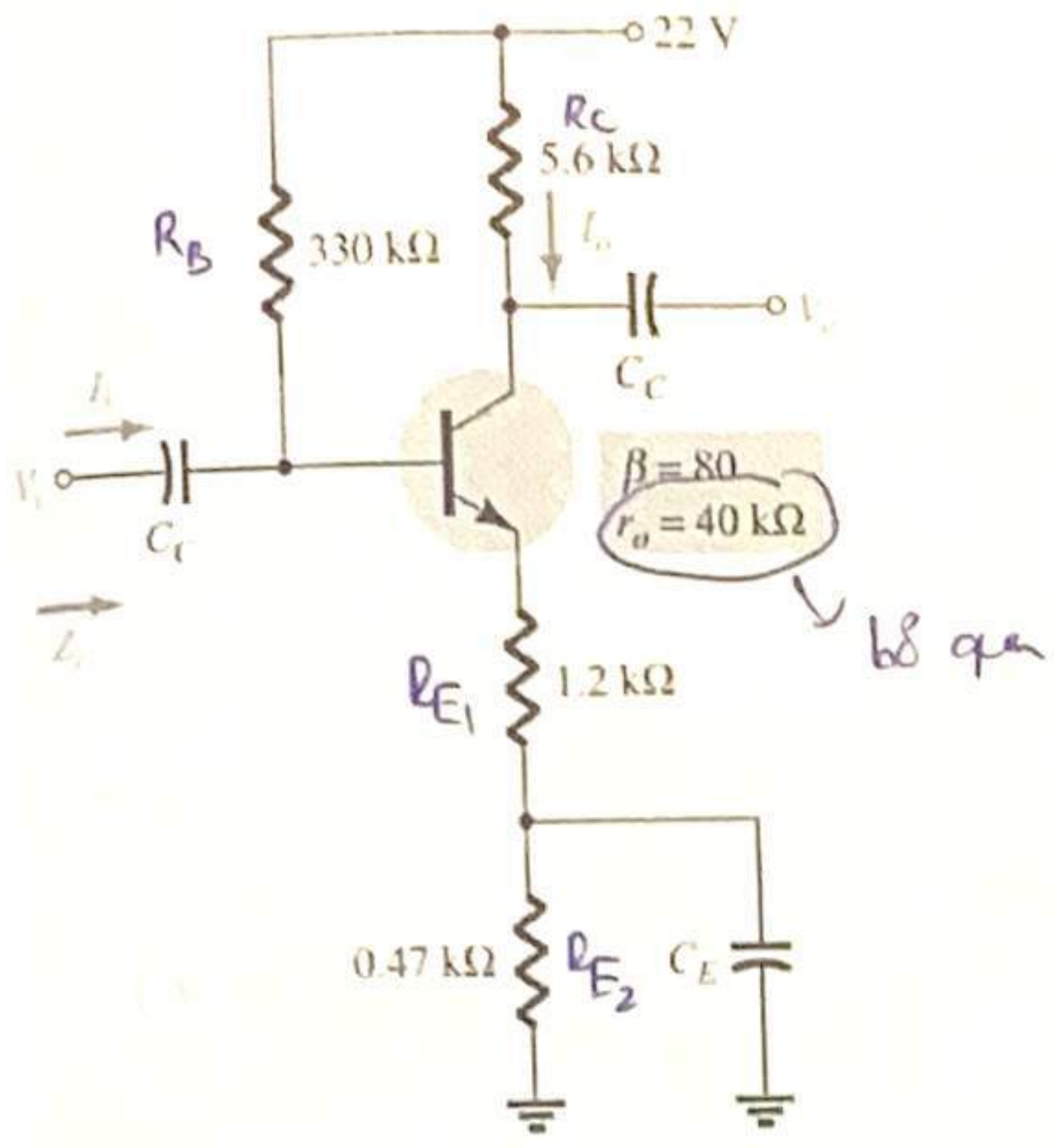
$$Z_{in} = \frac{v_i}{i_i} \Big|_{i_o=0} = \frac{I_E R_E}{I_B + I_E} = \frac{I_E R_E}{I_B + \frac{I_E R_E}{R_L}} = \frac{(\beta + 1) R_E}{1 + \frac{(\beta + 1) R_E}{R_L}}$$

$$\Rightarrow Z_{in} = 2878.6 \Omega$$

$$A_v = \frac{V_o}{V_i} = \frac{V_C}{V_B} = \frac{-i_c (R_C // r_o)}{I_E r_e} = \frac{-\beta}{\beta + 1} \cdot \frac{\frac{R_C // r_o}{R_E + r_e}}{r_e} = -240.97$$

$$-274.94$$

C1.9: For the network of Fig 1.3:
 a. Determine r_e .
 b. Determine Z_i and $A_v=V_o/V_i$.



KVL: $V_{CC} = V_{R_B} + V_{BE} + V_{R_{E1}} + V_{R_{E2}}$

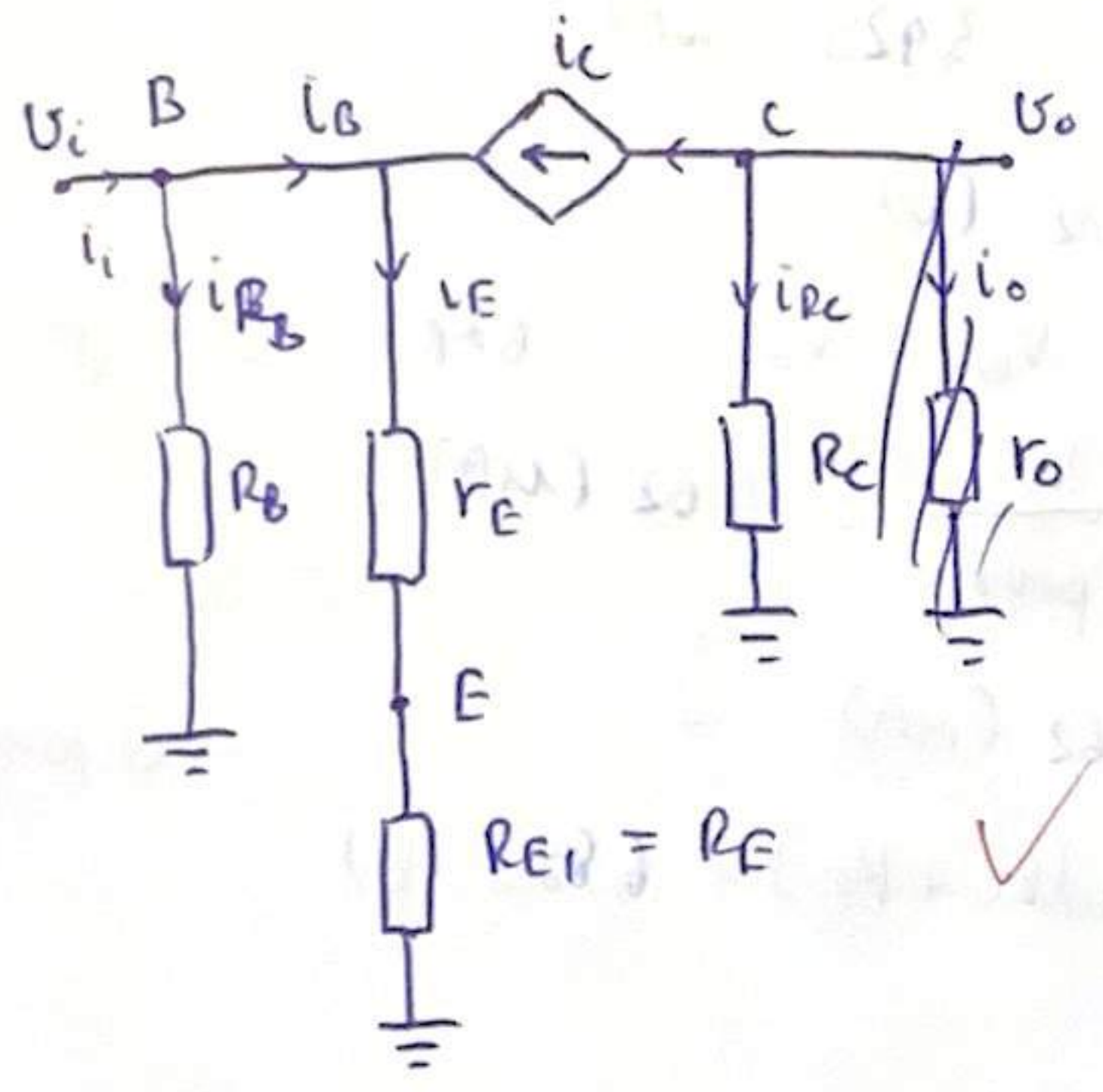
$\Rightarrow V_{CC} = I_B R_B + V_{BE} + I_B (\beta + 1) (R_{E1} + R_{E2})$

$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) (R_{E1} + R_{E2})} = 4.578 \cdot 10^{-5} (A)$

$\Rightarrow \begin{cases} I_C = \beta I_B = 3.66 (mA) \\ I_E = (\beta + 1) I_B = 3.81 (mA) \end{cases}$

a. Ta có: $r_e = \frac{V_T}{I_E} = \frac{26 mV}{3.81 mA} = 7 \Omega$

b. Ta có mô hình tương đương T model.



$Z_i = \frac{V_i}{I_i} \Big|_{I_o=0} = \frac{V_B}{I_B + I_{R_B}} = \frac{I_E (r_e + R_E)}{I_B + \frac{I_E (r_e + R_E)}{R_B}}$

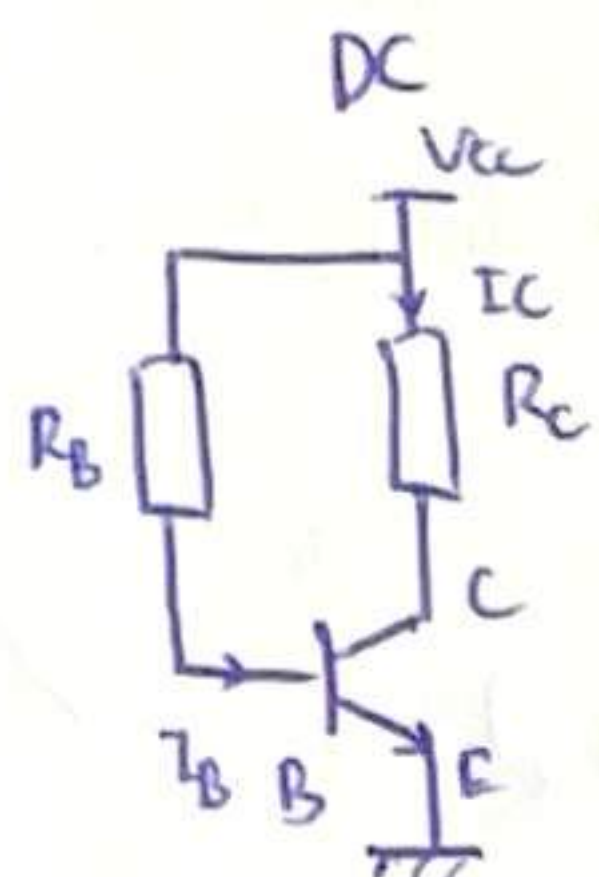
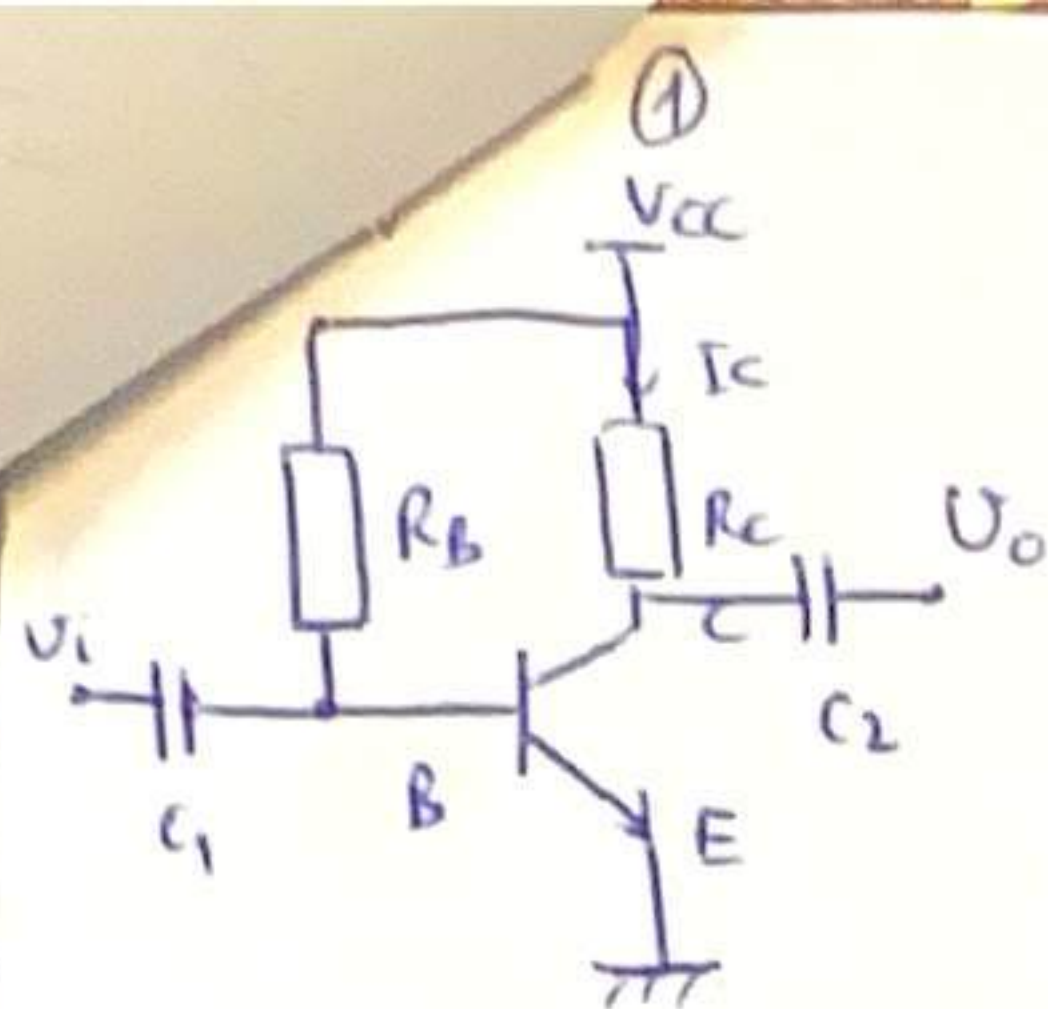
$(V_B = I_{R_B} R_B = I_E (r_e + R_E))$

$Z_i = \frac{(\beta + 1) (r_e + R_E)}{1 + \frac{(\beta + 1) (r_e + R_E)}{R_B}}$

đây cái này là kết quả tương đương

$Z_i = R_B \parallel ((\beta + 1) (r_e + R_E)) = \frac{1}{\frac{1}{R_B} + \frac{1}{(\beta + 1) (r_e + R_E)}} = \frac{1}{\frac{1}{330k} + \frac{1}{7 + 1.2k(81)}} = 75.42k \Omega$

$A_v = \frac{V_o}{V_i} = \frac{V_c}{V_i} = \frac{-i_c (R_C \parallel R_L)}{I_E (r_e + R_E)} = \frac{-\beta}{\beta + 1} \cdot \frac{\frac{R_C \cdot R_L}{R_C + R_L}}{r_e + R_E} \approx -4.02$



KVL: $V_{CC} = I_C R_C + V_{CE}$

$\Rightarrow V_{CC} = I_C R_C + V_{CE}$

$\Rightarrow \begin{cases} I_C = \frac{V_{CC} - V_{CE}}{R_C} \\ V_{CE} = V_{CC} - I_C R_C \end{cases}$

$I_C(\text{sat}) | V_{CE}=0 = \frac{V_{CC}}{R_C} = 4 \text{ (mA)}$

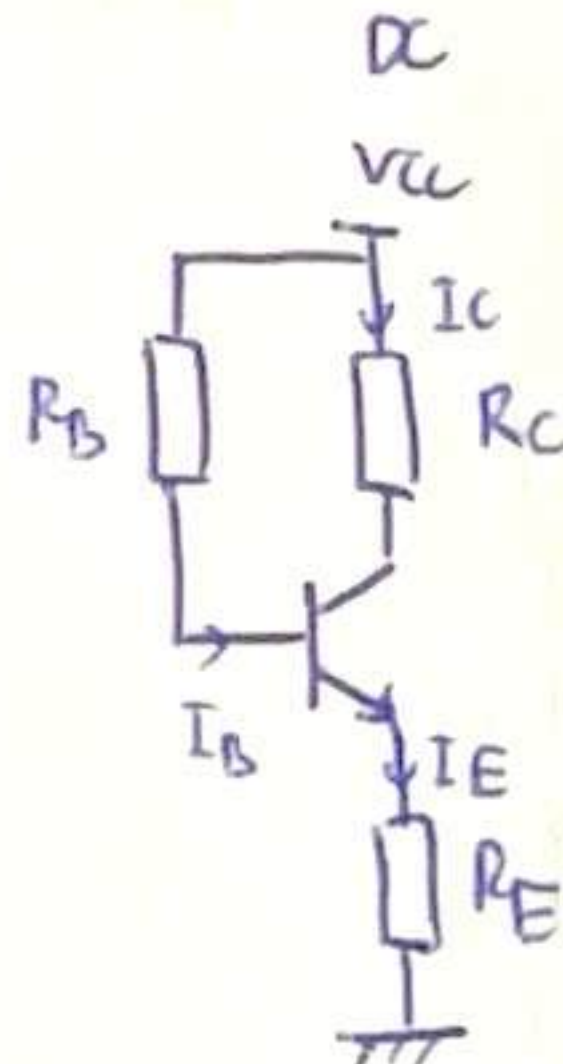
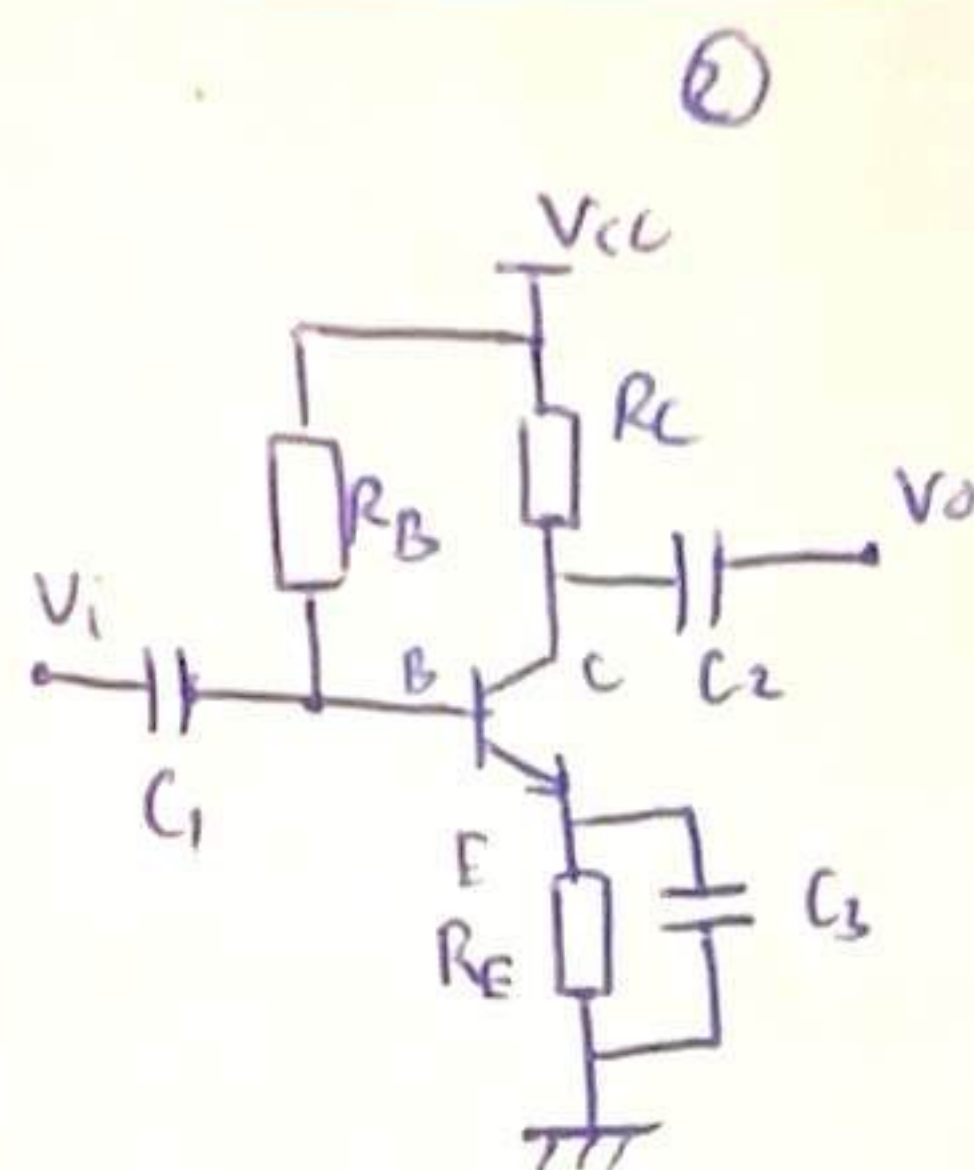
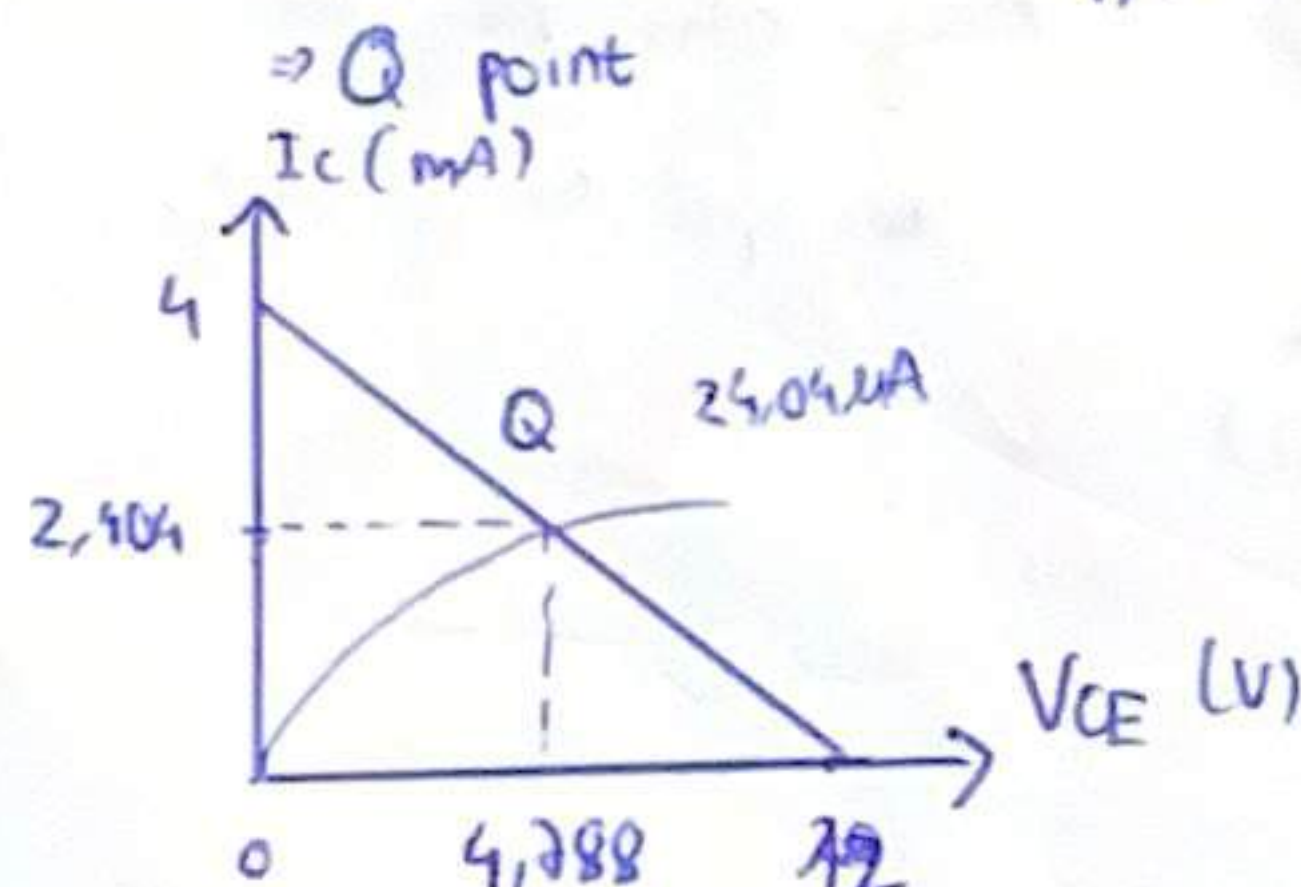
$V_{CE}(\text{cutoff}) | I_C=0 = V_{CC} = 12 \text{ (V)}$
 $\Rightarrow$ DC load line

$\Rightarrow I_B$ có Q point:

$V_{CC} = V_{RB} + V_{BE} = I_B R_B + V_{BE}$

$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{24.04} = 24.04 \text{ (}\mu\text{A)}$

$\Rightarrow \begin{cases} I_C(Q) = \beta I_B = 1.064 \text{ (mA)} \\ V_{CE}(Q) = V_{CC} - I_C(Q) R_C = 4.788 \text{ (V)} \end{cases}$



KVL: $V_{CC} = V_{RC} + V_{CE} + V_{RE}$

$\Rightarrow V_{CC} = I_C R_C + V_{CE} + I_E R_E$

$\Rightarrow V_{CC} = I_C (R_C + R_E) + V_{CE} \quad (I_C \approx I_E \text{ do } \beta = 100 \gg 1)$

$\Rightarrow \begin{cases} I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E} \\ V_{CE} = V_{CC} - I_C (R_C + R_E) \end{cases}$

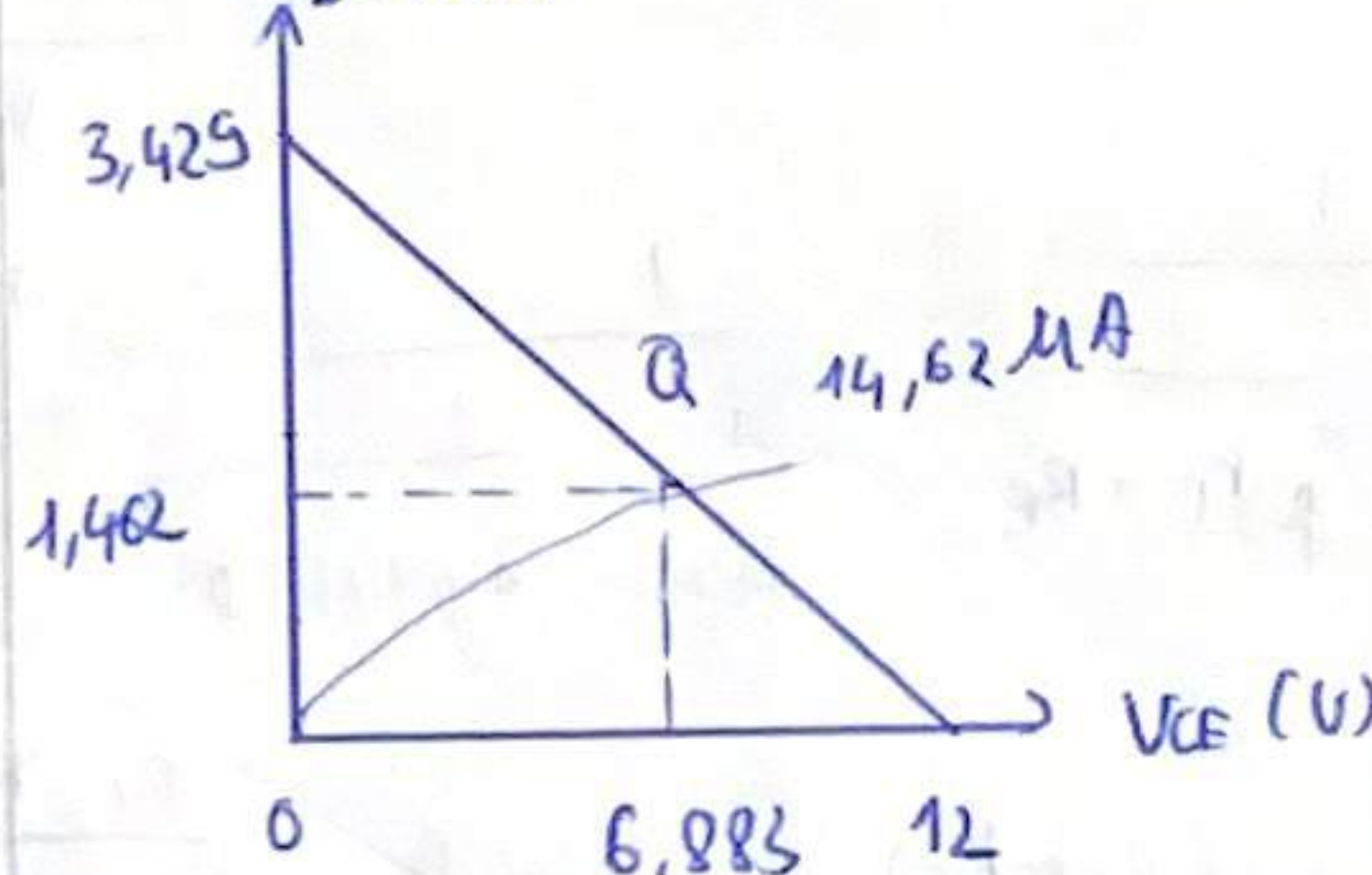
$I_C(\text{sat}) | V_{CE}=0 = \frac{V_{CC}}{R_C + R_E} = 3.429 \text{ (mA)}$

$V_{CE}(\text{cutoff}) | I_C=0 = V_{CC} = 12 \text{ (V)}$

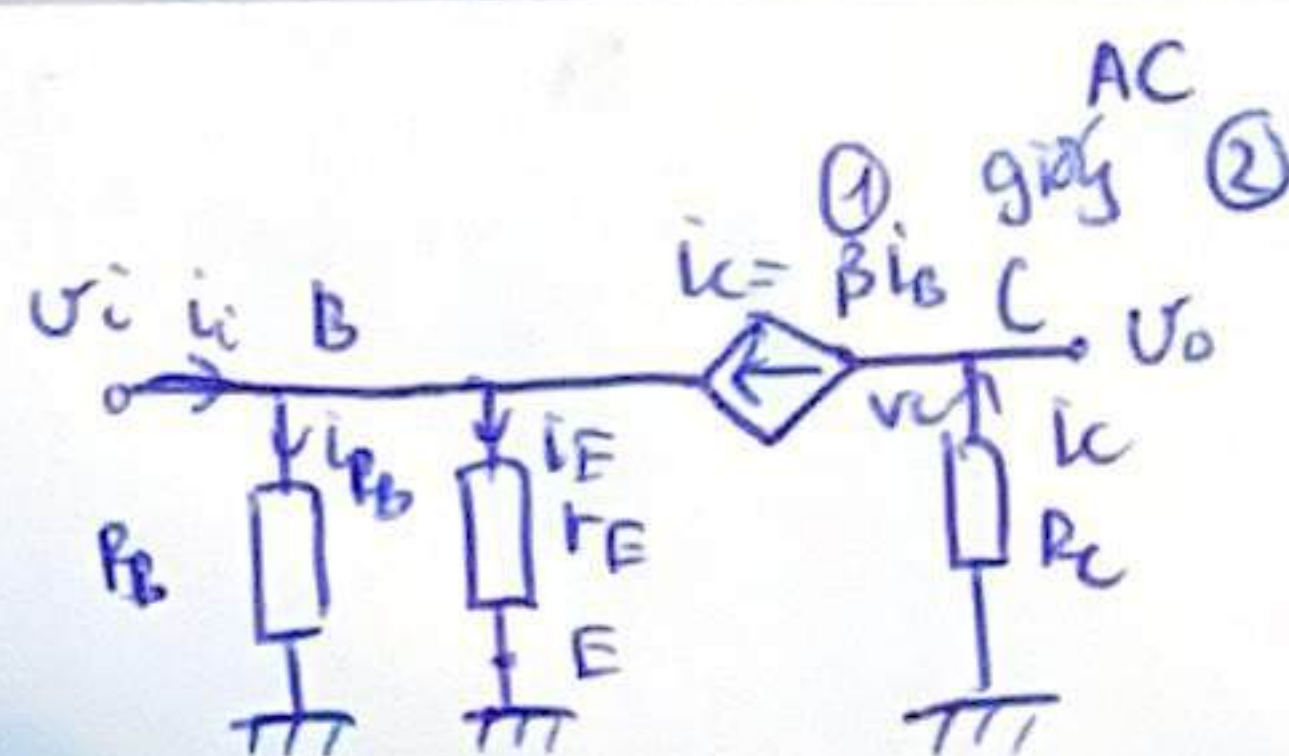
$\Rightarrow$ Q point $V_{CC} = V_{RB} + V_{BE} + V_{RE} = I_B R_B + V_{BE} + (\beta + 1) I_B R_E$

$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} = 14.62 \text{ (}\mu\text{A)}$

$\Rightarrow \begin{cases} I_C(Q) = \beta I_B = 1.462 \text{ (mA)} \\ V_{CE}(Q) = V_{CC} - I_C(Q) (R_C + R_E) = 6.883 \text{ (V)} \end{cases}$

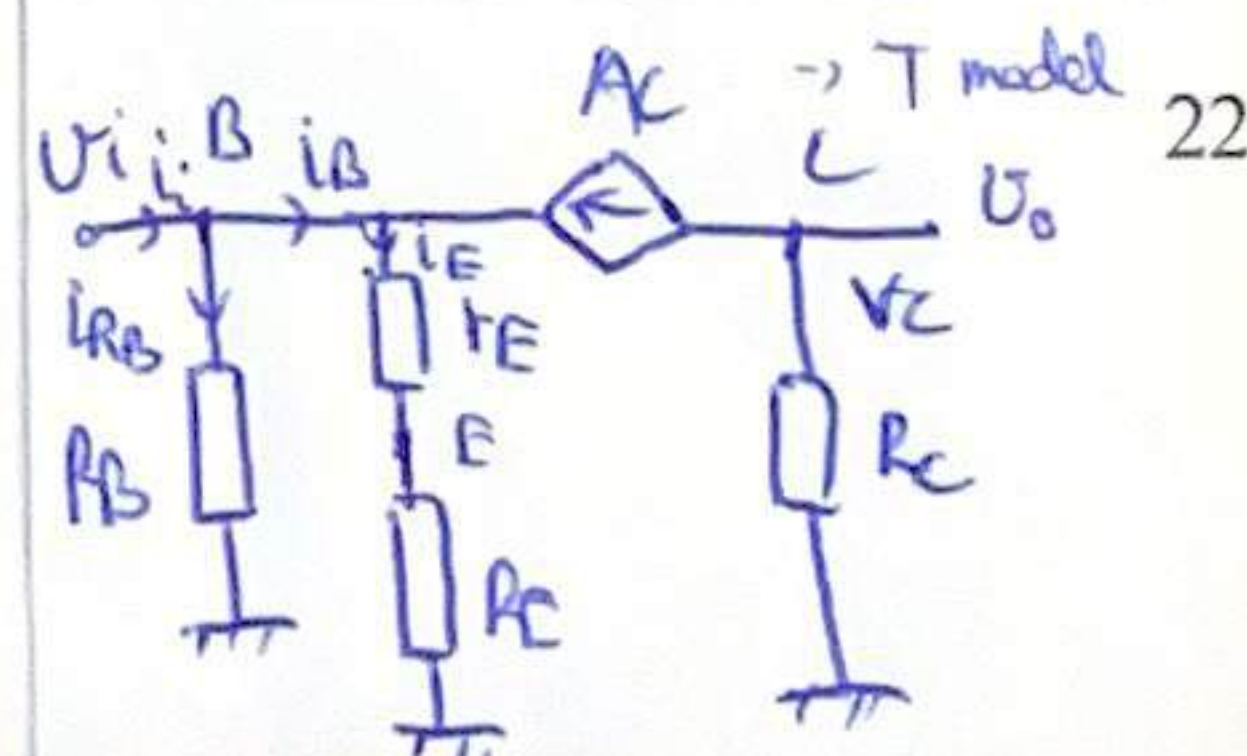


tray đó $r_E = \frac{26 \text{ mV}}{I_E} = 17.78 \text{ (}\Omega\text{)}$



$\Rightarrow$ T model

tray đó $r_E = \frac{26 \text{ mV}}{I_E} = 10.22 \text{ (}\Omega\text{)}$



① + ②

$$Z_0 = R_c = 3k\Omega$$

$$Z_i = R_B \parallel \beta r_E = \frac{R_B \cdot \beta r_E}{R_B + \beta r_E} = 1079,51 \Omega$$

$$A_{v0} = \frac{V_o}{V_i} = \frac{V_c}{V_b} = \frac{-i_c R_c}{i_E r_E} = \frac{-\beta}{\beta + 1} \frac{R_c}{r_E} \approx \frac{-R_c}{r_E} = -277,26$$

$$A_{v1} = \frac{i_o}{i_i} = \frac{i_c}{i_b + i_{R_B}} = \frac{i_c}{i_b + \frac{r_E \cdot i_E}{R_B}} = \frac{\beta}{1 + \frac{r_E (\beta + 1)}{R_B}}$$

$$A_{v1} \approx 99,97$$

$$I_E \approx I_C \quad (\beta = 100 \gg 1)$$

① un ② : $V_B = i_{R_B} R_B = r_E \cdot i_E \Rightarrow i_{R_B} = \frac{r_E \cdot i_E}{R_B}$

③ : $V_b = i_{R_B} R_B = (r_E + R_E) \cdot i_E \Rightarrow i_{R_B} = \frac{(r_E + R_E) \cdot i_E}{R_B}$

③

$$Z_0 = R_c = 3k\Omega$$

$$Z_i = R_B \parallel \beta (R_E + r_E) = \frac{R_B \cdot \beta (R_E + r_E)}{R_B + \beta (R_E + r_E)} = 46639,82 \Omega$$

$$A_{v0} = \frac{V_o}{V_i} = \frac{V_c}{V_b} = \frac{-i_c R_c}{i_E (r_E + R_E)} = \frac{\beta}{\beta + 1} \frac{-I_C R_c}{r_E + R_E}$$

$$A_{v0} \approx \frac{-R_c}{r_E + R_E} = -5,79$$

$$A_{v1} = \frac{i_o}{i_i} = \frac{i_c}{i_b + i_{R_B}} = \frac{i_c}{i_b + \frac{i_E (r_E + R_E)}{R_B}} = \frac{i_c}{i_b + \frac{i_E (r_E + R_E)}{R_B}}$$

$$R_B \gg 1$$

$$A_{v1} = \frac{\beta}{1 + \frac{(\beta + 1)(R_E + r_E)}{R_B}} = \frac{\beta R_B}{R_B + (\beta + 1)(R_E + r_E)}$$

$$A_{v1} = 99,99$$

①

②

③

 $I_E(\omega)$ 24,04 μA 14,62 μA 14,62 μA

DC

 $I_C(\omega)$

2,404 mA

1,462 mA

1,462 mA

 $V_{CE}(\omega)$

4,788 V

6,885 V

6,885 V

AC

 Z_i 1029,51 Ω 1029,51 Ω 46639,87 Ω Z_o 3 k Ω 3 k Ω 3 k Ω A_v

-277,26

-277,26

-5,779

 A_i

99,77

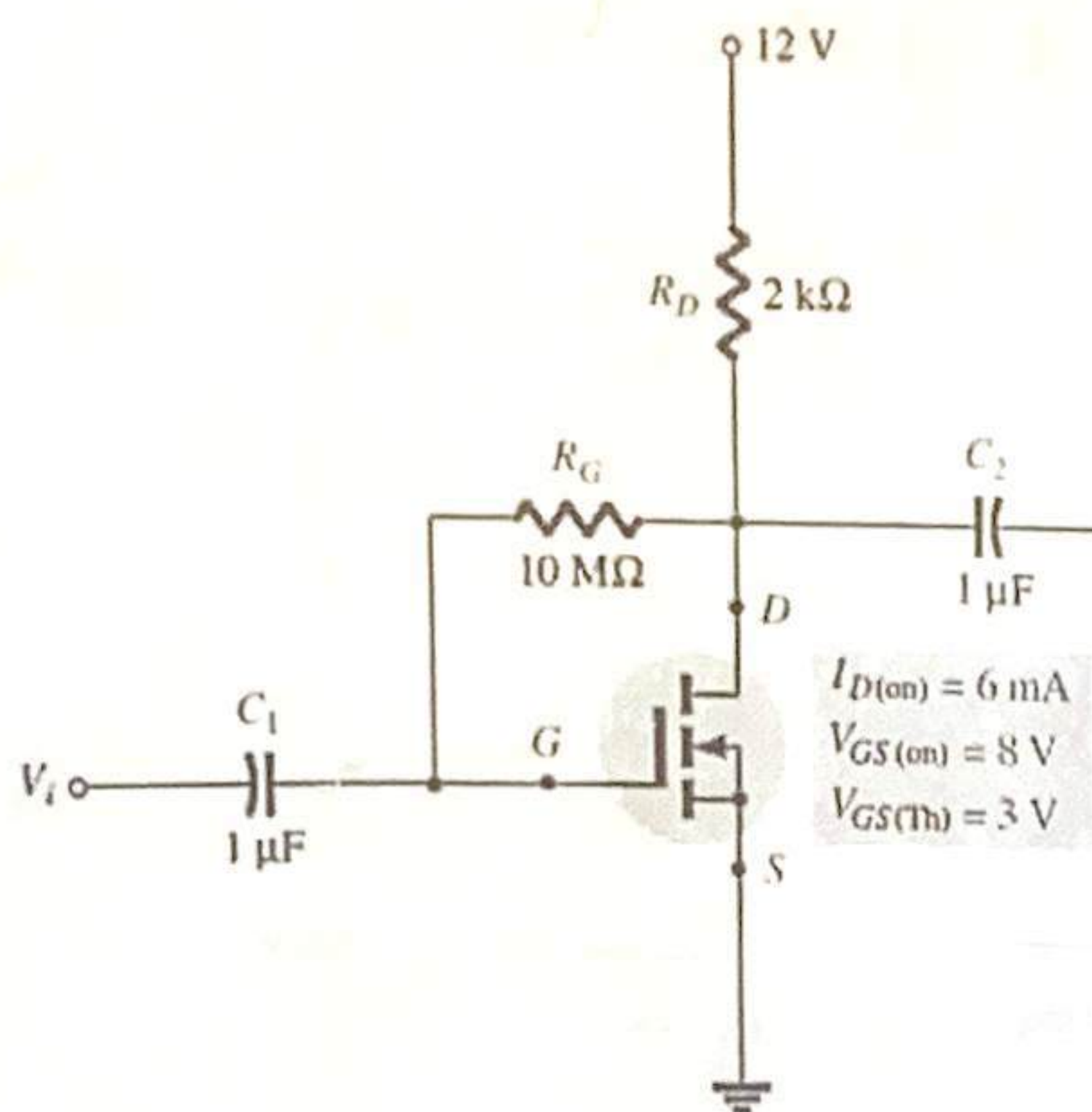
99,77

89,99

WEEK 6+7

HOMEWORK of Chapter 2. MOSFET

C2.1 Determine Q-point and DC loadline?



Ta có: $I_D = k(V_{GS} - V_{th})^2$

$$\Rightarrow k = \frac{I_D}{V_{GS} - V_{th}} = \frac{6}{(8-3)^2} = 0,24 \text{ (mA/V}^2\text{)}$$

Mạch hồi tiếp $\Rightarrow I_G = 0$

$$V_G = V_D = 0$$

Ta có tiếp: $12 = I_D \cdot R_D + V_{DS}$

$$\Rightarrow 12 = I_D R_D + V_{GS}$$

$$\Rightarrow I_D = \frac{12 - V_{GS}}{R_D} = 0,24 k(V_{GS} - V_{th})^2$$

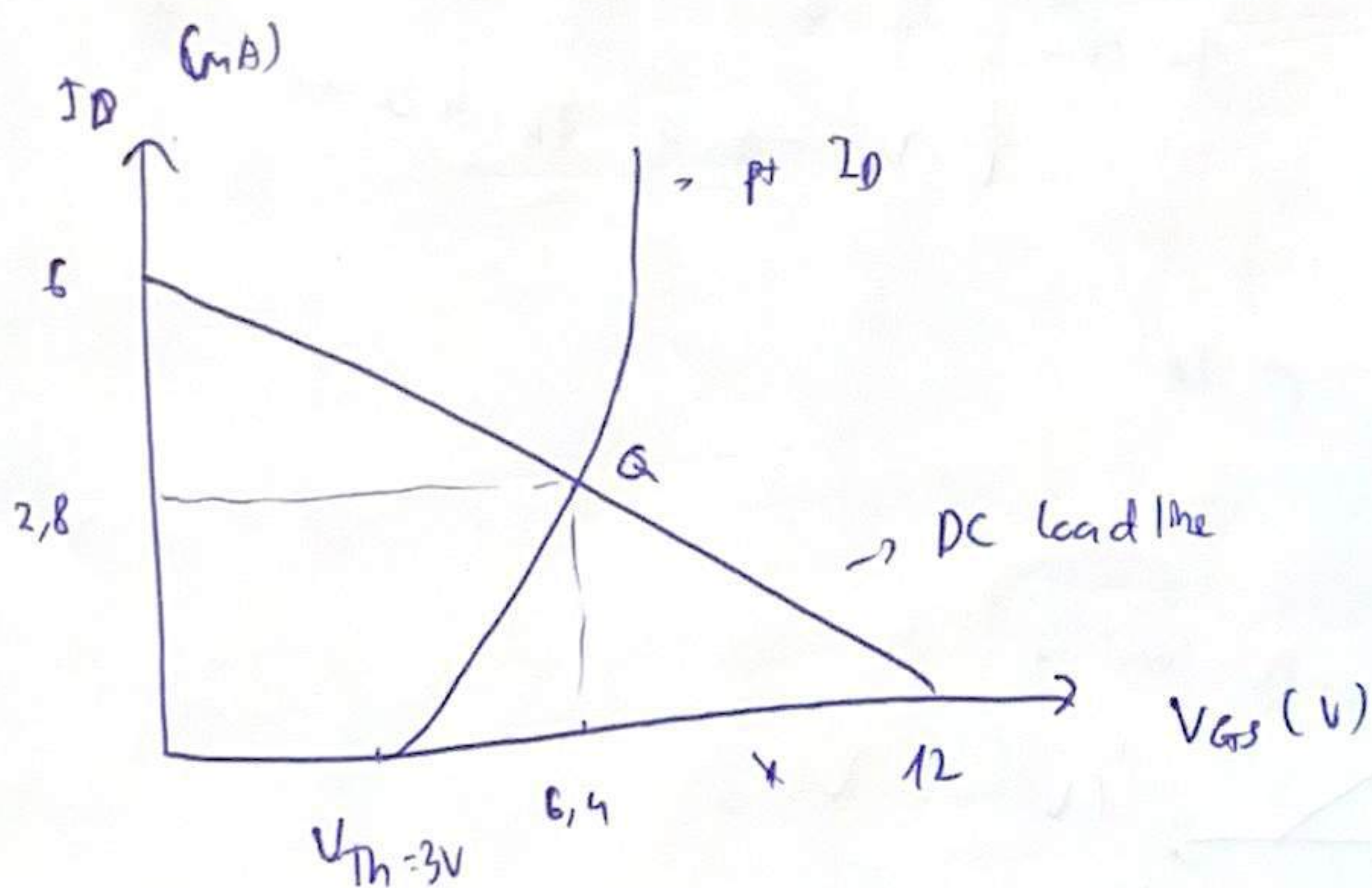
$$\Rightarrow 6 - \frac{V_{GS}}{2} \text{ (mA)} = 0,24 (V_{GS} - 3)^2$$

$$\Rightarrow V_{GS} = 6,4 \text{ V}$$

$$\Rightarrow I_D = 2,8 \text{ mA}$$

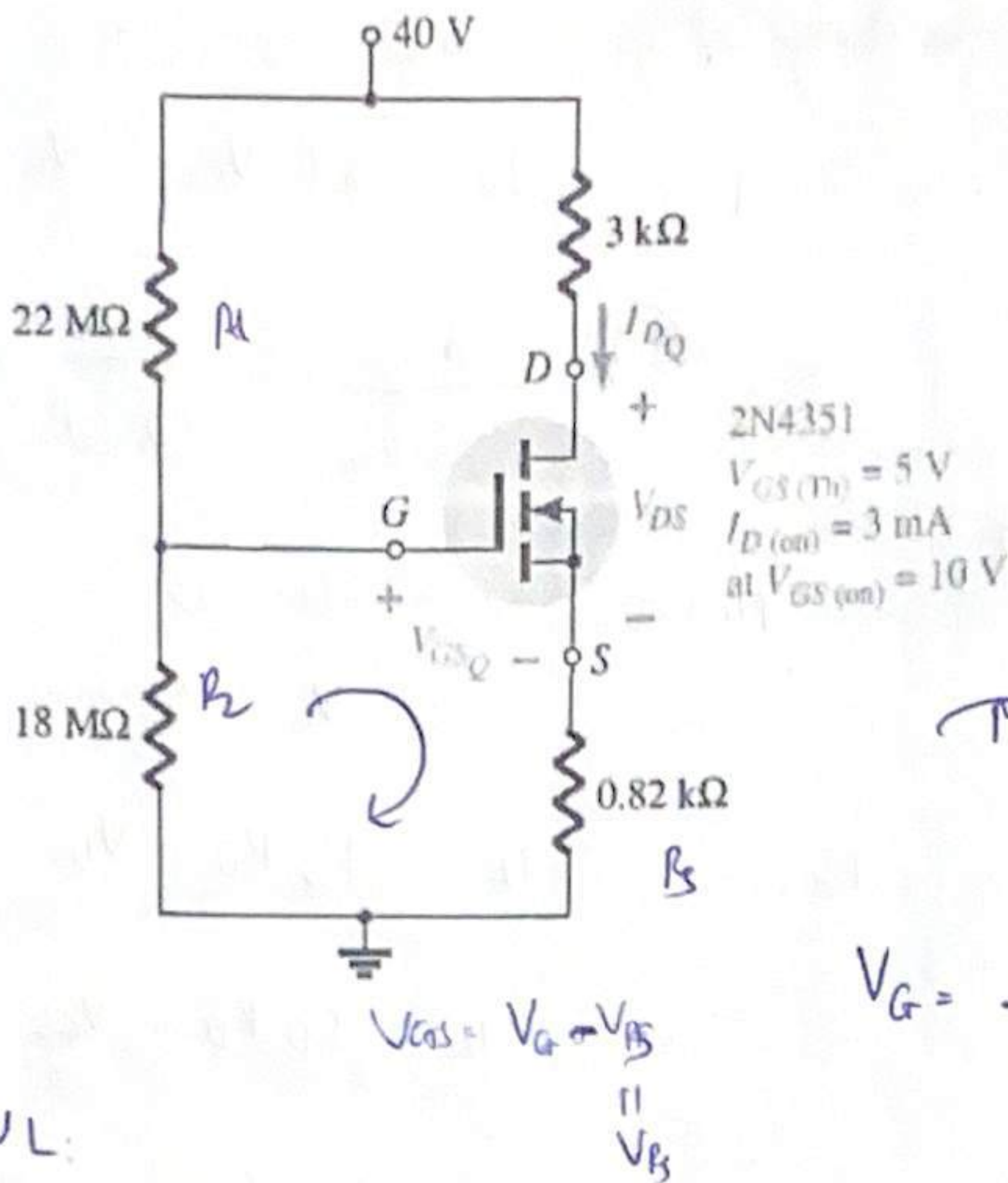
$$\Rightarrow \begin{cases} I_D = 0 \Rightarrow V_{GS} = 12 \text{ (V)} \\ V_{GS} = 0 \Rightarrow I_D = 6 \text{ mA} \end{cases}$$

$\Rightarrow$ DC load line



-0,25*

C2.2 Determine Q-point and DC loadline?



Ta có:

$$I_D = \frac{1}{2} k (V_{GS} - V_{th})^2$$

$$\Rightarrow k = \frac{I_D}{(V_{GS} - V_{th})^2} = \frac{3}{(10 - 5)^2} = 0,12 \text{ (mA/V}^2\text{)}$$

Mạch hồi tiếp $\rightarrow$ $\begin{cases} I_G = 0 \\ V_G = V_D \end{cases}$

$$V_G = \frac{R_2}{R_1 + R_2} \cdot 40 = \frac{18}{18 + 22} \cdot 40 = 18\text{ V}$$

Ta có từ KVL:

$$V_{GS} + V_{RS} - V_G = 0$$

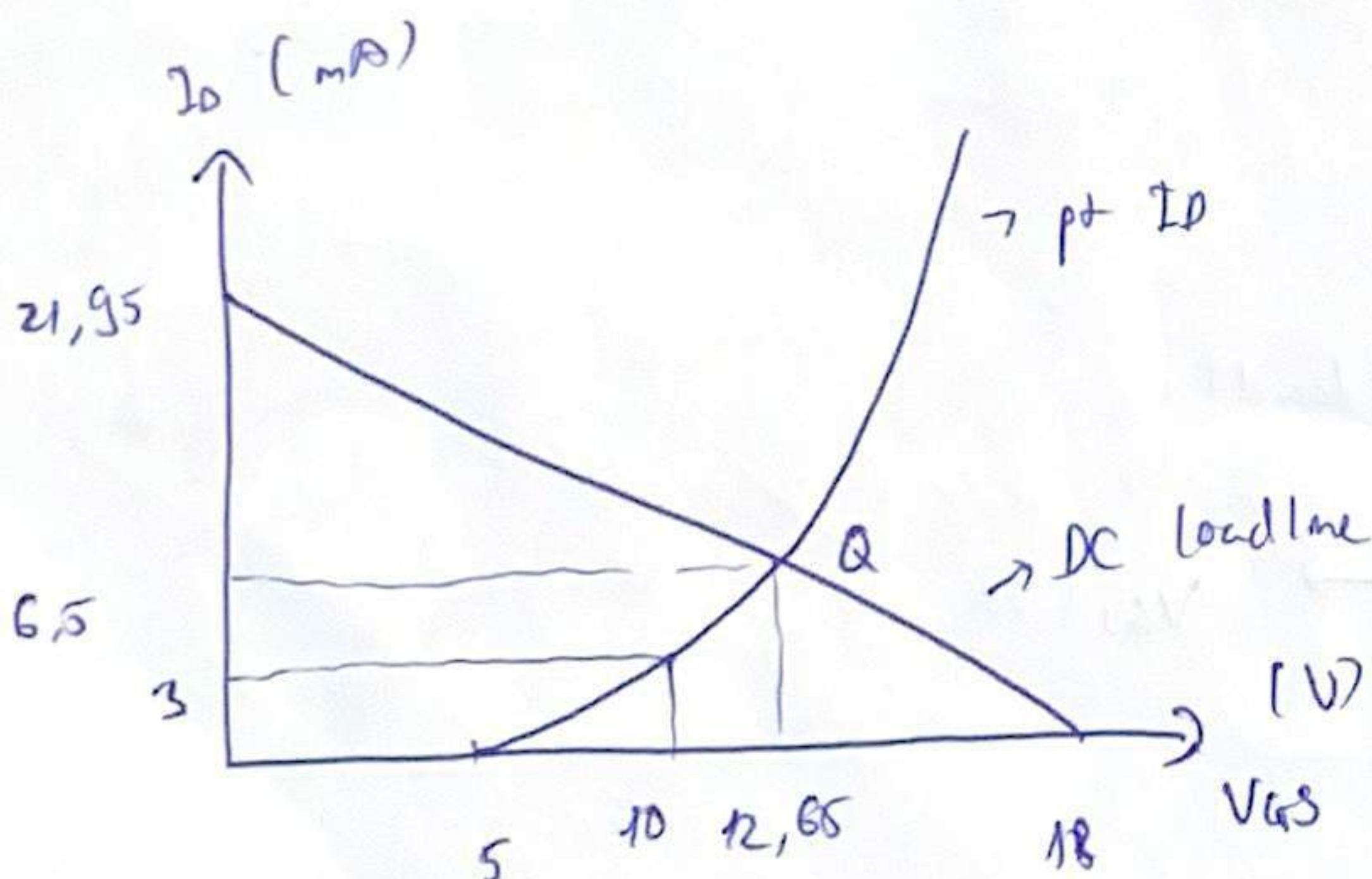
$$\Rightarrow V_{GS} + I_D R_S - V_G = 0 \Rightarrow I_D = \frac{V_G - V_{GS}}{R_S}$$

nên $I_D = \frac{1}{2} k (V_{GS} - V_{th})^2$

$$\Rightarrow 0,12 \cdot 10^{-3} (V_{GS} - 5)^2 = \frac{18 - V_{GS}}{0,82 \cdot 10^{-3}} \Rightarrow V_{GS} = 12,65\text{ V}$$

$$\Rightarrow I_D = 6,5\text{ mA}$$

Ta có: $\begin{cases} I_D = 0 \Rightarrow V_{GS} = 18\text{ V} \\ V_{GS} = 0 \Rightarrow I_D = 21,95\text{ mA} \end{cases}$



Trên đó: $V_{GS} = V_G - V_S = \frac{R_2}{R_1 + R_2} V_{DD} - I_D R_S$

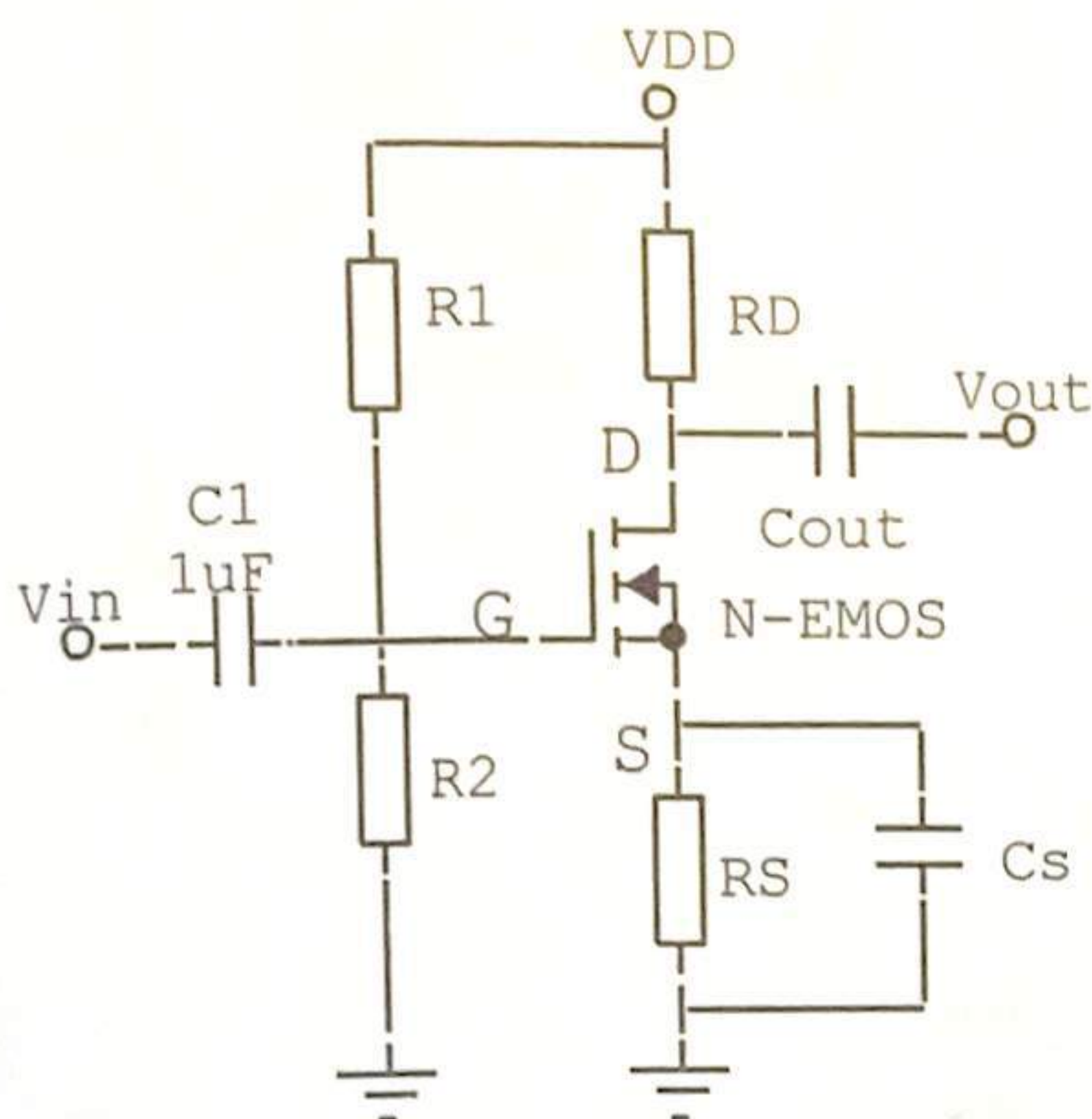
a. $\Rightarrow I_D = \frac{V_{DD} R_2}{R_2 + R_1} - V_{GS} \frac{R_2 + R_1}{R_S} = \frac{5 R_2}{R_2 + R_1} - V_{GS}$

mà $I_D = \frac{1}{2} k (V_{GS} - V_{TH})^2$

C2.3 Name of the circuit?: CS - Voltage divider ✓

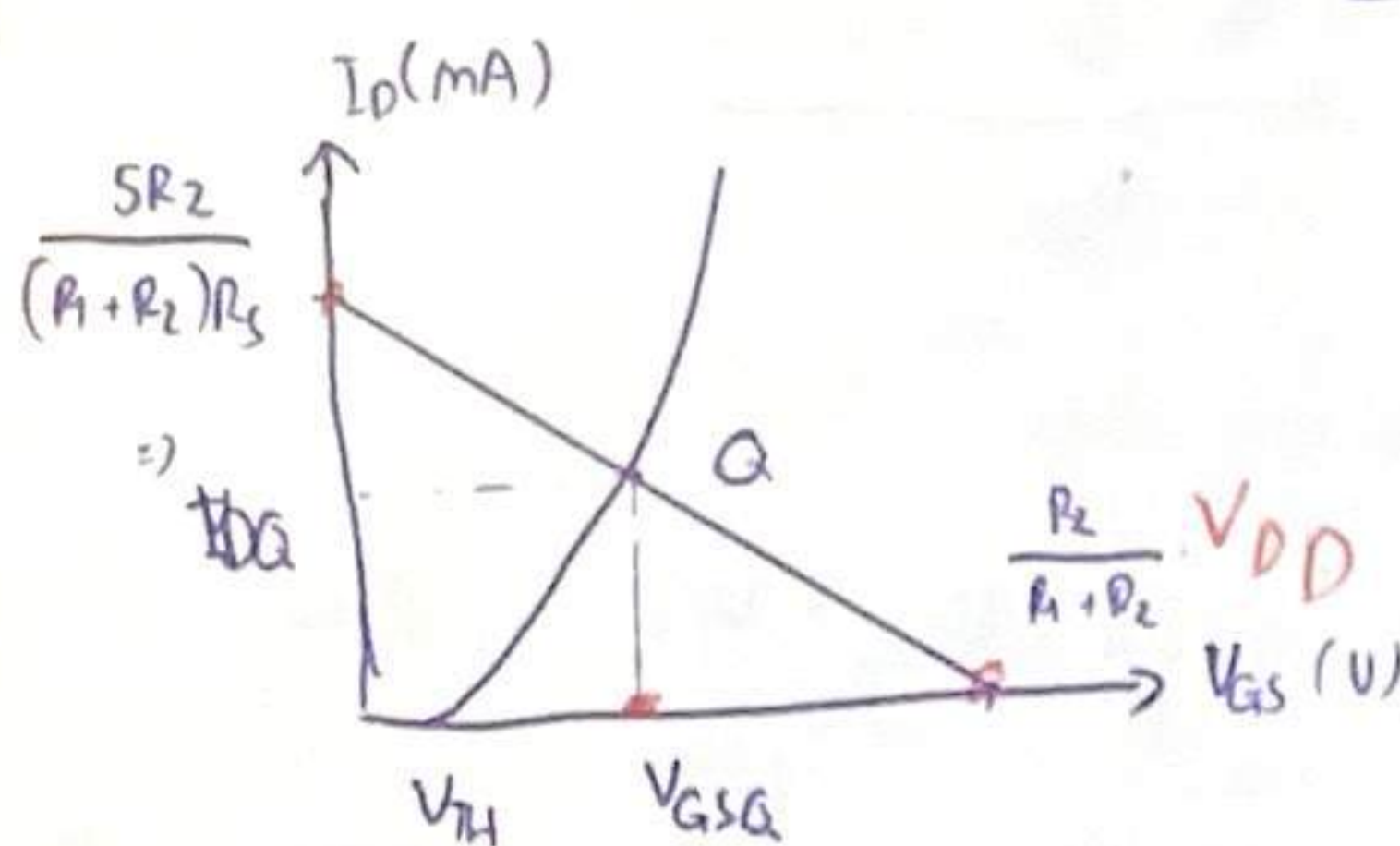
Given $V_{DD} = 5V$, $V_{TH} = 1V$, $k = 1.10^{-5} A/V^2$.

- Determine R_1 , R_2 , R_D , R_S so that Q-point locates in the middle of DC loadline?
- Determine voltage gain, current gain, input and output impedances for the circuit without load, with load equal to $4 k\Omega$ and for 4Ω ?
- Comments?



Đặt Q nằm giữa đường tải và tải

$\Rightarrow \begin{cases} I_{DQ} = \frac{1}{2} I_{Dmax} = \frac{5}{2} \frac{R_2}{(R_1 + R_2) R_S} \\ V_{GSQ} = \frac{1}{2} V_{GSmax} = \frac{5}{2} \frac{R_2}{(R_1 + R_2) R_S} \end{cases}$



$I_{DQ} = \frac{1}{2} \cdot 10^{-5} (V_{GSQ} - 1)^2 \left(= \frac{1}{2} k (V_{GSQ} - V_{TH})^2 \right) \frac{x}{R_S} \Rightarrow \begin{cases} R_S = \frac{2 \cdot 10^5 \cdot x}{(x-1)^2} \\ I_{DQ} = \frac{(x-1)^2}{2 \cdot 10^5} \end{cases}$

Đặt $\frac{5 R_2}{2(R_1 + R_2) R_S} = x \Rightarrow V_{GSQ} = x$

Để dễ tính toán, chọn $V_{GS} = 2V \Rightarrow x = 2 \Rightarrow$ thay vào, ta có các biến

$\begin{cases} R_S = \frac{2 \cdot 10^5}{(2-1)^2} = 2 \cdot 10^5 = 200 k\Omega \\ I_{DQ} = \frac{(2-1)^2}{2 \cdot 10^5} = \frac{1}{2 \cdot 10^5} = 0,005 mA \\ \frac{5 R_2}{2(R_1 + R_2)} = x = 2 \Rightarrow \frac{R_2}{R_1 + R_2} = \frac{4}{5} \end{cases}$

$\Rightarrow$ chọn $\begin{cases} R_1 = 1 M\Omega \\ R_2 = 4 M\Omega \end{cases}$ (tổ kháng MOSFET tuy nhỏ nhưng cỡ $10^6 \Omega$)

* Ta có $V_{DS sat} = V_{GS} - V_{TH} = 1V \Rightarrow V_{DS} \geq V_{DS sat} = 1V$

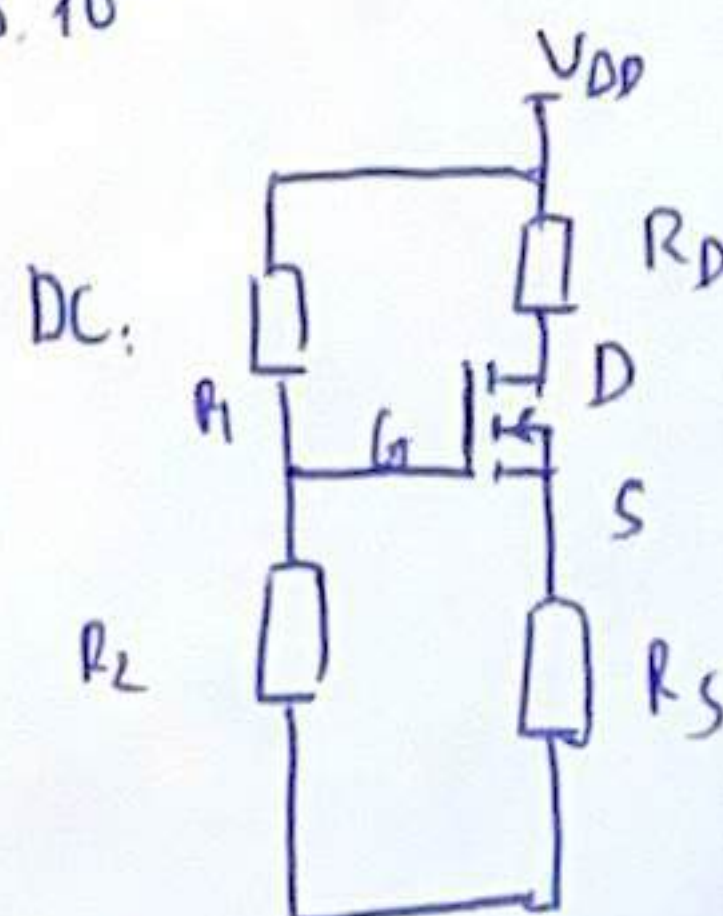
$V_{DD} = I_{DQ} (R_S + R_D) + V_{DS} = 0,005 \cdot 10^{-3} (400 \cdot 10^3 + R_D) + V_{DS} = 0,005 \cdot 10^{-3} R_D + V_{DS} + 2 = 5$

$\Rightarrow 0,005 \cdot 10^{-3} R_D + V_{DS} = 3 \Rightarrow R_D \leq \frac{3-1}{0,005 \cdot 10^{-3}} = 400 k\Omega$

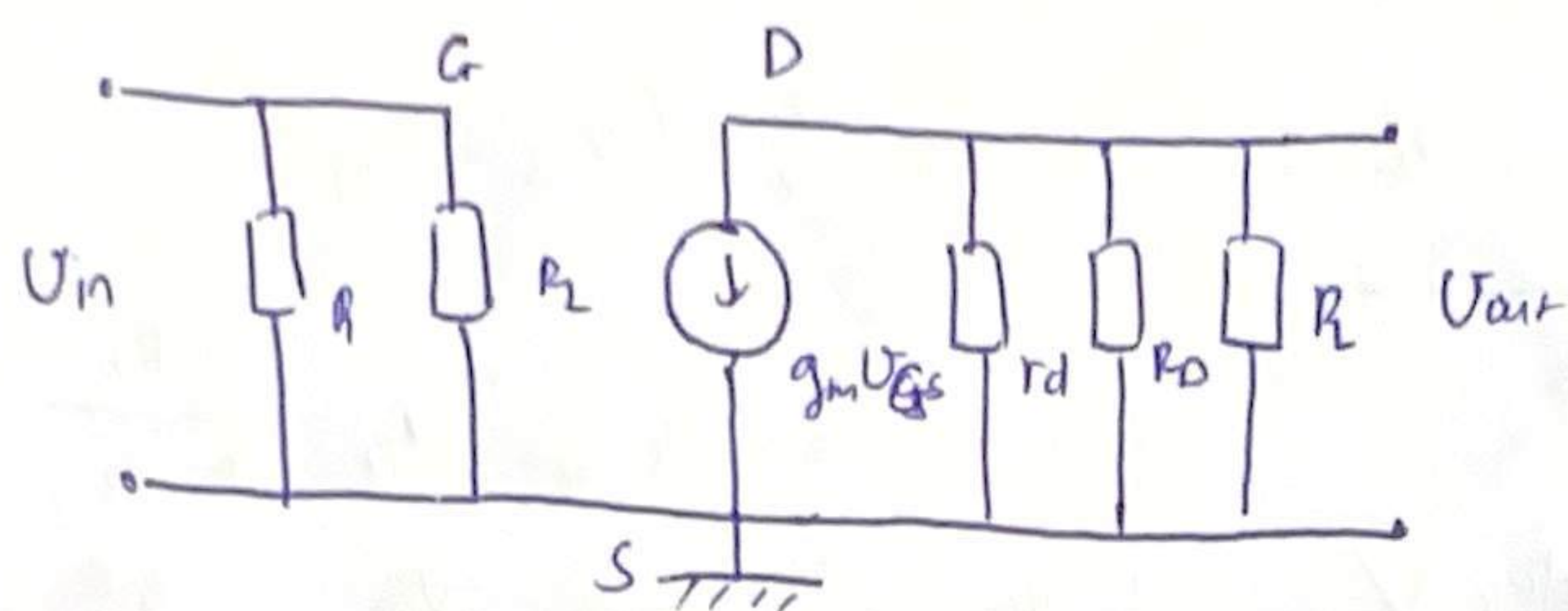
mà $V_{DS} \geq 1$

$\Rightarrow$ chọn $R_D = 100 k\Omega$

Vậy chọn $\begin{cases} R_1 = 1 M\Omega \\ R_2 = 4 M\Omega \\ R_D = 100 k\Omega \\ R_S = 400 k\Omega \end{cases}$



b. AC: $\Rightarrow$ ngắn tụ, thay MOSFET = mô hình nhỏ hiện như dưới đây



Và $R_L = 0$, khi là 4Ω , $Z_{in} = R_1 \parallel R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} = 0,8M\Omega$ ✓

$Z_o = \frac{U_o}{I_o} \Big|_{U_{in}=0} = R_D \parallel R_L \parallel r_d \Bigg\} \Rightarrow Z_o = R_D \parallel R_L$, với TH không có R_L
 $Z_o = R_D$ ✓
 mà $r_d \rightarrow \infty$

$A_v = \frac{U_o}{U_i} = \frac{-g_m U_{gs} (R_D \parallel R_L)}{U_{gs}} = -g_m (R_D \parallel R_L)$. Với TH không có R_L
 $A_v = -g_m R_D$ ✓

$A_i = \frac{I_o}{I_i} \Rightarrow$ không ảnh hưởng; $I_G = 0$

Trong đó $g_m = 2k(V_{GSQ} - V_T) = 2 \cdot 10^{-5} \cdot (2 - 1) = 2 \cdot 10^{-5} \text{ (A/V)}$

$\Rightarrow$ ta có:

	Ko R_L	$R_L = 4\Omega$	$R_L = 4\Omega$
Z_i	$0,8M\Omega$	$0,8M\Omega$	$0,8M\Omega$
Z_o	$100k\Omega$	3846Ω	$3,99\Omega$ ✓
A_v	-2	-0,027	$-8 \cdot 10^{-5}$

c. Nhận xét: R_L ảnh hưởng đến A_v ; Z_o

R_L giảm thì Z_o giảm; $|A_v|$ giảm - 0,25. Để giảm ảnh hưởng của nó, ta mắc thêm Stage đệm (buffer) hoặc follower

Source-follower (Common-drain) ✓
 Op-amp đệm (voltage-follower)

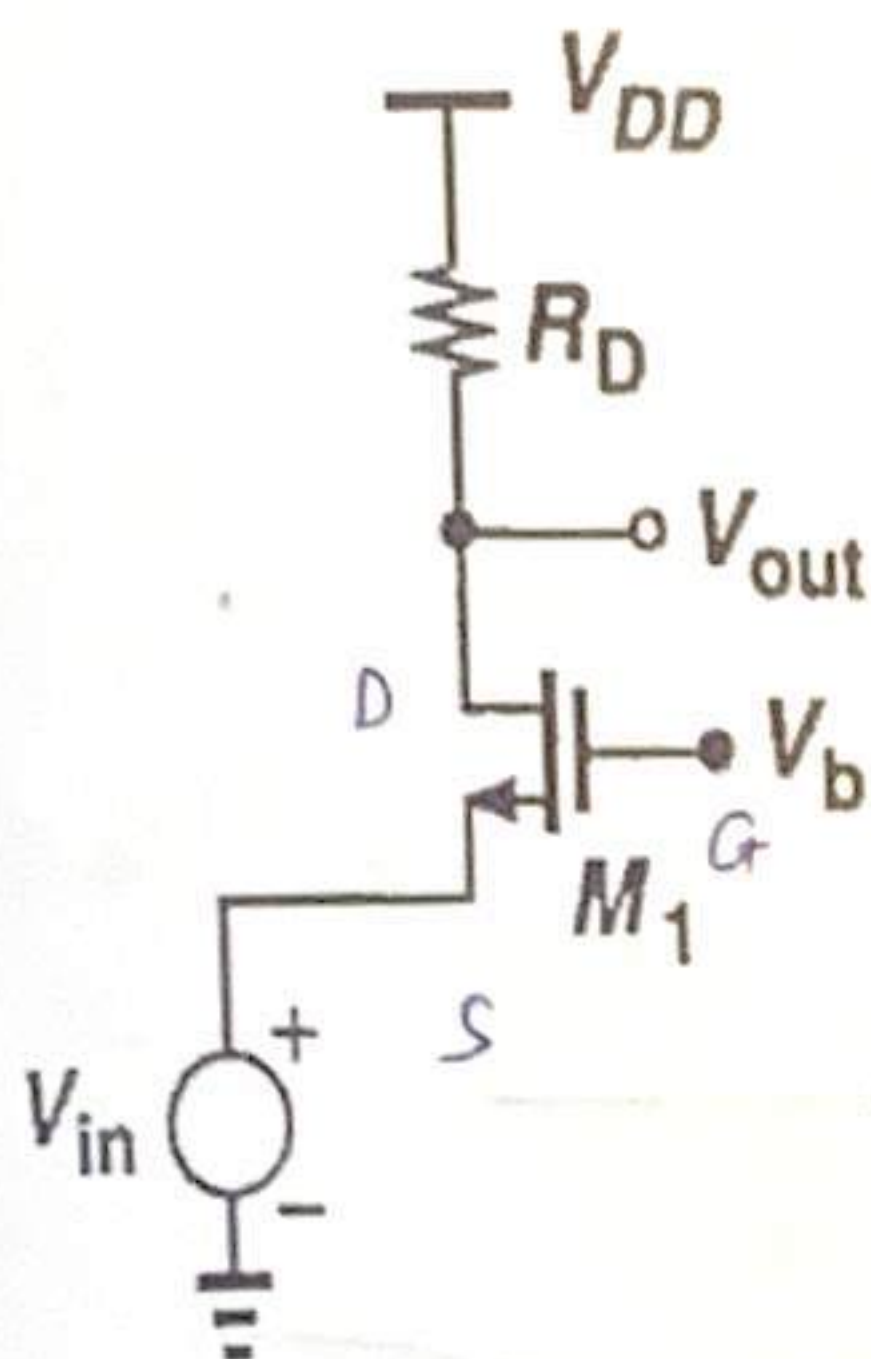
để $Z_{tải}$ sẽ không ảnh hưởng hoặc ảnh hưởng tới A_v

C2.4 Name of the circuit?: CG

- Determine Q point, DC loadline?
- Draw small equivalent circuit?
- Determine A_v of the circuit?

($V_m = \text{const}$)

→ pt DC load line

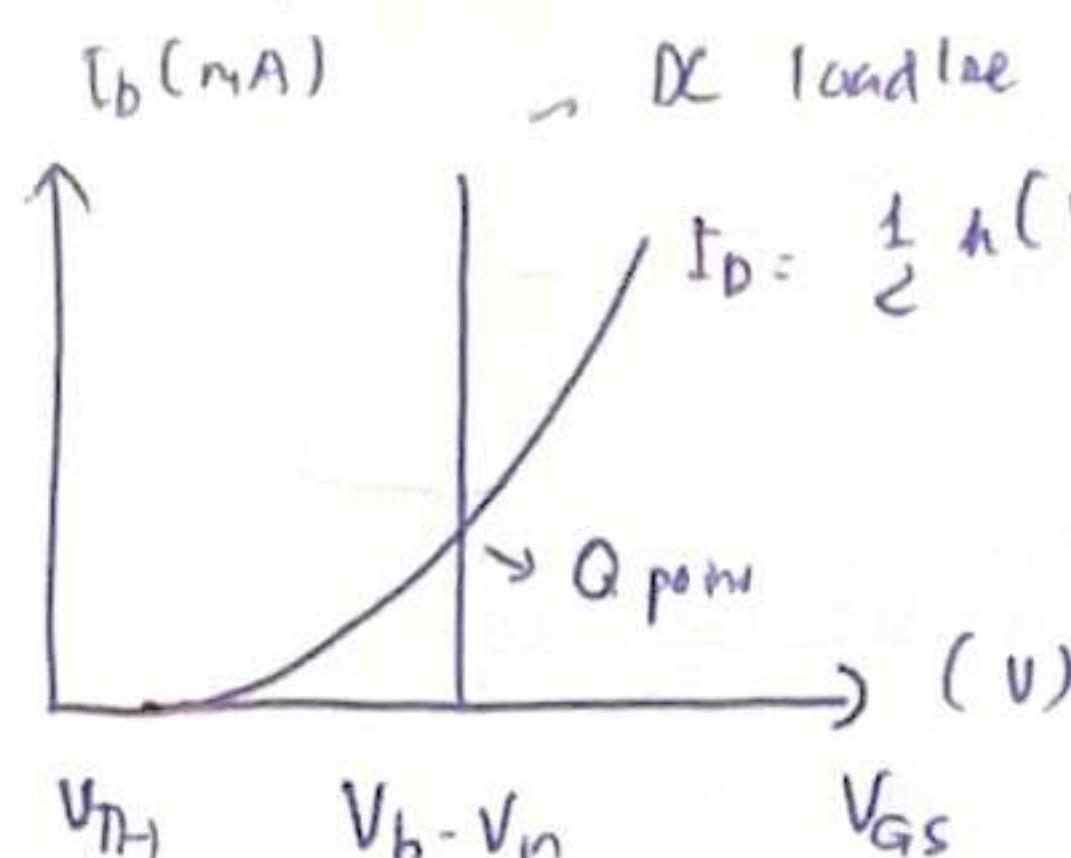


a. Ta có: $V_{GS} = V_G - V_S = V_b - V_{in}$

$$I_D = \frac{1}{2} \mu C_{ox} (V_{GS} - V_{th})^2 = \frac{1}{2} \mu C_{ox} (V_b - V_{in} - V_{th})^2$$

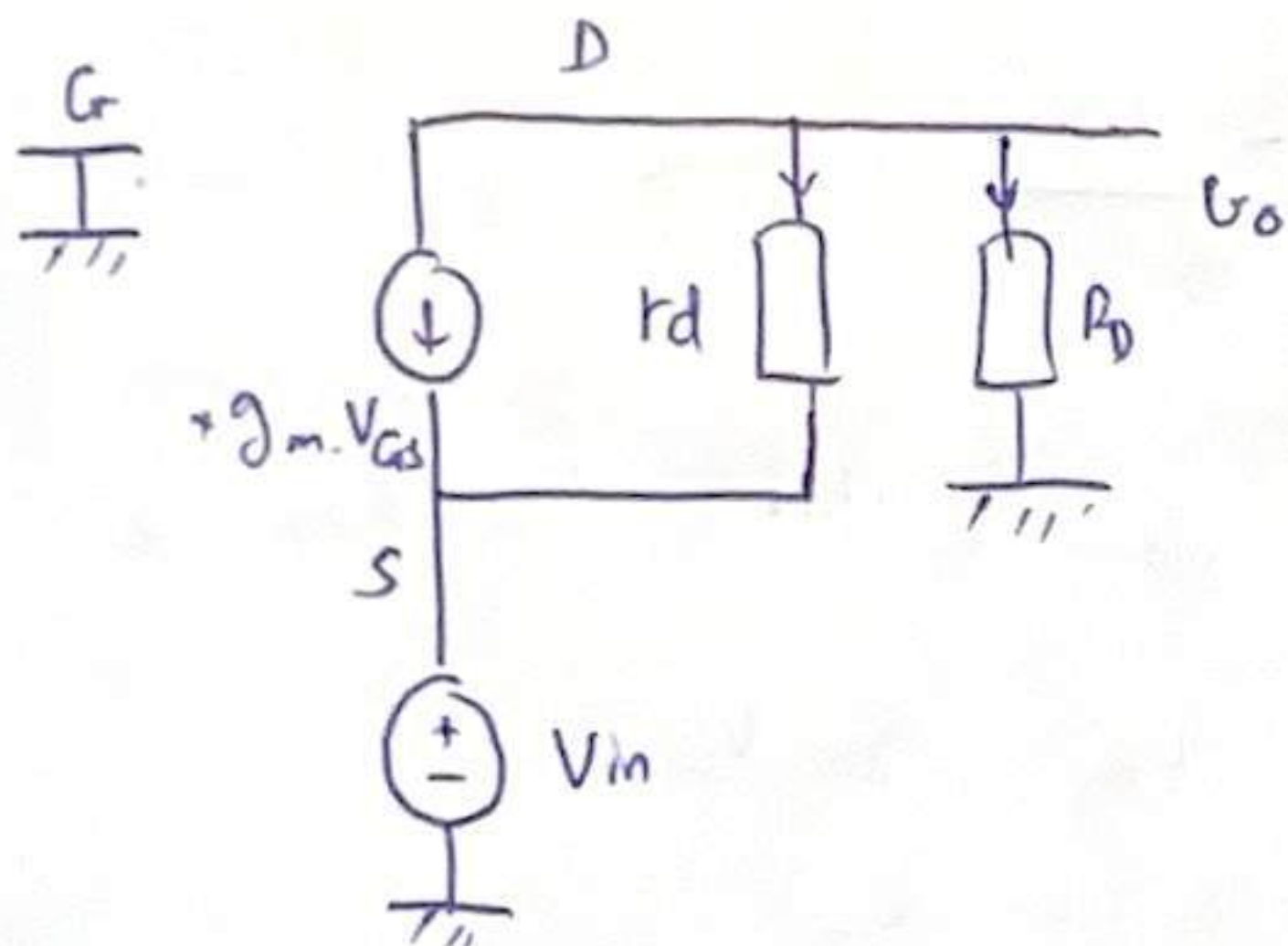
→ pt I_D

→ ta có Q point là giao của DC loadline và pt I_D



→ nên V_m là AC
thì $V_{th} = 0$

b. AC:



c. Ta có: $A_v = \frac{V_o}{V_i}$

$$V_o = I_{RD} \cdot R_D$$

$$V_S = V_m$$

$$V_{GS} = V_G - V_m = 0 - V_m = -V_{in}$$

$$I_{RD} = -(g_m V_{GS} + I_{rd}) = -(g_m (-V_{in}) + \frac{V_o - V_m}{r_d})$$

$$= g_m V_m - \frac{I_{RD} R_D - V_m}{r_d} = g_m V_m + \frac{V_m}{r_d} - \frac{I_{RD} R_D}{r_d}$$

$$\Rightarrow I_{RD} \left(1 + \frac{R_D}{r_d}\right) = g_m V_m + \frac{V_m}{r_d} \quad \Rightarrow I_{RD} = \frac{g_m V_m + \frac{V_m}{r_d}}{1 + \frac{R_D}{r_d}}$$

$$\Rightarrow A_v = \frac{V_o}{V_m} = \frac{I_{RD} R_D}{V_m} = \frac{\left(g_m + \frac{1}{r_d}\right) R_D}{1 + \frac{R_D}{r_d}}$$

a. Mạch CD; phân cực bằng bộ chia áp

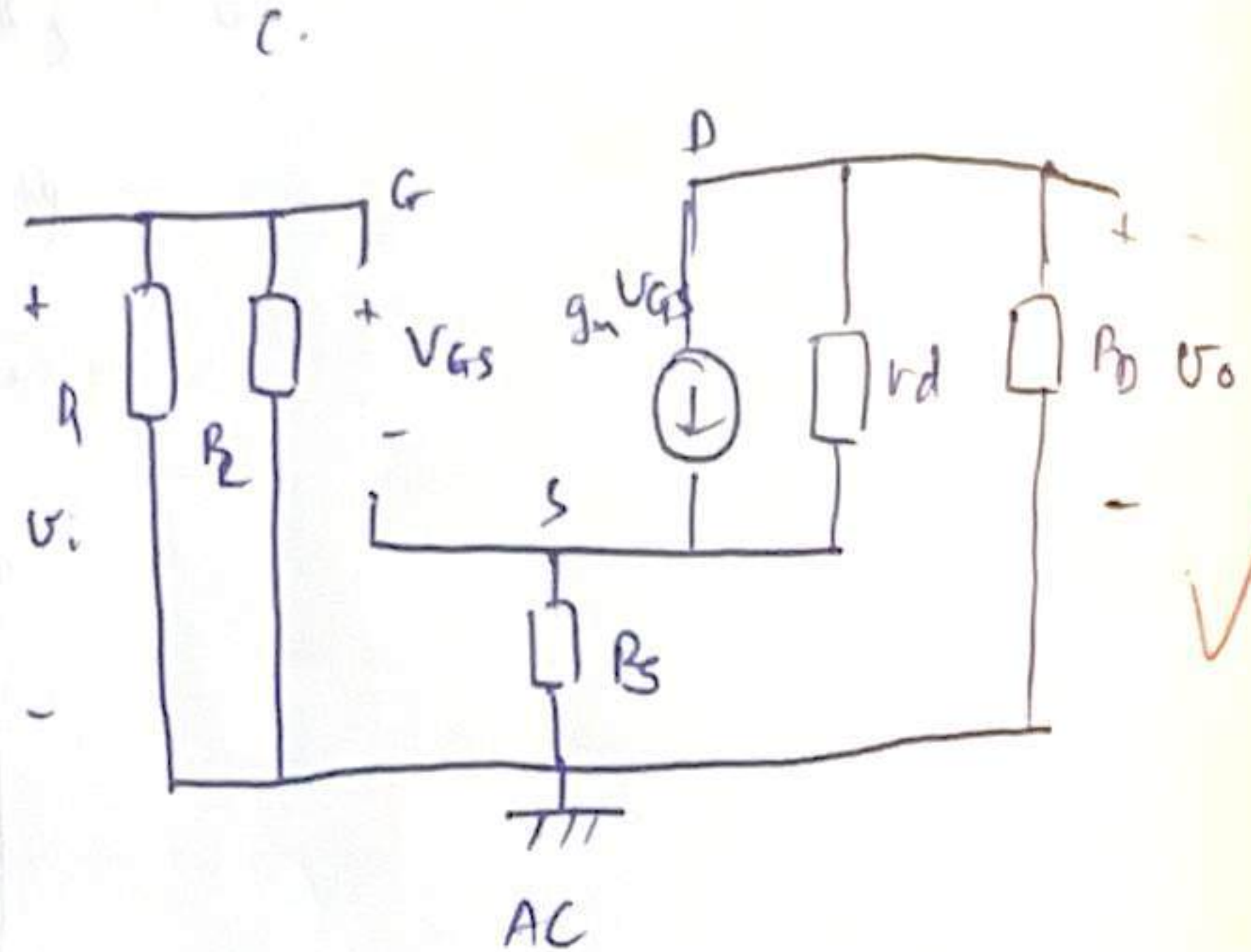
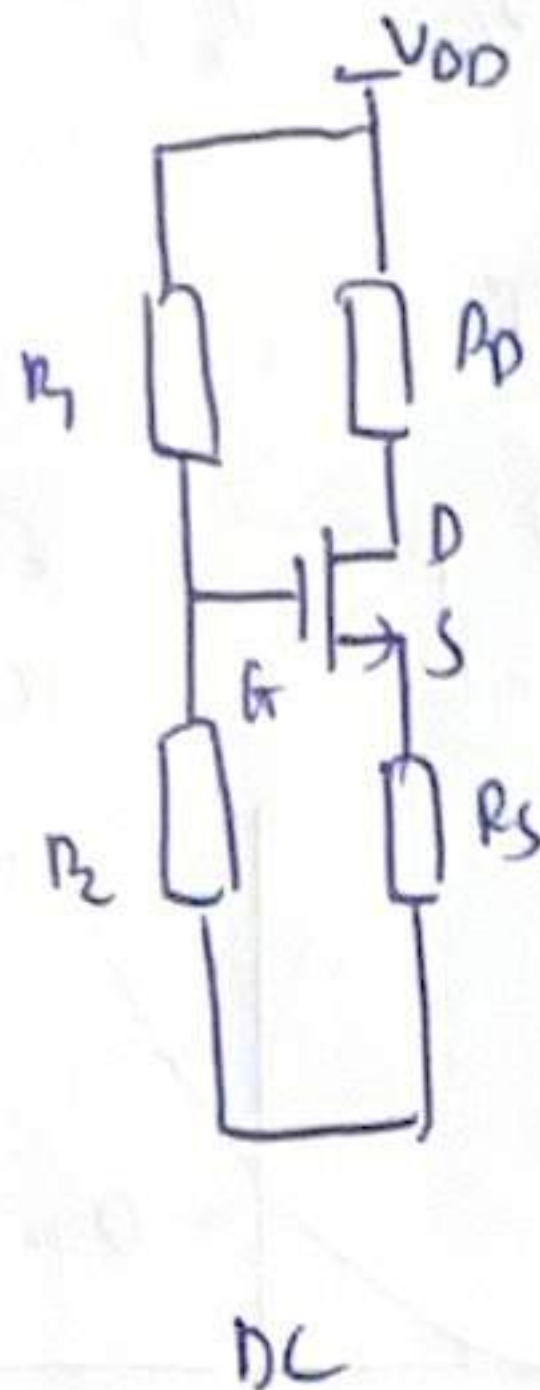
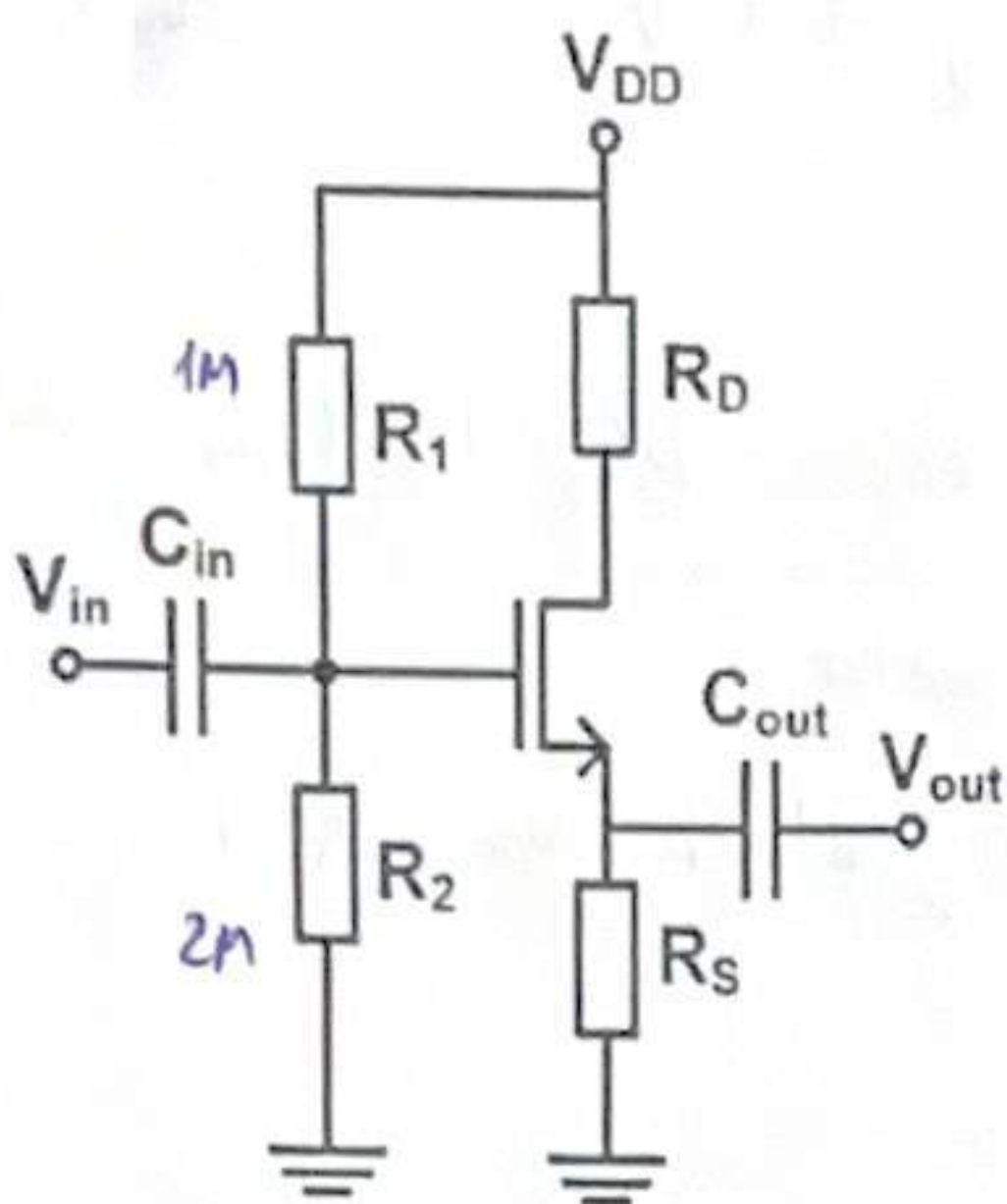
b. Ta có $I_D = \frac{1}{2} k (V_{GS} - V_{TH})^2 = \frac{1}{2} \cdot 10^{-3} (V_{GS} - 0,5)^2 = 10^{-3}$

$\Rightarrow V_{GS} = 1,91 \text{ (V)}$ (Lý do loại TH còn lại?)

Ta có $V_{DD} = I_D (R_S + R_D) + V_{GS} \Rightarrow R_S = \frac{V_{DD} - V_{GS}}{I_D} = \frac{3 - 1,91}{10^{-3}} = 1090 \Omega$

C2.5 Cho $V_{DD} = 3V$, $V_{TH} = 0.5 V$, $k = 10^{-3} A/V^2$, $R_1 = 1 M\Omega$ & $R_2 = 2 M\Omega$; $R_D = 10 k\Omega$;

- Nêu tên của mạch khuếch đại và phương pháp phân cực được sử dụng ở mạch bên.
- Biết dòng điện I_{DS} của MOSFET bằng 1 mA, xác định điện áp phân cực V_{GS} của MOSFET. Từ đó xác định giá trị điện trở R_S .
- Vẽ mô hình tín hiệu nhỏ của mạch và xác định hệ số khuếch đại của mạch.
- Từ đó xác định hệ số khuếch đại của mạch cho 3 trường hợp của tải như sau: (1) không có tải (2) $R_{t\grave{a}i} = 4 k\Omega$ và (3) $R_{t\grave{a}i} = 4\Omega$. (Lập bảng so sánh).



Ta có $V_G = \frac{R_2}{R_1 + R_2} \cdot V_{DD} = \frac{2}{1+2} \cdot 3 = 2V$

$\Rightarrow V_G = V_{GS} + I_D \cdot R_S \Rightarrow R_S = \frac{V_G - V_{GS}}{I_D} = \frac{2 - 1,91}{10^{-3}} = 90 \Omega$

c. $g_m = k(V_{GS} - V_{TH}) = 2,82 \cdot 10^3 \Rightarrow$ Sử dụng công thức $g_m = k(V_{GS} - V_{TH})$

$V_O = V_S \Rightarrow V_{GS} = V_G - V_S = V_{in} - V_O$; $r_d \rightarrow \infty$ do $I_{DQ} \rightarrow 0$
 $V_i = V_G \Rightarrow I_{R_S} = g_m \cdot V_{GS}$

$\Rightarrow \frac{V_O}{R_S} = g_m (V_i - V_O) \Rightarrow V_O = g_m R_S (V_i - V_O)$

$\Rightarrow V_O (1 + g_m R_S) = g_m R_S V_{in} \Rightarrow \frac{V_O}{V_i} = A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{2,82 \cdot 10^3 \cdot 90}{1 + 2,82 \cdot 10^3 \cdot 90}$

$\Rightarrow A_v = 0,2024$ (0,2024) - 0,25

d. Khi có tải $\rightarrow$ thay R_S bằng $R_S \parallel R_L$

Khi đó $A_v = \frac{g_m (R_S \parallel R_L)}{1 + g_m (R_S \parallel R_L)}$

	không tải	$R_L = 4k\Omega$	$R_L = 4\Omega$
A_v	0,2024	0,1989	0,0107
	X	X	X

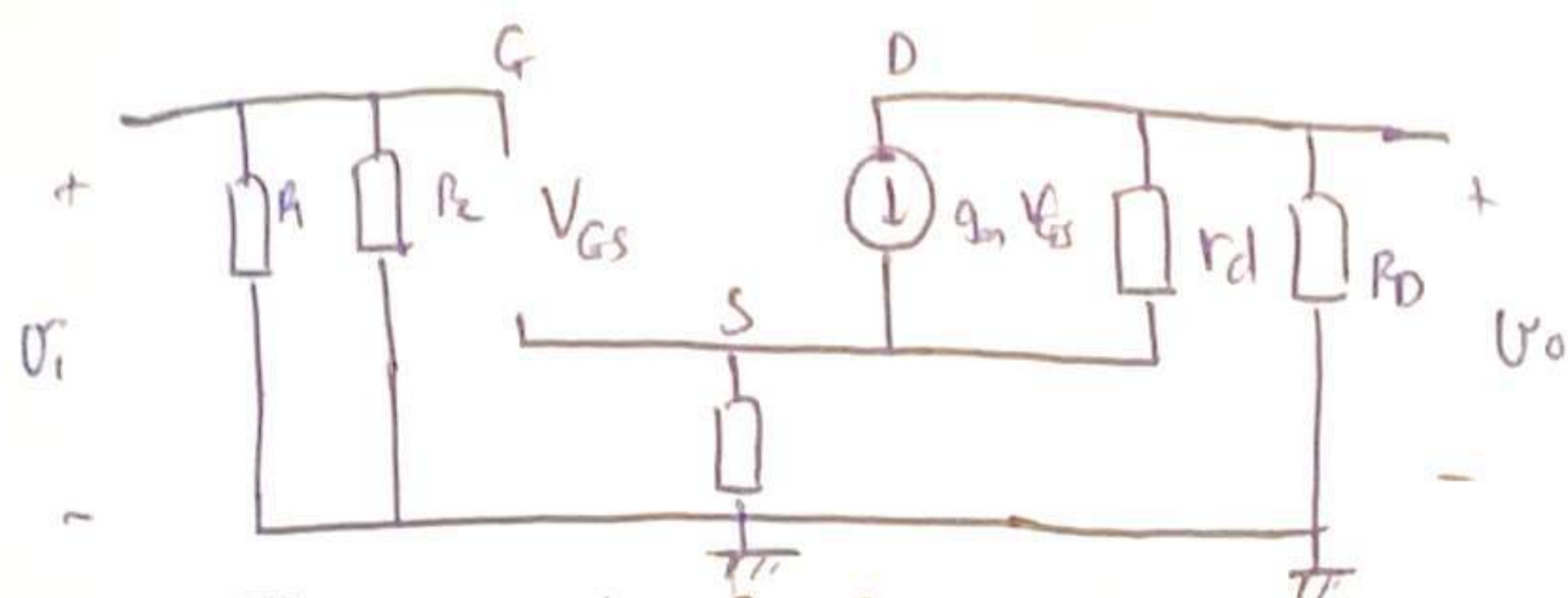
$\rightarrow$ ta có bảng

- 0,5

(Sử dụng $g_m \Rightarrow$ sử dụng)

$0,1126$ $0,11$ $5,378 \cdot 10^{-3}$

b. AC



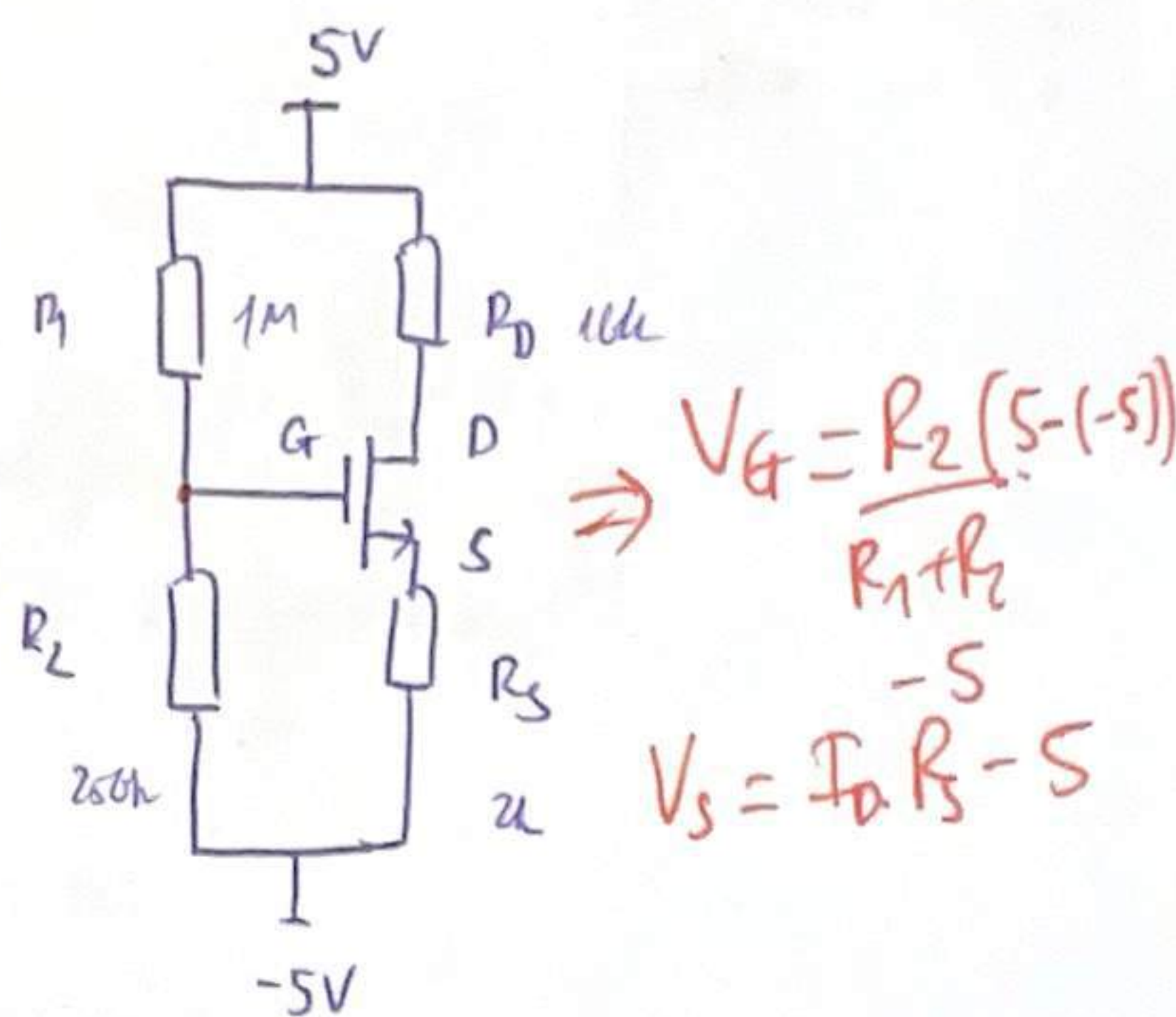
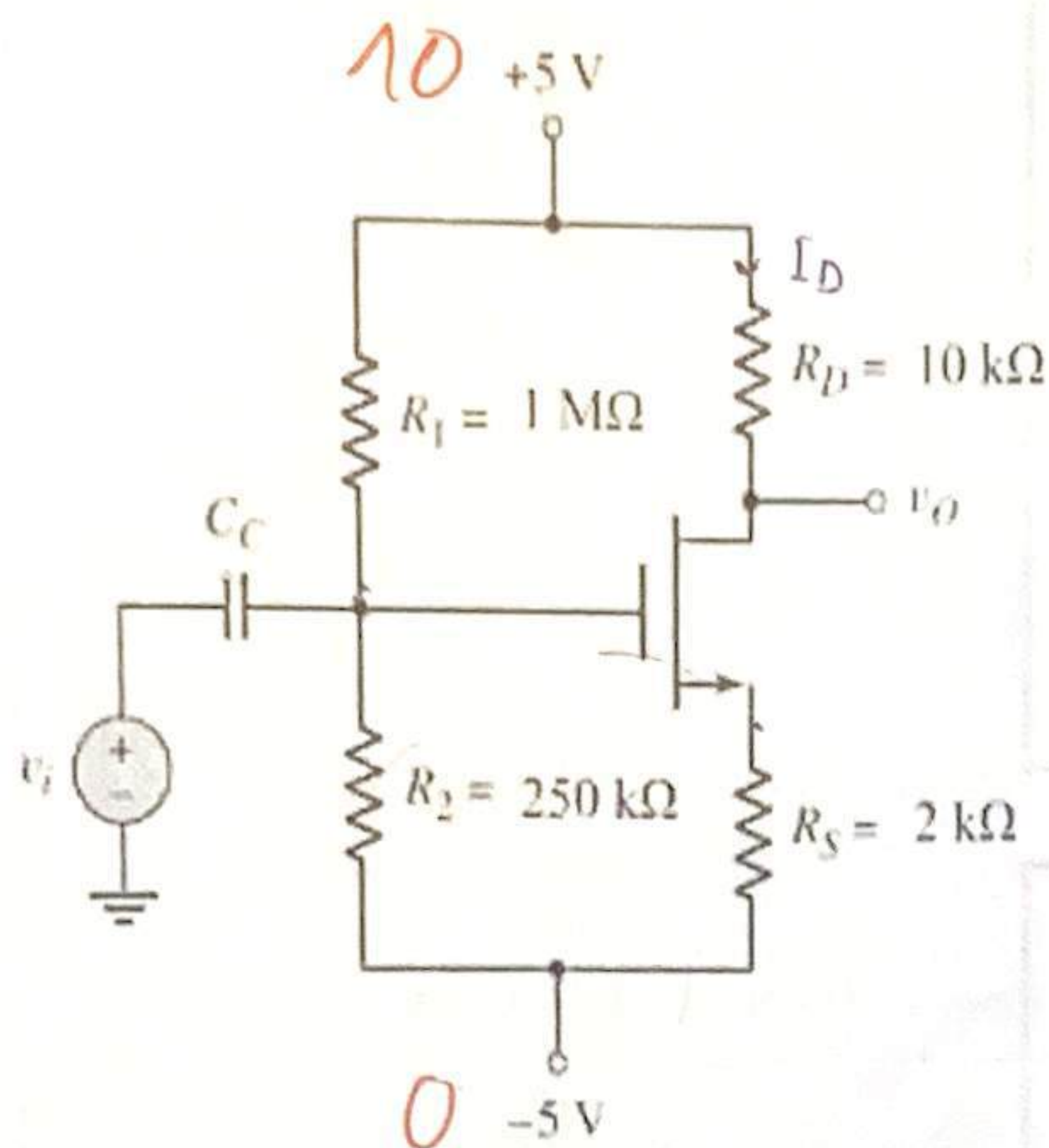
$$g_m = 2k(V_{GSQ} - V_{th})$$

$$g_m = 2 \cdot 0,5 \cdot 10^{-3} (1,385 - 0,6)$$

$$g_m = 7,85 \cdot 10^{-4}$$

$$A_v = \frac{v_o}{v_i}$$

C2.6 The parameters for the transistor in the circuit shown in Figure Q3 are $V_T = 0.6$ V, $K_n = 0.5$ mA/V², and $\lambda = 0$. (a) Determine the quiescent values of I_{DQ} and V_{DSQ} , (b) Find the small signal voltage gain v_o/v_i .



a. Ta có: $\frac{5 - (-5)}{R_1 + R_2} \cdot R_2 = V_G \Rightarrow V_G = \frac{R_2}{R_1 + R_2} (5 - (-5)) = \frac{250k \cdot 10}{250k + 1M} = 2V$

Ta lại có $V_G = V_{GSQ} + V_S = V_{GSQ} + I_{DQ} R_S$ (Tính đúng nhưng ở chuyển về 10V và 0V)
 $-0,25$

mà $I_{DQ} = \frac{1}{2} k (V_{GSQ} - V_{th})^2$ (*)

$\Rightarrow V_G = V_{GSQ} + \frac{1}{2} R_S k (V_{GSQ} - V_{th})^2$

$\Rightarrow 2 = V_{GSQ} + \frac{1}{2} 2k \cdot 0,5 \cdot 10^{-3} (V_{GSQ} - 0,6)^2$

$\Rightarrow \begin{cases} V_{GSQ} = 1,385 \text{ V} \\ \text{thay vào (*) } I_{DQ} = 3,08 \cdot 10^{-4} \text{ (A)} = 0,308 \text{ mA} \end{cases}$ (lý do loại TL còn lại!)

$V_{DD} = 5 - I_{DQ} R_D + V_{GS} + R_S I_{DQ} \Rightarrow 5 - (-5) = (10k + 2k) I_{DQ} + V_{GS} = 0,25$
 $\Rightarrow V_{GS} = 6,304 \text{ V}$ $7,2925 \text{ (V)}$

b (tiếp)

$v_o = -(g_m v_{gs}) \cdot R_D$

$\Rightarrow v_{gs} = \frac{v_{in}}{1 + g_m R_S}$

$\Rightarrow A_v = \frac{v_o}{v_i} = \frac{-g_m R_D}{1 + g_m R_S}$

$V_{GS} = v_{in} - g_m v_{gs} R_S$

$\Rightarrow v_{in} = V_{GS} (1 + g_m R_S)$

$\Rightarrow v_o = -g_m R_D \cdot \frac{v_i}{1 + g_m R_S}$

$= \frac{-7,85 \cdot 10^{-4} \cdot 10k}{1 + 7,85 \cdot 10^{-4} \cdot 2k} = -3,05$

$g_m = k (V_{GS} - V_{th})$

$\Rightarrow A_v = -3,05$

ĐK để P-Mosfet ở vùng hoạt động

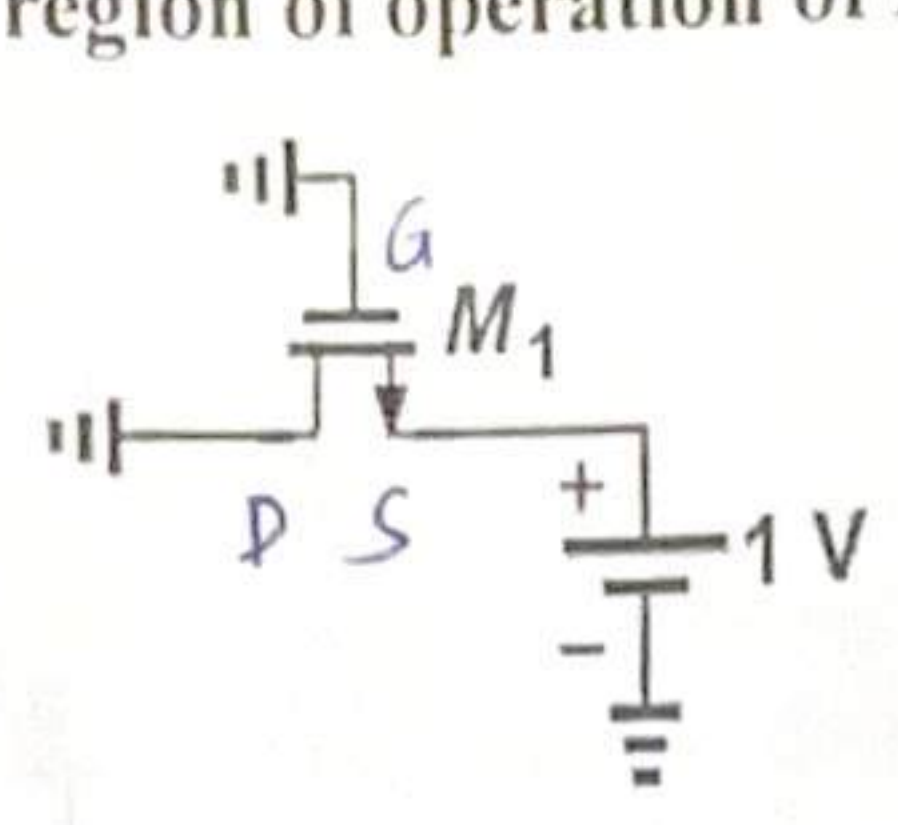
$$\begin{cases} V_{SG} > |V_{th}| \\ V_{SD} < V_{SG} - |V_{th}| \end{cases}$$

$$V_S > V_D \rightarrow \begin{cases} S \rightarrow D \\ D \rightarrow S \end{cases}$$

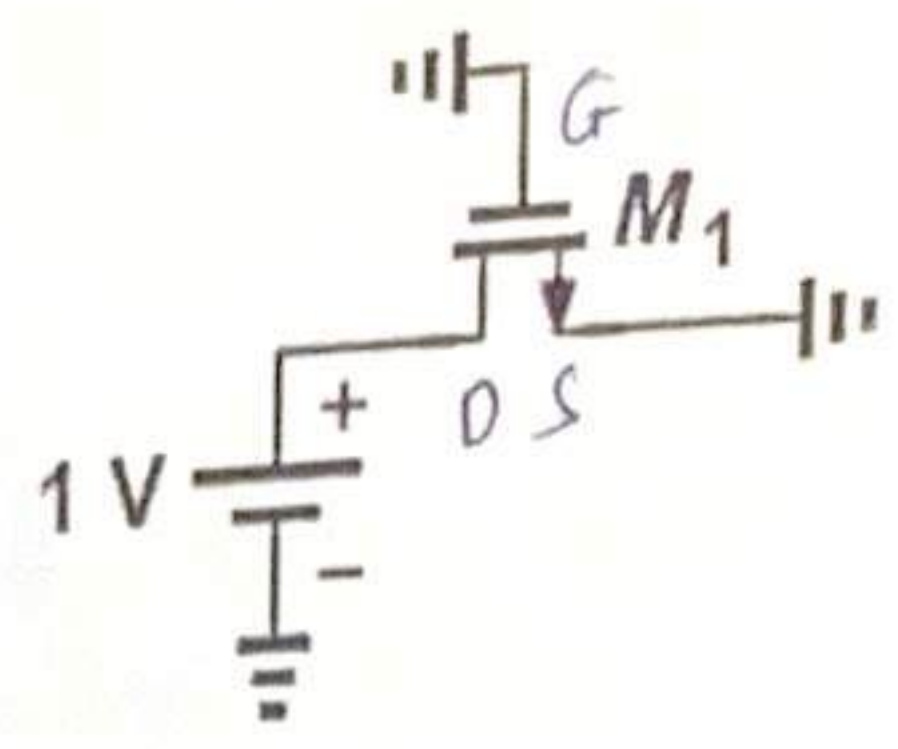
-1,5A

C2.7 Determine the region of operation of M1 in each of the circuits shown in fig:

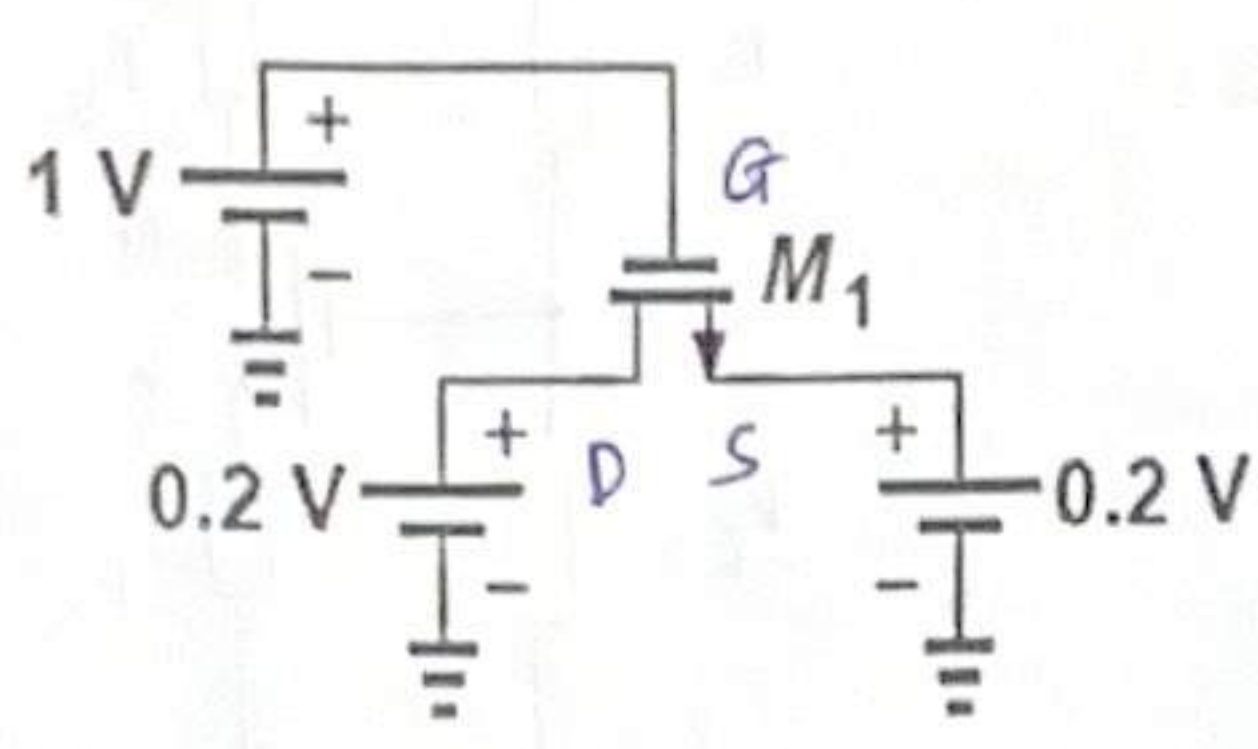
Lưu ý: NMOS 2 cách vẽ



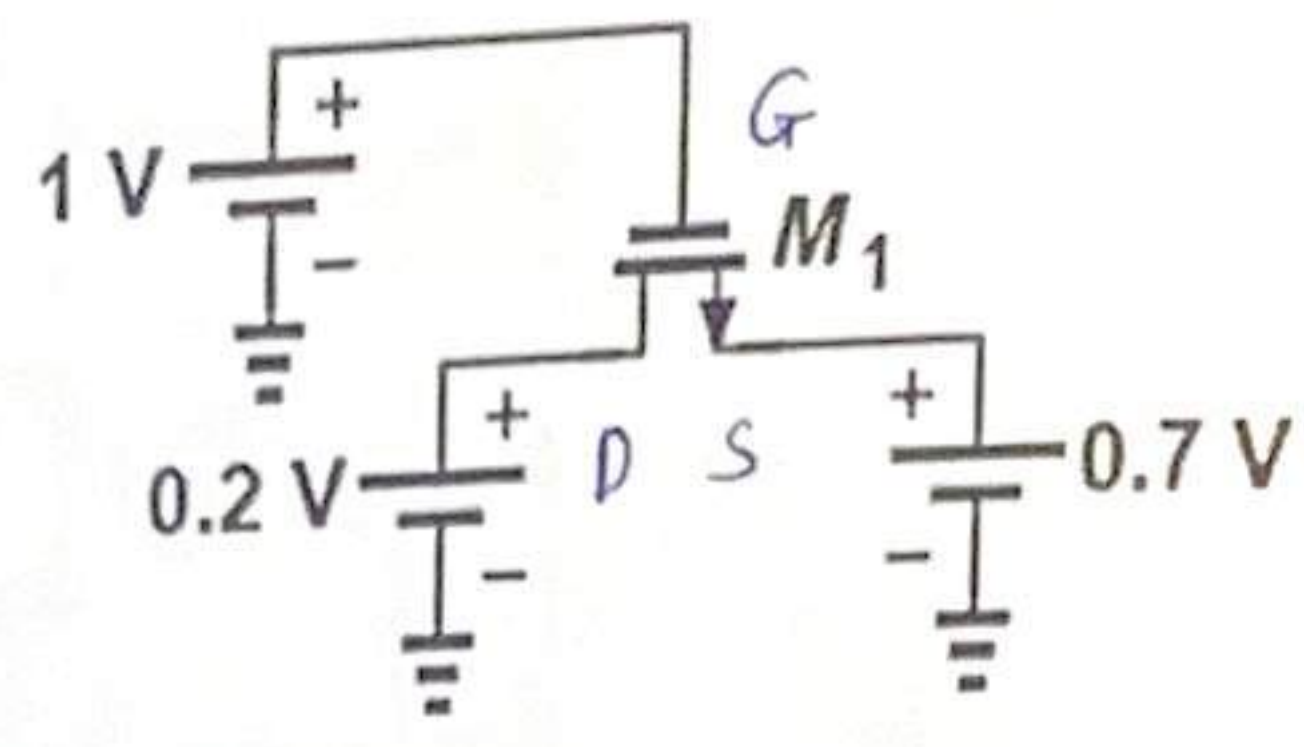
(a)



(b)



(c)



(d)

Tất cả đều là Enhancement mode P-MOSFET, (chọn $V_{th} = 0,5V$)

a. $V_S = 1V$
 $V_G = 0V \rightarrow V_{SG} = 1V > |V_{th}| = 0,5V$
 $V_D = 0V \rightarrow V_{SD} = 1V > V_{SG} - |V_{th}| = 1 - 0,5 = 0,5V$
 $\Rightarrow$ vùng hoạt động

b. $V_S = 0V$
 $V_G = 0V \rightarrow V_{SG} = 0 < |V_{th}| = 0,5V$
 $V_D = 1V \rightarrow V_{SD} = -1 < V_{SG} - |V_{th}| = -0,5V$
 $\Rightarrow$ vùng cutoff

c. $V_S = 0,2V$
 $V_G = 1V \rightarrow V_{SG} = 0,2 - 1 = -0,8V < |V_{th}| = 0,5V$
 $V_D = 0,2V \rightarrow V_{SD} = 0 > V_{SG} - |V_{th}| = -1,3V$
 $\Rightarrow$ vùng cutoff

d. $V_S = 0,7V$
 $V_G = 1V \rightarrow V_{SG} = 0,7 - 1 = -0,3V < |V_{th}| = 0,5V$
 $V_D = 0,2V \rightarrow V_{SD} = 0,5 > V_{SG} - |V_{th}| = -1,3V$
 $\Rightarrow$ vùng hoạt động

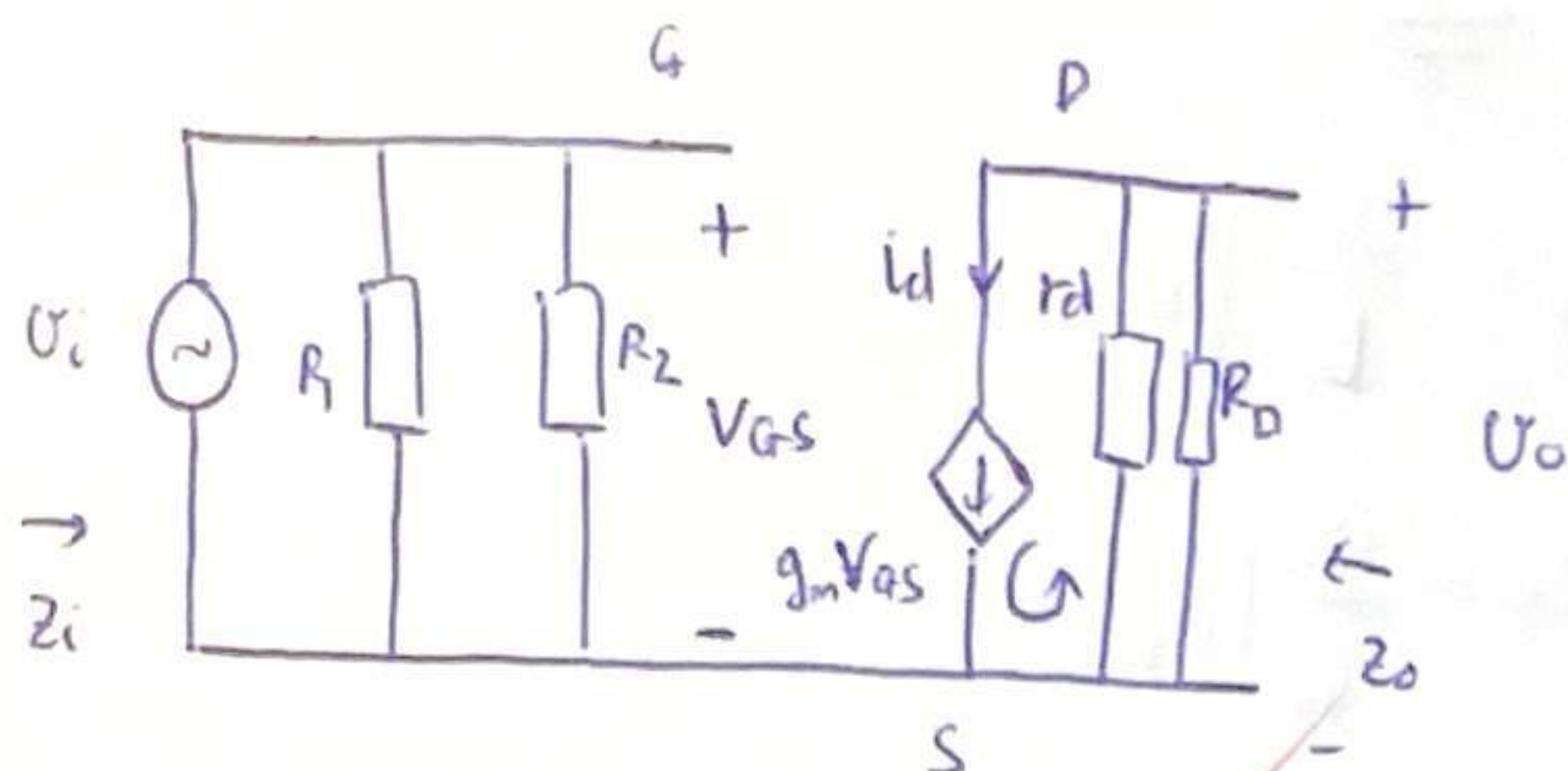
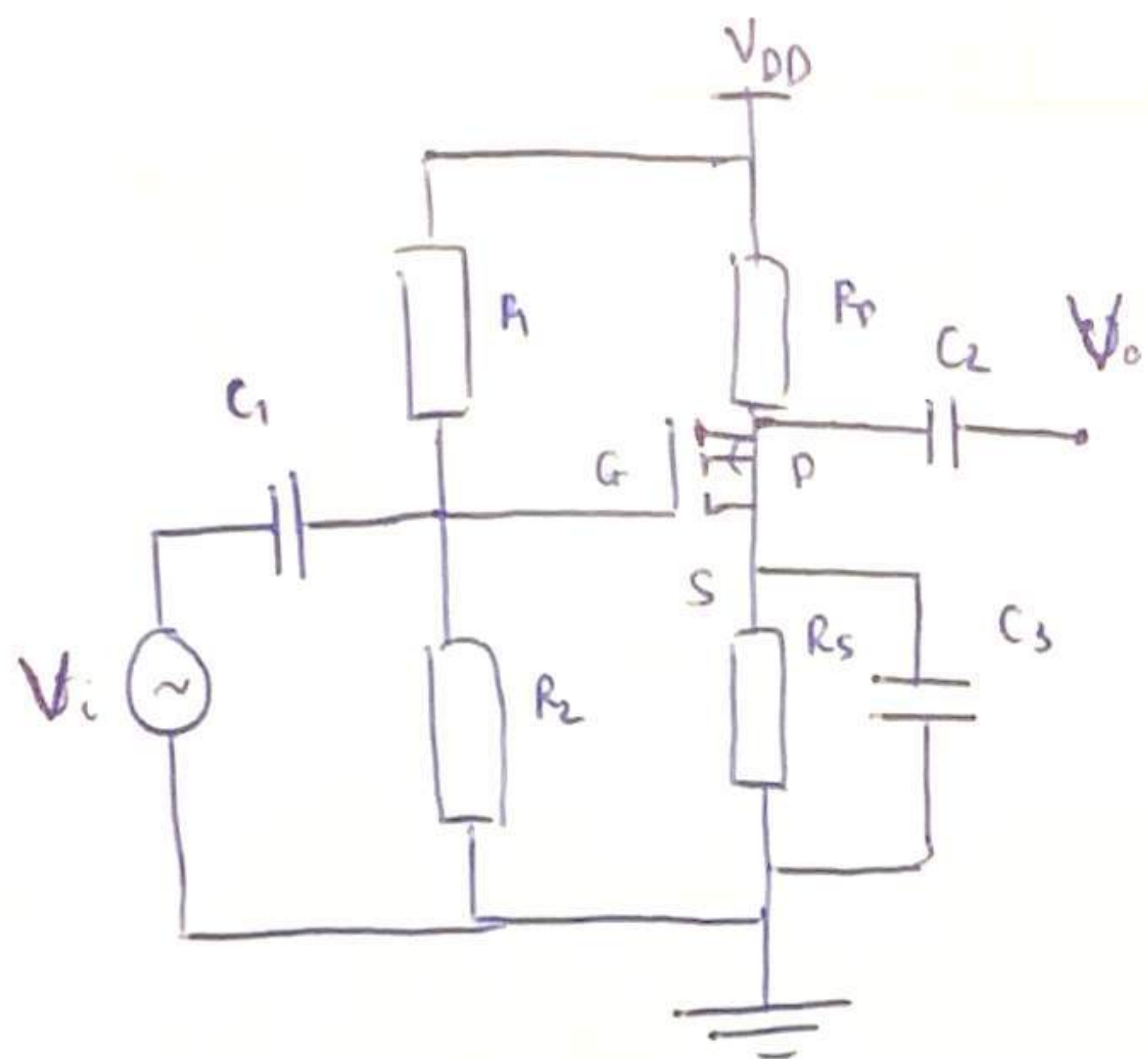
key: a) hoạt động
 b) cutoff
 c) triode
 d) hoạt động

So sánh AC, lập bảng cho CS, CG, CD

1

TH1: CS (có tụ)

AC



Ta có: $Z_{in} = \frac{U_{in}}{I_{in}} \Big|_{I_o=0} = R_1 \parallel R_2$

$Z_{out} = \frac{U_o}{I_o} \Big|_{U_i=0} = r_d \parallel R_D$

Chứng minh rõ

Trong miền hợp lý: $r_d \rightarrow \infty \Rightarrow r_d \parallel R_D = R_D \Rightarrow Z_{out} = R_D$

$A_U = \frac{U_o}{U_i}$

Ta có: $U_i = V_{GS}$

Áp dụng KVL: $U_o = -i_d \cdot (r_d \parallel R_D) \Rightarrow U_o = -g_m V_{GS} (r_d \parallel R_D)$

$i_d = g_m V_{GS}$

mà $V_{GS} = U_i$

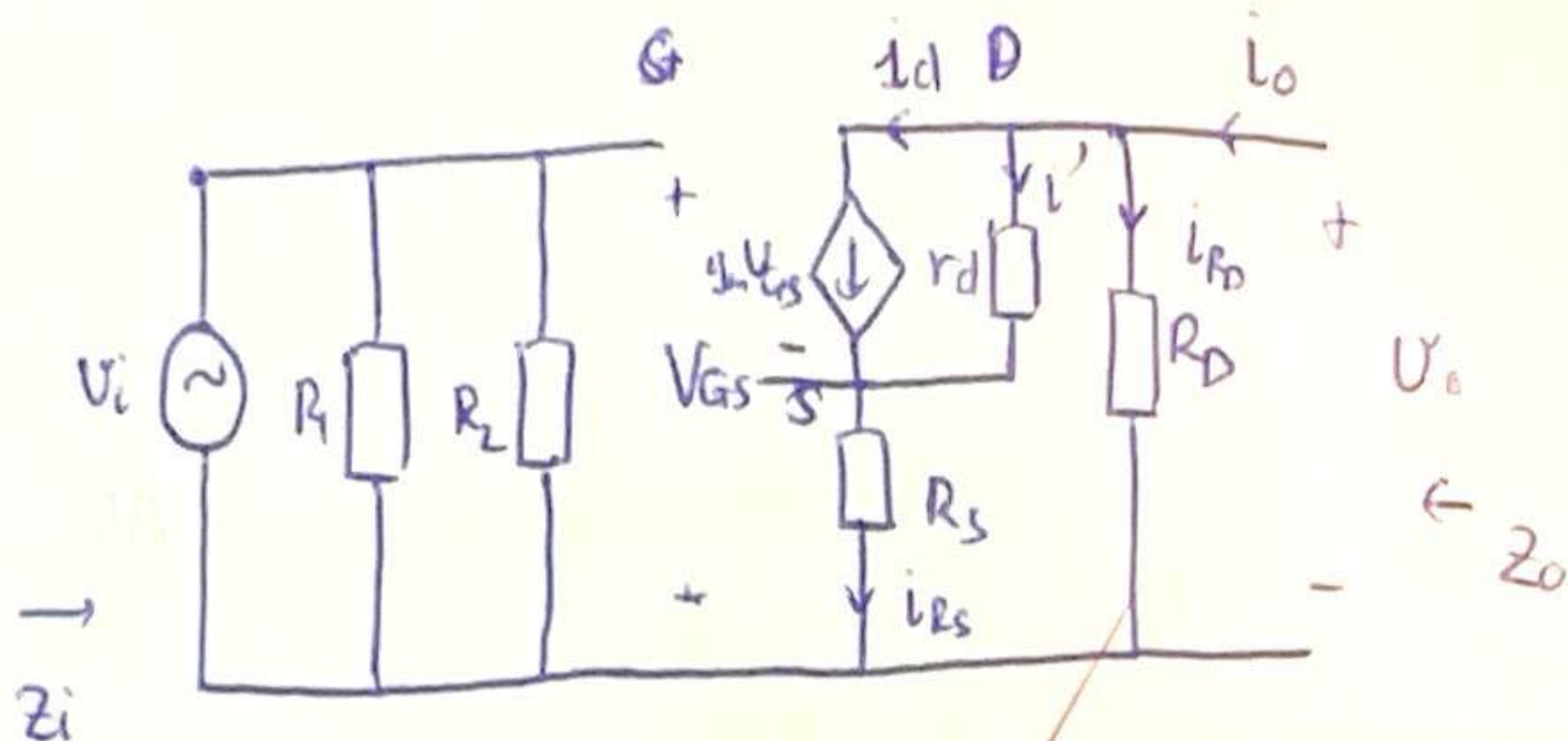
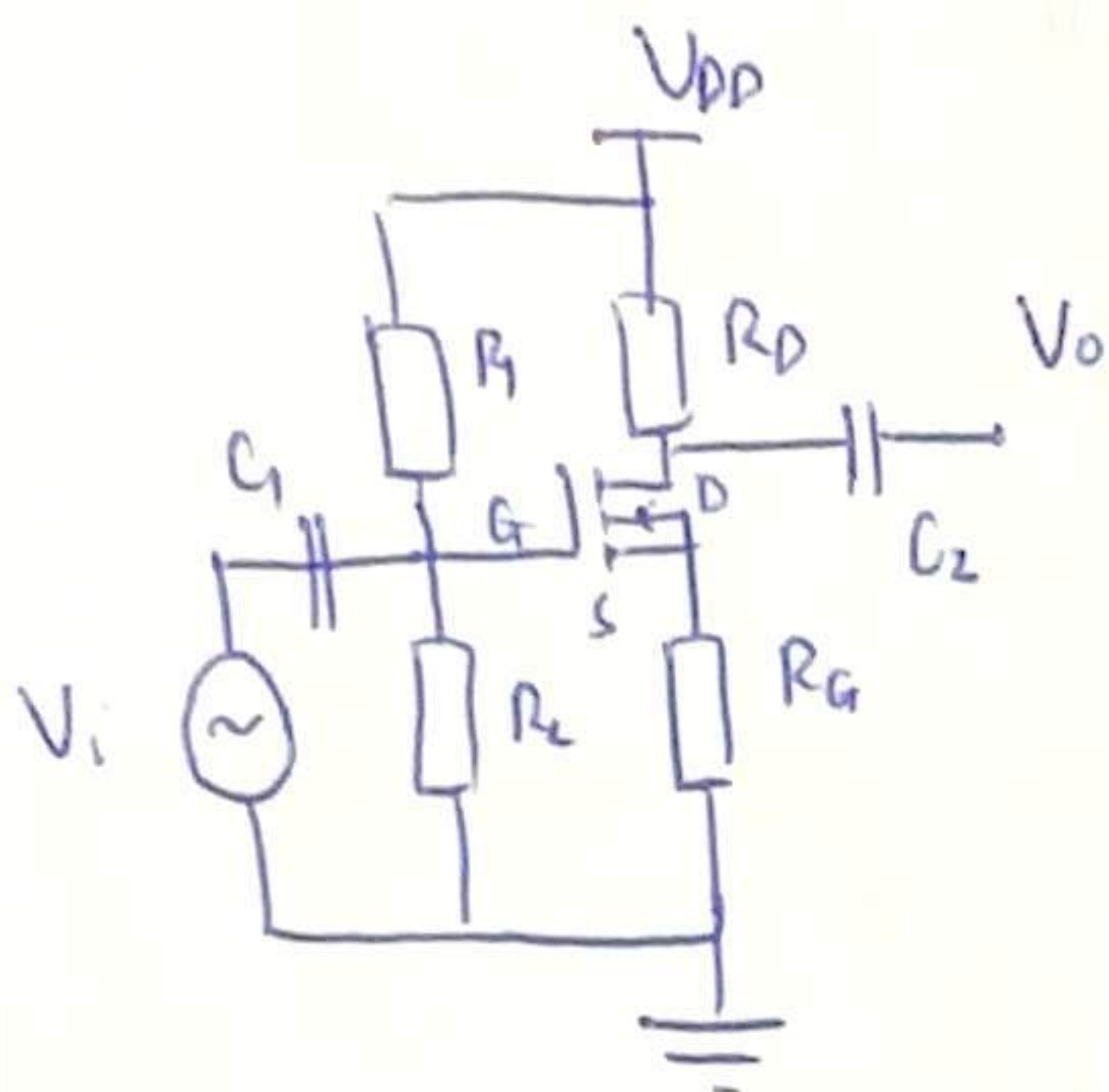
$\Rightarrow U_o = -g_m \cdot U_i (r_d \parallel R_D)$

$\Rightarrow A_U = \frac{U_o}{U_i} = \frac{-g_m \cdot U_i (r_d \parallel R_D)}{U_i} = -g_m (r_d \parallel R_D)$

Trong miền hợp lý: $r_d \rightarrow \infty \Rightarrow r_d \parallel R_D = R_D \Rightarrow A_U = -g_m R_D$

TH2: CS (khuấy m)

AC



Ta có: $Z_i = \frac{V_i}{I_i} \Big|_{V_o=0} = R_1 \parallel R_2$

$Z_o = \frac{V_o}{I_o} \Big|_{V_i=0}$

Ta có: $V_i = V_{GS} + V_{RS}$

$V_i = 0 \Rightarrow V_{GS} + V_{RS} = 0 \Leftrightarrow V_{RS} = -V_{GS}$

$i' = \frac{V_o - V_{RS}}{r_d} = \frac{V_o + V_{GS}}{r_d}$

$i_d = g_m V_{GS}$

$i_{RD} = \frac{V_o}{R_D}$

$\Rightarrow I_o = i_{RD} + i' + i_d$

$I_o = \frac{V_o}{R_D} + \frac{V_o + V_{GS}}{r_d} + g_m V_{GS}$

$I_o = V_o \left(\frac{1}{R_D} + \frac{1}{r_d} \right) + V_{GS} \left(\frac{1}{r_d} + g_m \right)$

$V_{GS} = -V_{RS} = -i_{RS} R_S = -(i_d + i') R_S = -(I_o - i_{RD}) R_S$

$\Rightarrow I_o = - (I_o - i_{RD}) \left(\frac{1}{r_d} + g_m \right) R_S + V_o \left(\frac{1}{R_D} + \frac{1}{r_d} \right) \quad (V_o = I_{RD} \cdot R_D)$

$\Rightarrow I_o \left(1 + \frac{R_S}{r_d} + g_m R_S \right) = i_{RD} \left(\frac{R_S}{r_d} + g_m R_S + \frac{R_D}{r_d} + 1 \right)$

$\Rightarrow -I_o = \frac{i_{RD} \left(1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d} \right)}{1 + \frac{R_S}{r_d} + g_m R_S}$

$Z_o = \frac{V_o}{I_o} \Big|_{V_i=0} = \frac{i_{RD} \cdot R_D}{i_{RD} \left(1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d} \right)} = \frac{1 + g_m R_S + \frac{R_S}{r_d}}{1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}} \cdot R_D$

Lưu ý: $r_d \rightarrow \infty \Rightarrow Z_o \approx \frac{1 + g_m R_S}{1 + g_m R_S} \cdot R_D = R_D$

$[R_D (1 + g_m R_S) + R_S] \parallel R_D$

$$\frac{U_o}{U_i} = \frac{i_{RD} \cdot R_D}{U_i}$$

$$U_i = V_{GS} + V_{RS} \Rightarrow V_{GS} = U_i - V_{RS} = U_i + i_{RD} R_S$$

Ta có KVL: $i' = \frac{U_o - U_{RS}}{r_d}$

KCL: $i_{RD} = -i_d - i' = -g_m V_{GS} - \frac{U_o - U_{RS}}{r_d}$

Thay V_{GS} , U_o và U_{RS}

$$i_{RD} = -g_m (U_i + i_{RD} R_S) - \frac{i_{RD} R_D - i_{RD} R_S}{r_d}$$

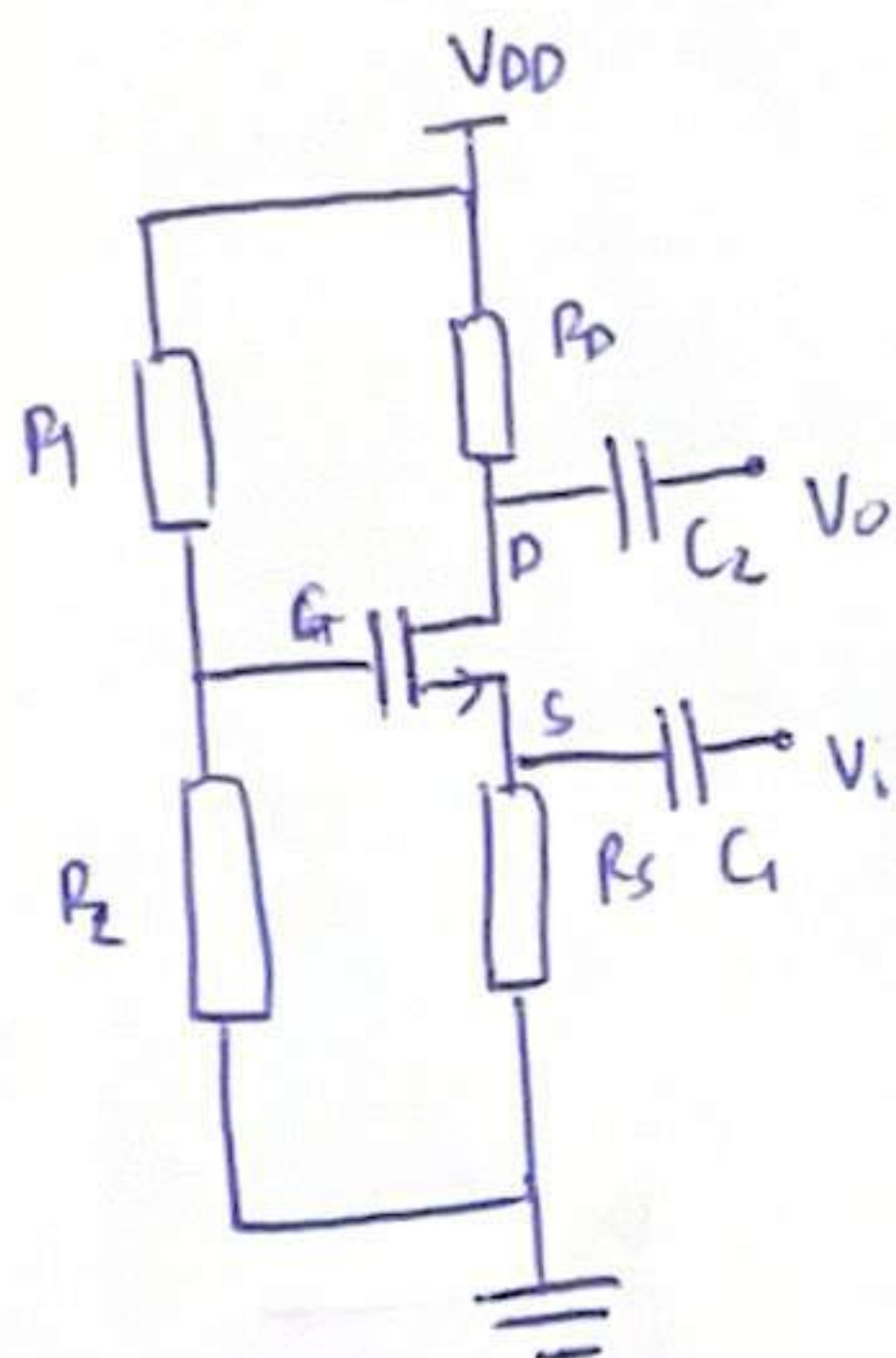
$$\Rightarrow i_{RD} \left(1 + g_m R_S + \frac{R_D + R_S}{r_d} \right) = -g_m U_i$$

$$\Rightarrow i_{RD} = \frac{-g_m U_i}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

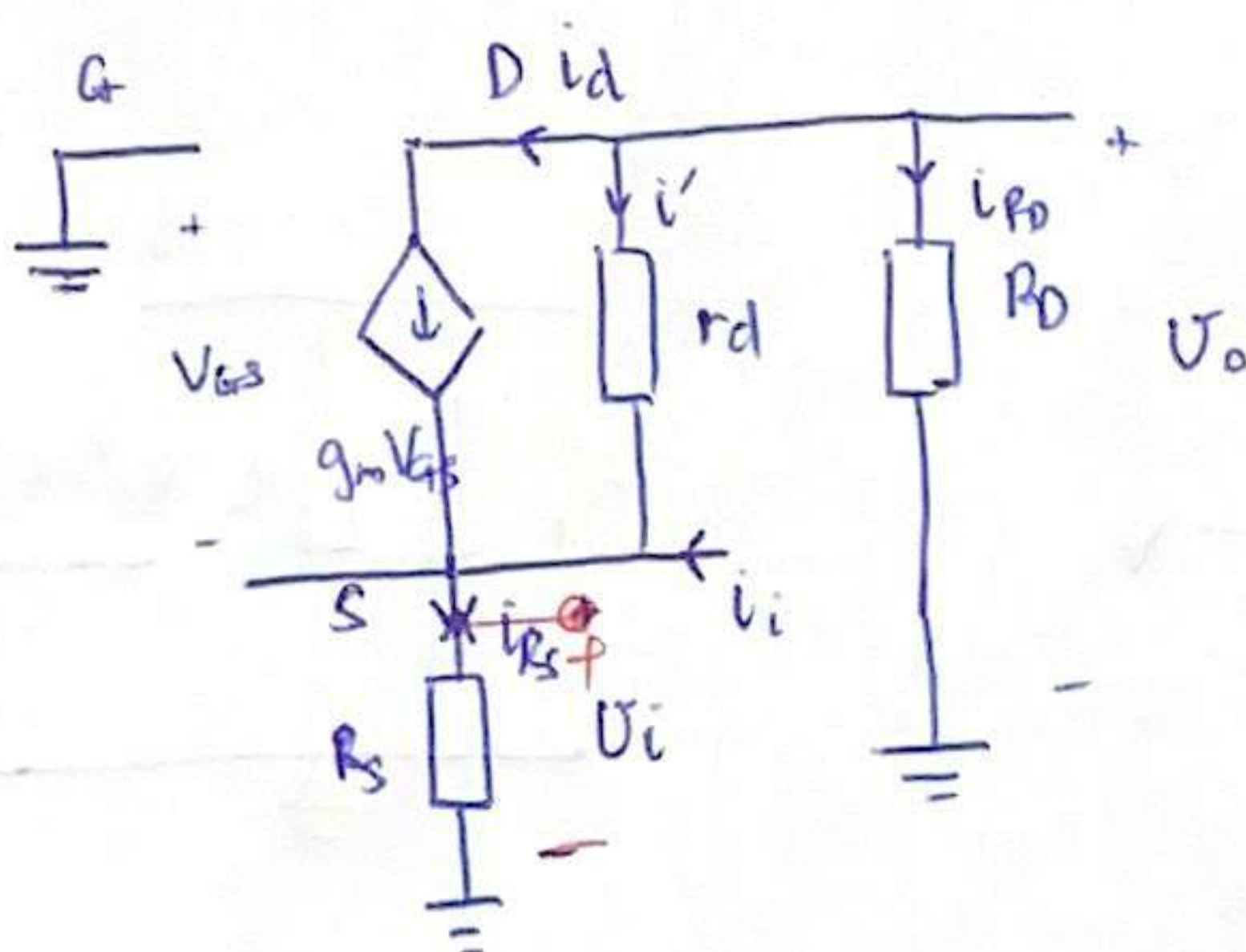
$$\Rightarrow U_o = +i_{RD} R_D = \frac{-g_m U_i R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}} \quad \rightarrow A_v = \frac{U_o}{U_i} = \frac{-g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

Lý tưởng: $r_d \rightarrow \infty \rightarrow A_v \approx \frac{-g_m R_D}{1 + g_m R_S}$ ✓

TH3: CG



AC



$$V_G = \frac{V_{DD} R_2}{R_1 + R_2} \Rightarrow AC: V_G = 0$$

$$Z_i = \frac{U_i}{i_i} \Big|_{i_o=0} = \frac{U_S}{i_i} = R_S$$

$$V_{GS} = V_G - V_S = -U_i$$

$$Z_o = \frac{U_o}{i_o} \Big|_{U_i=0} = \frac{i_{RD} R_D}{i_{RD}} = R_D \quad \text{đúng mới}$$

$$Z_i = \frac{U_i}{i_i} \Big|_{i_o=0} = \frac{-V_{GS}}{i_d + i' - i_{RS}} = \frac{-V_{GS}}{i_d - i_d - i_{RD} - i_{RS}} = \frac{-V_{GS}}{i_{RD} + i_{RS}} \quad A_v = \frac{U_o}{U_i}$$

$$\left. \begin{aligned} i_d + i' &= g_m V_{GS} + \frac{U_o - U_i}{r_d} \\ \text{mà KVL: } i_d + i' &= -i_{RP} = -\frac{U_o}{R_D} \end{aligned} \right\} \Rightarrow -\frac{U_o}{R_D} = g_m V_{GS} + \frac{U_o - U_i}{r_d}$$

$$\text{mà } V_{GS} = -U_i$$

$$\Rightarrow -g_m U_i + \frac{U_o}{r_d} - \frac{U_i}{r_d} + \frac{U_o}{R_D} = 0$$

$$\Rightarrow U_i \left(g_m + \frac{1}{r_d} \right) = U_o \left(\frac{1}{R_D} + \frac{1}{r_d} \right) \Rightarrow A_v = \frac{U_o}{U_i} = \frac{g_m + \frac{1}{r_d}}{\frac{1}{R_D} + \frac{1}{r_d}}$$

$$\text{lý ngy } r_d \rightarrow \infty \Rightarrow A_v \approx g_m R_D$$

$$\left. \begin{aligned} \text{Ta có: } i_i &= i_d + i' - i_{R_S} \\ i_d + i' &= -\frac{U_o}{R_D} \end{aligned} \right\} \Rightarrow i_i - \frac{U_i}{R_S} = \frac{U_o}{R_D} \Rightarrow i_i - \frac{U_i}{R_S} = \frac{A_v \cdot U_i}{R_D}$$

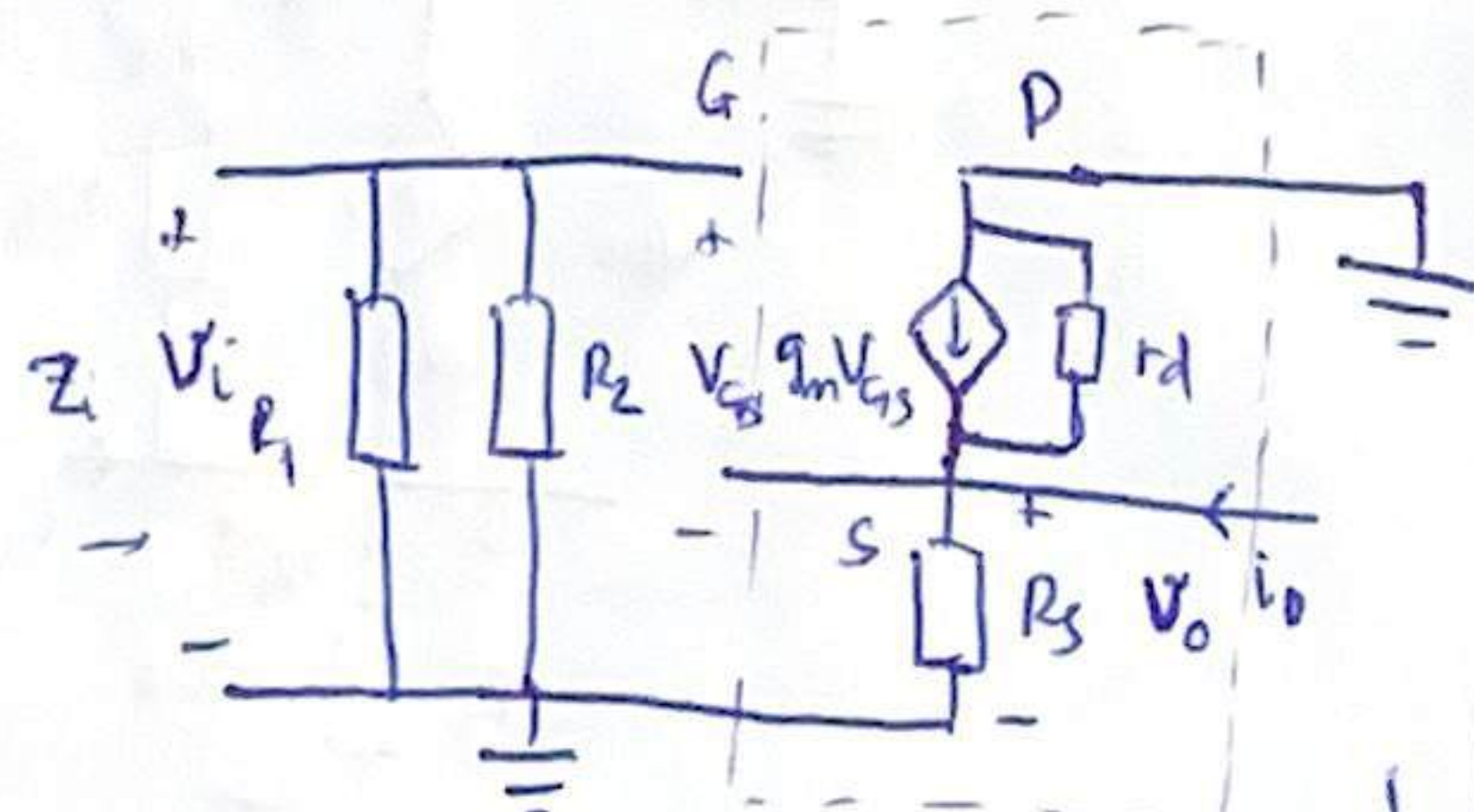
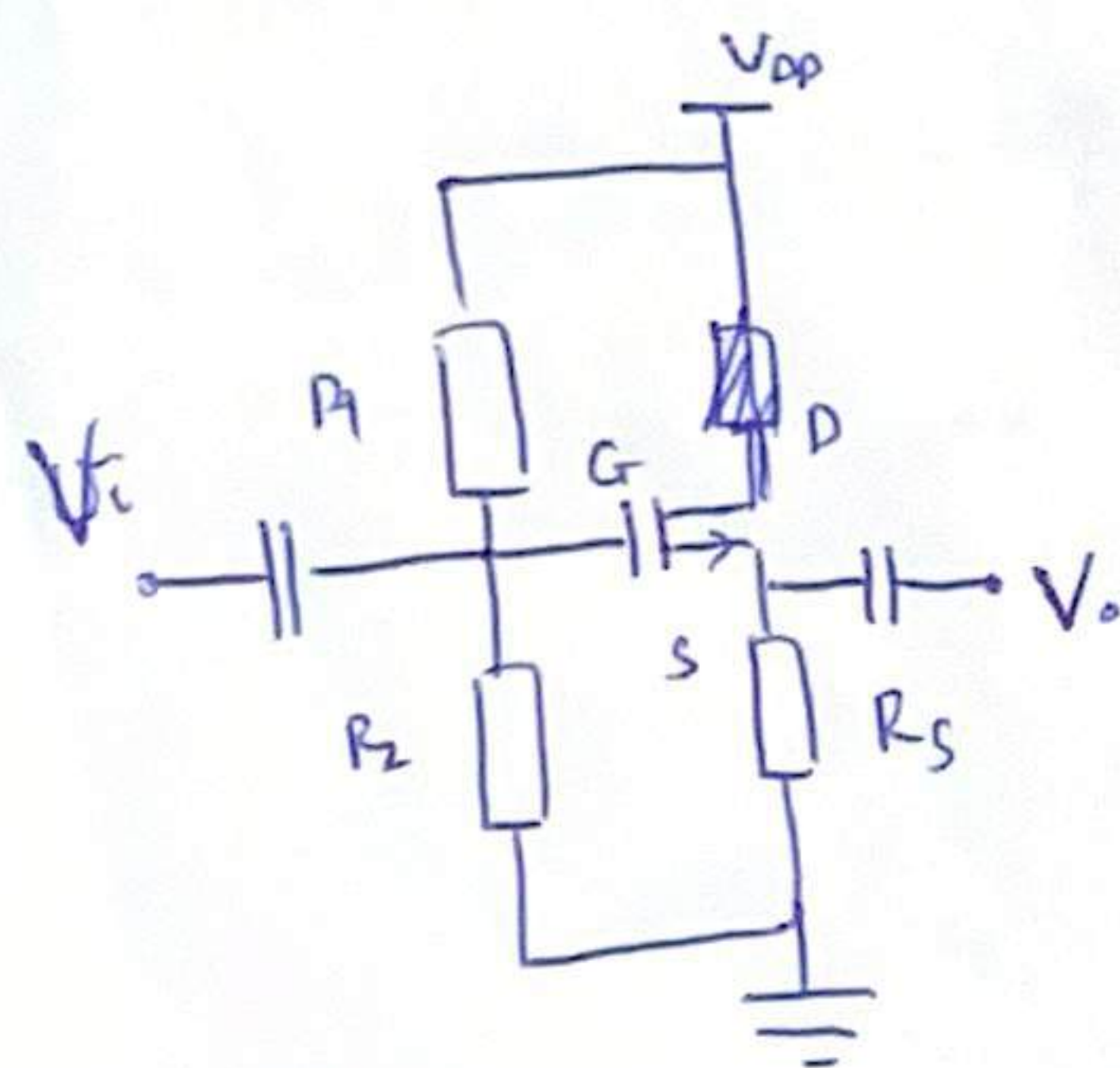
$$\Rightarrow i_i = \frac{U_i}{R_S} = \frac{U_i}{R_D} \left(\frac{g_m + \frac{1}{r_d}}{\frac{1}{R_D} + \frac{1}{r_d}} \right) = U_i \left(\frac{g_m + \frac{1}{r_d}}{1 + \frac{R_D}{r_d}} \right) \Rightarrow i_i = U_i \left(\frac{g_m + \frac{1}{r_d}}{1 + \frac{R_D}{r_d}} + \frac{1}{R_S} \right)$$

$$\Rightarrow \frac{i_i}{U_i} = \frac{g_m + \frac{1}{r_d}}{1 + \frac{R_D}{r_d}} + \frac{1}{R_S} = \frac{1}{Z_i} \Rightarrow Z_i = \frac{r_d + R_D}{r_d g_m + r_d} \parallel R_S$$

$$\text{lý ngy } r_d \rightarrow \infty \Rightarrow Z_i \approx \frac{1}{g_m} \parallel R_S$$

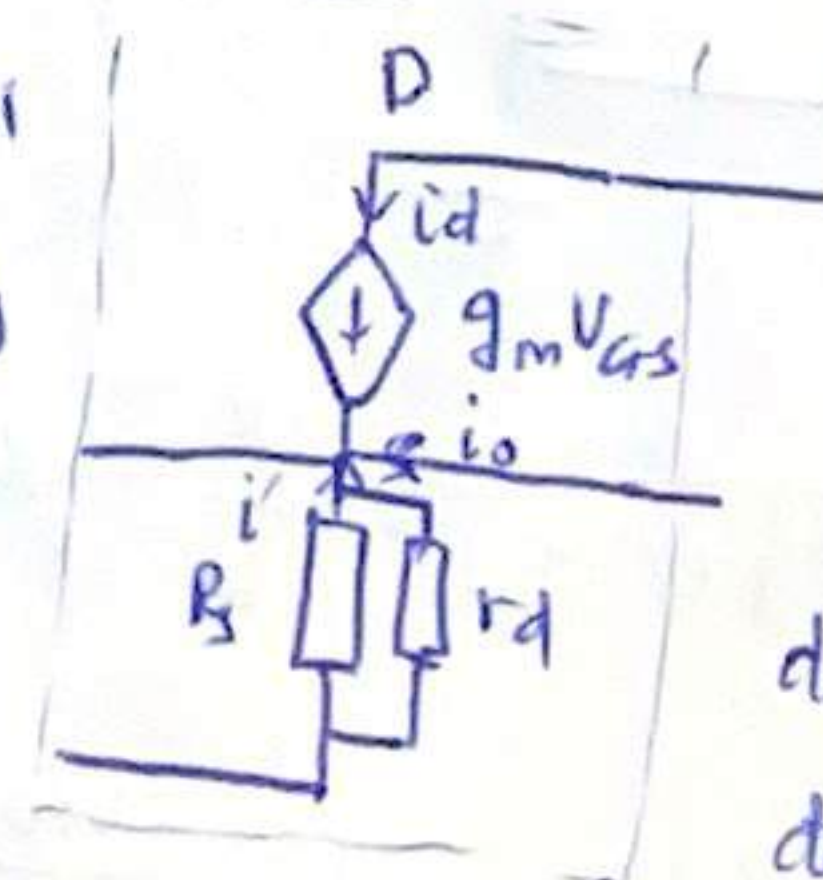
TH4: CD (Source follower)

Ac



$$Z_i = \frac{U_i}{i_i} \Big|_{i_o=0} = R_1 \parallel R_2$$

$$Z_o = \frac{U_o}{i_o} \Big|_{U_i=0} = R_S$$



$$\text{KVL, ta có: } U_i = V_{GS} + U_o \Rightarrow V_{GS} = U_i - U_o$$

$$\text{KCL: } i_d = i' \text{ (c)} \quad g_m V_{GS} = \frac{U_o}{R_S \parallel R_D} = g_m (U_i - U_o)$$

do R_S, r_d
đều có 1 đầu S,
1 đầu cửa

$$\frac{V_o}{R_S \parallel r_d} + g_m V_o = g_m V_i \quad (1) \quad \left(\frac{1}{R_S \parallel r_d} + g_m \right) V_o = g_m V_i$$

$$\Rightarrow A_V = \frac{V_o}{V_i} = \frac{\left(\frac{1}{R_S \parallel r_d} + g_m \right)}{1 + g_m (R_S \parallel r_d)} = \frac{g_m (R_S \parallel r_d)}{1 + g_m (R_S \parallel r_d)}$$

↳ khi $r_d \rightarrow \infty \Rightarrow A_V = \frac{g_m R_S}{1 + g_m R_S}$ ✓

$$Z_o = \frac{V_o}{i_o} \Big|_{V_i=0}$$

$$\text{KCL: } i_d + i' + i_o = 0$$

$$\Rightarrow i_o = -i_d - i' = -g_m V_{gs} - \frac{0 - V_o}{R_S \parallel r_d} = g_m V_o + \frac{V_o}{R_S \parallel r_d}$$

$$(V_{gs} = -V_o) = V_o \left(g_m + \frac{1}{R_S \parallel r_d} \right)$$

$$\Rightarrow Z_o = \frac{V_o}{i_o} \Big|_{V_i=0} = \frac{1}{g_m + \frac{1}{R_S \parallel r_d}} = \frac{1}{g_m + \frac{1}{R_S \parallel r_d}}$$

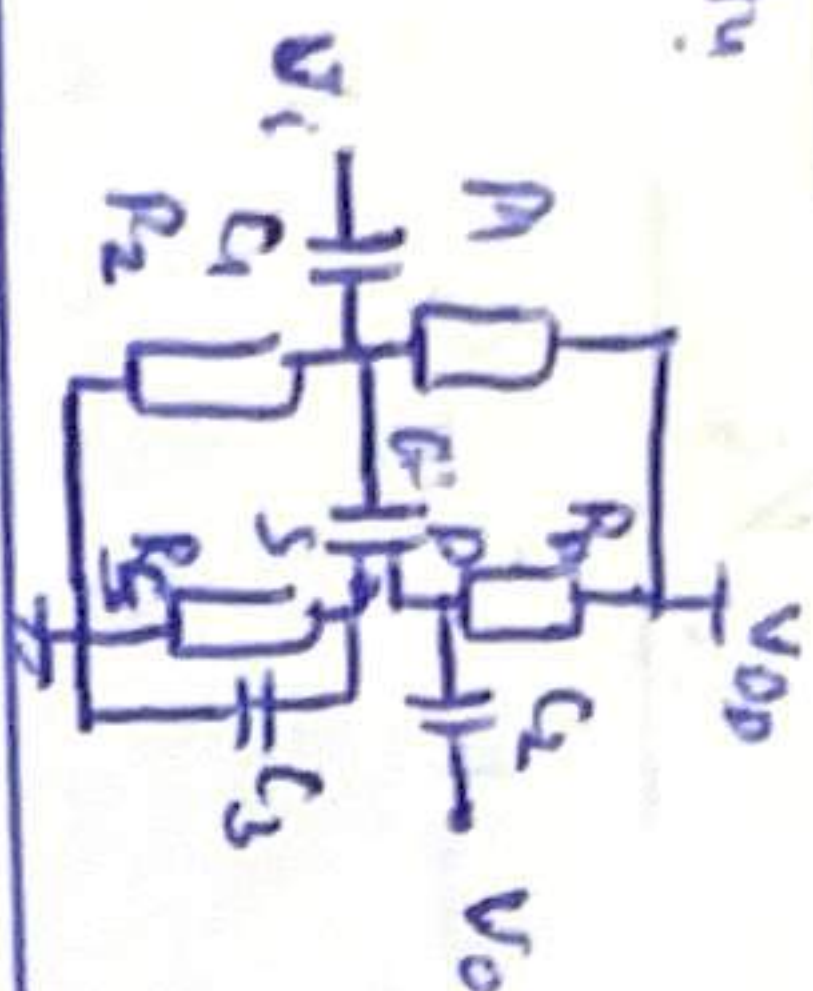
$$\Rightarrow Z_o = \left(\frac{1}{g_m} \right) \parallel (R_S \parallel r_d)$$

↳ khi $r_d \rightarrow \infty \Rightarrow Z_o = \frac{1}{g_m} \parallel R_S$

Biasing so seth

Match

CS (6th)



$$R_1 \parallel R_2$$

Z_i

Z_o

$$r_d \parallel R_D$$

$r_d \rightarrow \infty$

$$R_D$$

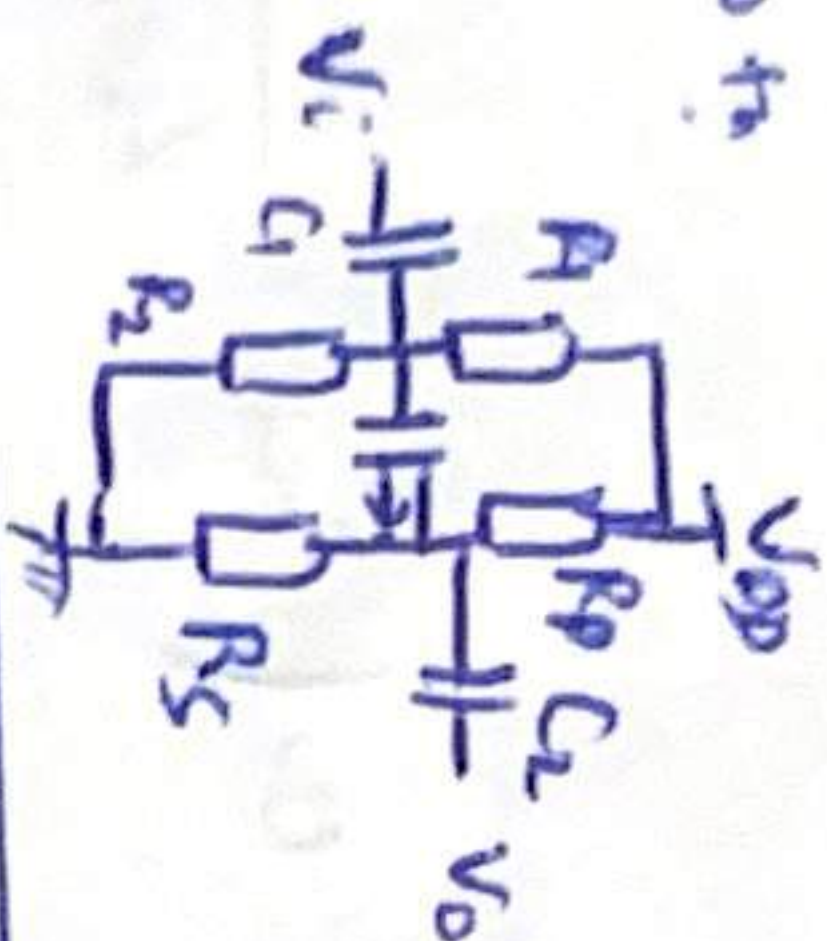
$$-g_m (r_d \parallel R_D)$$

$r_d \rightarrow \infty$

$$-g_m R_D$$

AC

CS 40th



$$R_1 \parallel R_2$$

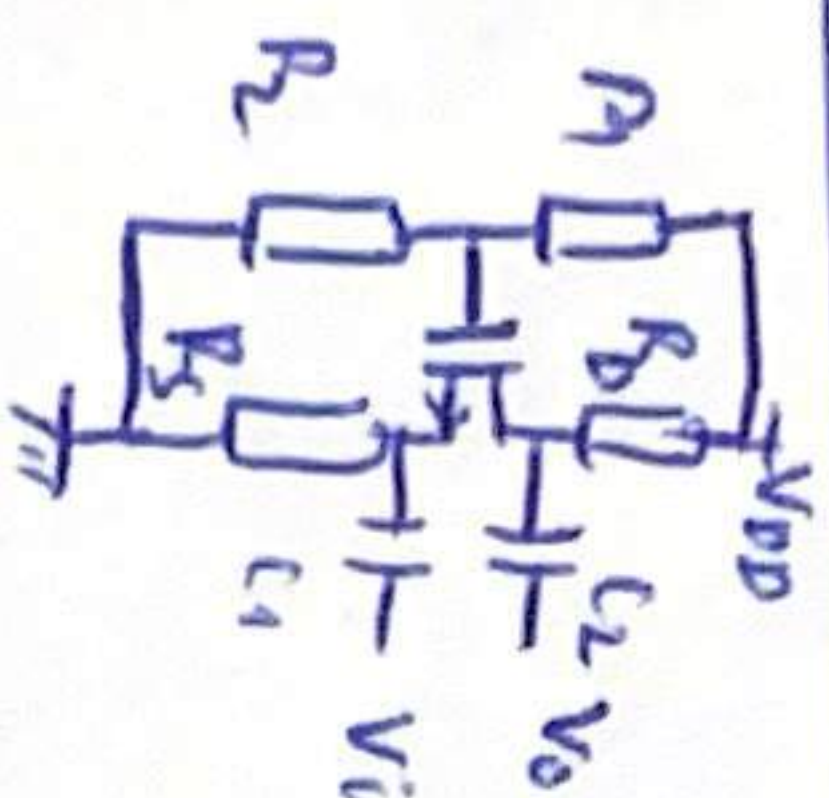
$r_d \rightarrow \infty$

$$R_D$$

$r_d \rightarrow \infty$

$$\frac{-g_m R_D}{1 + g_m R_S}$$

CG



$$\frac{r_d + R_D}{r_d \cdot g_m + 1} \parallel R_S$$

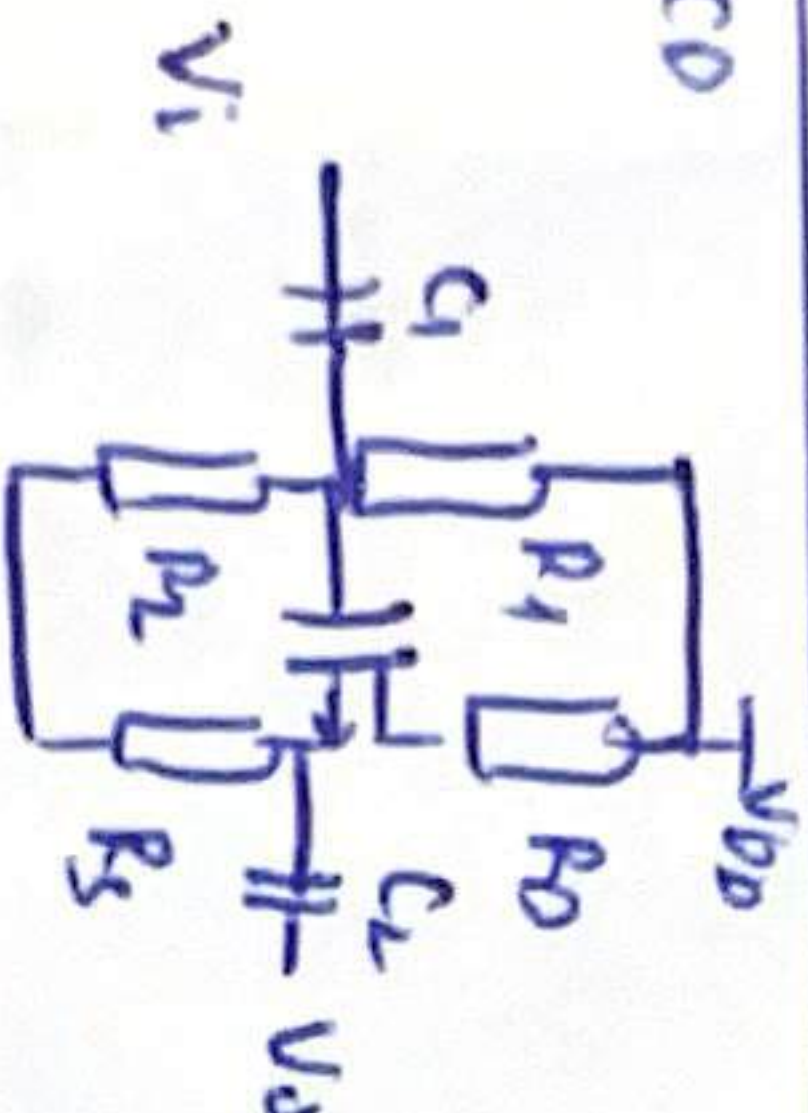
$$r_d \rightarrow \infty: \frac{1}{g_m} \parallel R_S$$

$$R_D$$

$r_d \rightarrow \infty$

$$g_m \cdot R_D$$

CD



$$R_1 \parallel R_2$$

$$R_S \parallel R_D \parallel \frac{1}{g_m}$$

$$r_d \rightarrow \infty: R_S \parallel \frac{1}{g_m}$$

$$\frac{R_S \parallel R_D}{\frac{1}{g_m} + R_S \parallel R_D}$$

$r_d \rightarrow \infty$

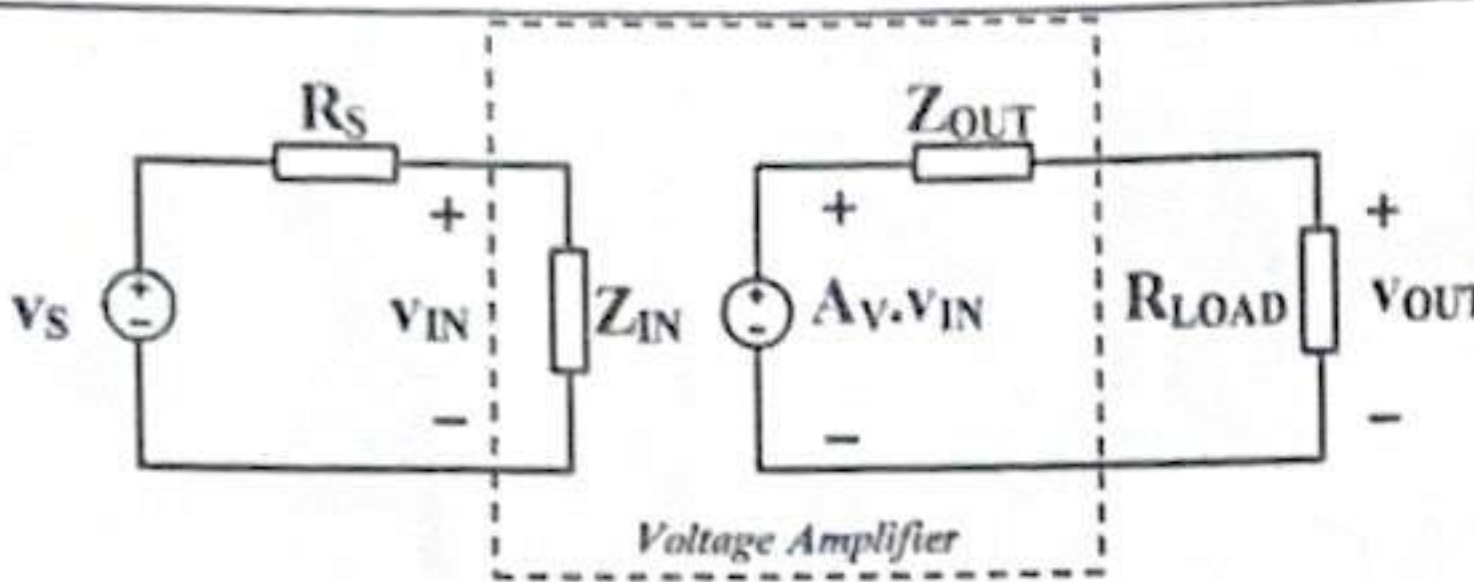
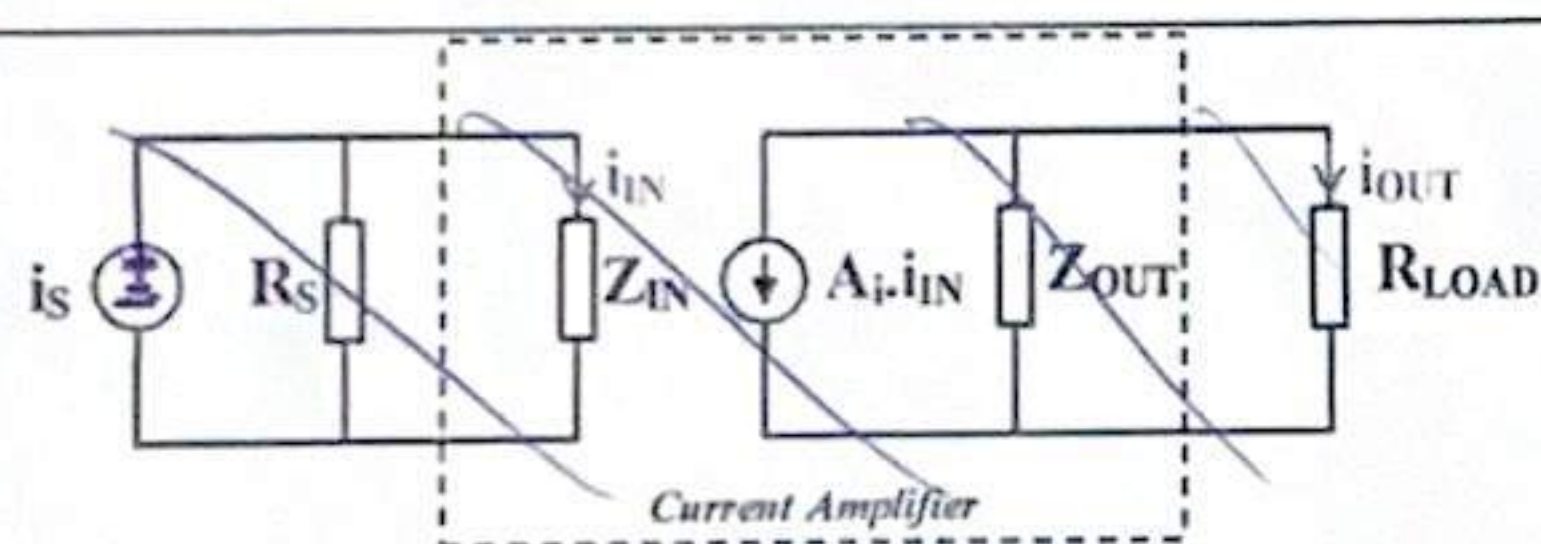
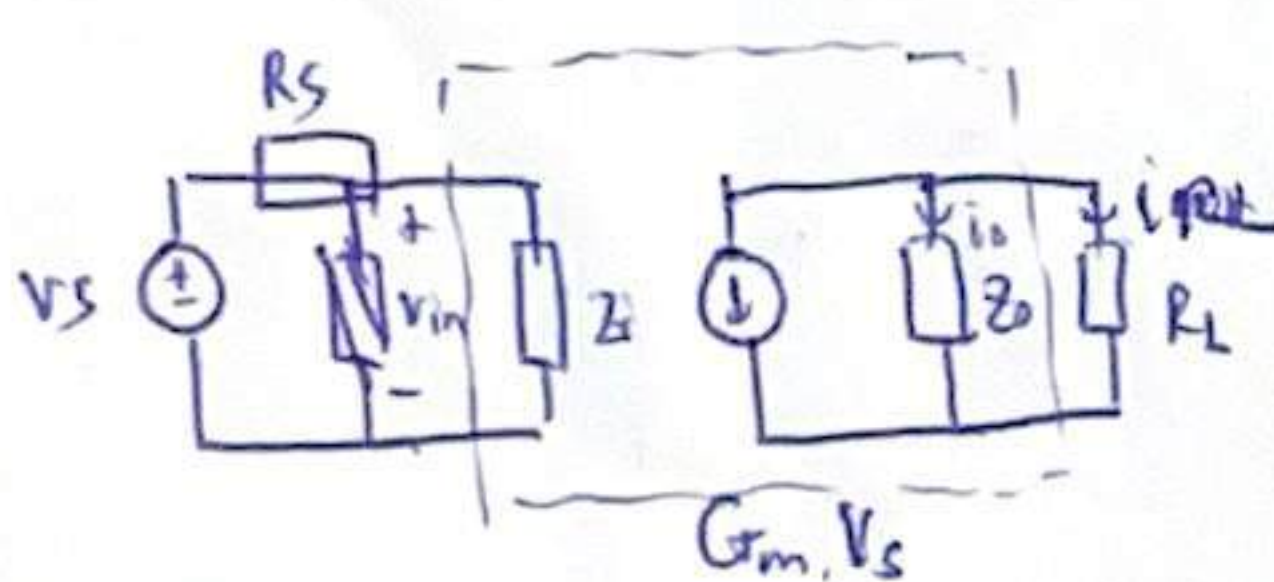
$$\frac{g_m \cdot R_S}{1 + g_m R_S}$$

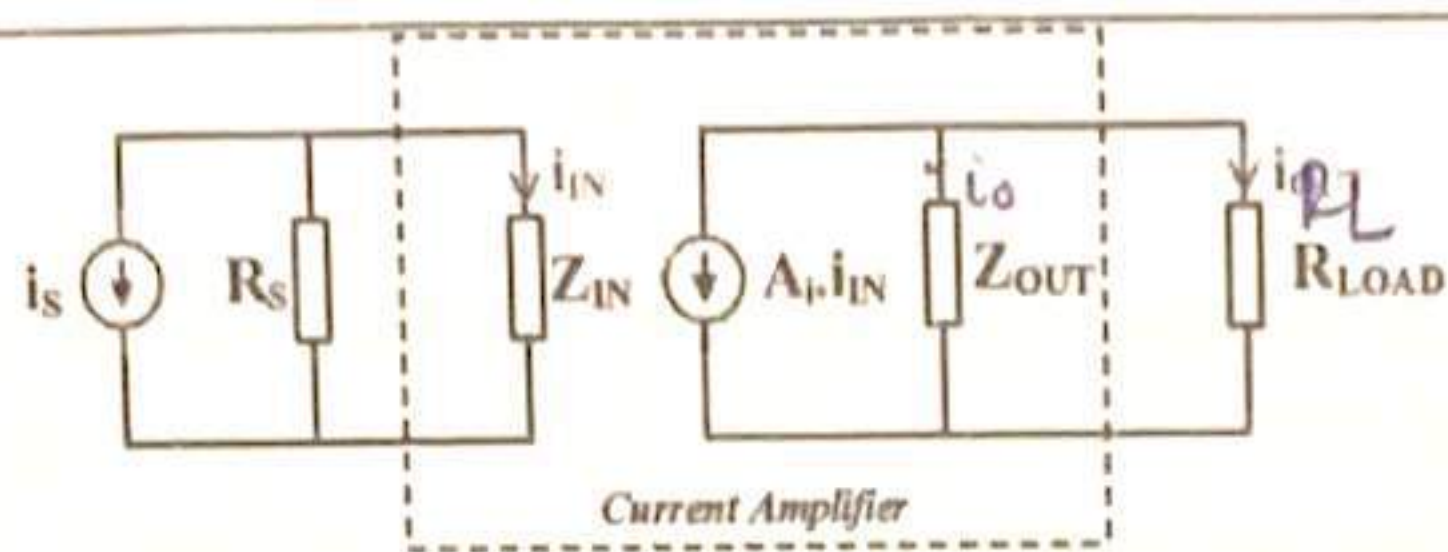
→ ko đồng đơn R_s, R_L

WEEK 8

HOMework of Chapter 3. Impact of R_s and R_{load}

C3.1 Determine again the conditions to have the total gain of these amplifiers get closed to the gain without load and source resistors?

	Gain without R_s, R_{load}	Total gain	Conditions of $Z_{in}; Z_{out}$?
 Voltage Amplifier $\frac{v_{RL}}{v_o} = \frac{R_L}{R_L + Z_o}$ $\frac{v_i}{v_s} = \frac{Z_i}{R_s + Z_i}$	$A_v^{NL} = \frac{v_{out}}{v_{in}}$	$A_v^L = \frac{v_{RL}}{v_i} = \frac{R_L}{Z_o + R_L} \cdot A_v^{NL}$ $= \frac{v_o}{v_i} \cdot \frac{R_L}{Z_o + R_L}$ $A_v^T = \frac{v_{RL}}{v_s} = \frac{v_{RL}}{v_i} \cdot \frac{v_i}{v_s}$ $= A_v^L \cdot \frac{Z_i}{R_s + Z_i}$ $= \frac{R_L}{Z_o + R_L} \cdot A_v^T \cdot \frac{Z_i}{R_s + Z_i}$	$A_v^{NL} \approx A_v^T$ $\Rightarrow \begin{cases} \frac{R_L}{Z_o + R_L} \rightarrow 1 \\ \frac{Z_i}{R_s + Z_i} \rightarrow 1 \end{cases}$ $\Rightarrow \begin{cases} Z_o \rightarrow 0 \\ Z_i \rightarrow \infty \end{cases}$
 Current Amplifier  $G_m \cdot v_s$ $\frac{i_{RL}}{i_o} = \frac{Z_o}{Z_o + R_L}$ $\frac{v_i}{v_s} = \frac{Z_i}{Z_i + R_s}$	$G_m^{NL} = \frac{i_o}{v_i}$	$G_m^L = \frac{i_{RL}}{v_i} = \frac{Z_o}{Z_o + R_L} \cdot \frac{i_o}{v_i}$ $= G_m^{NL} \cdot \frac{Z_o}{Z_o + R_L}$ $G_m^T = \frac{i_{RL}}{v_s} = \frac{i_{RL}}{v_i} \cdot \frac{v_i}{v_s}$ $= \frac{Z_o}{Z_o + R_L} \cdot G_m^{NL} \cdot \frac{Z_i}{Z_i + R_s}$	$G_m^T \approx G_m^{NL}$ $\Rightarrow \begin{cases} \frac{Z_o}{Z_o + R_L} \rightarrow 1 \\ \frac{Z_i}{Z_i + R_s} \rightarrow 1 \end{cases}$ $\Rightarrow \begin{cases} Z_o \rightarrow \infty \\ Z_i \rightarrow 0 \end{cases}$



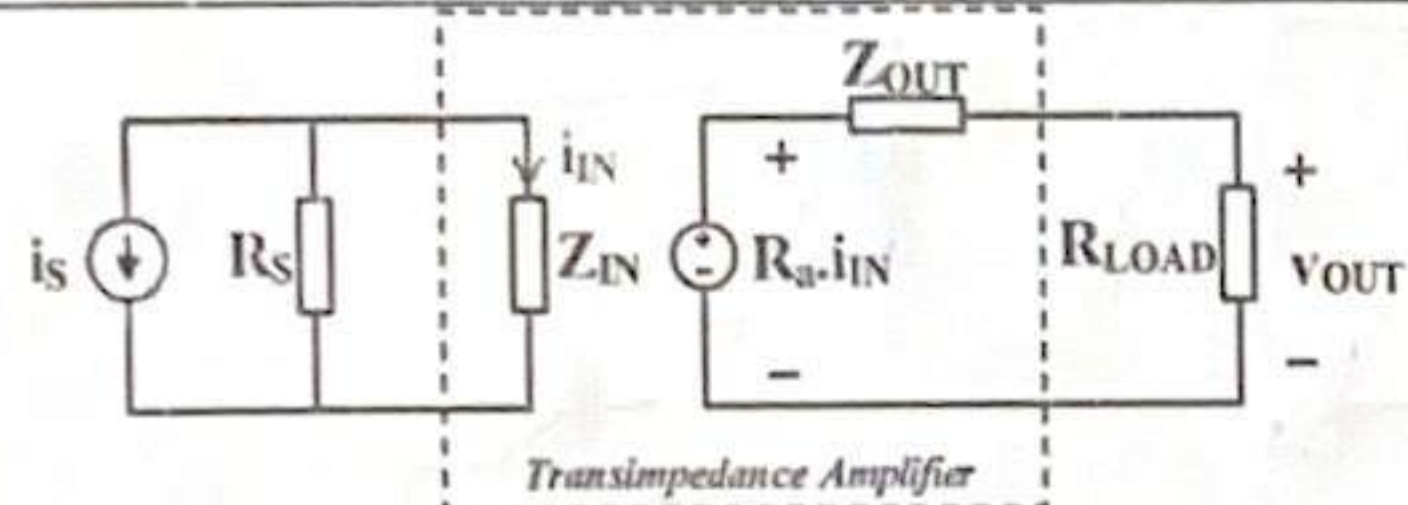
$$\frac{i_{PL}}{i_o} = \frac{Z_o}{Z_o + R_L}$$

$$\frac{i_i}{i_s} = \frac{R_s}{R_s + Z_i}$$

$$A_i^{NL} = \frac{i_o}{i_i}$$

$$\begin{aligned} A_i^T &= \frac{i_{PL}}{i_s} \\ &= \frac{i_o}{i_i} \cdot \frac{Z_o}{Z_o + R_L} \cdot \frac{Z_i}{Z_i + R_s} \\ &= A_i^{NL} \cdot \frac{Z_o}{Z_o + R_L} \cdot \frac{R_s}{Z_i + R_s} \end{aligned}$$

$$A_i^T \approx A_i^{NL} \Rightarrow \begin{cases} \frac{Z_o}{Z_o + R_L} \rightarrow 1 \\ \frac{R_s}{Z_i + R_s} \rightarrow 1 \\ Z_o \rightarrow \infty \\ Z_i \rightarrow 0 \end{cases}$$



$$\frac{i_i}{i_s} = \frac{R_s}{R_s + Z_i}$$

$$\frac{V_{RL}}{V_o} = \frac{R_L}{R_L + Z_o}$$

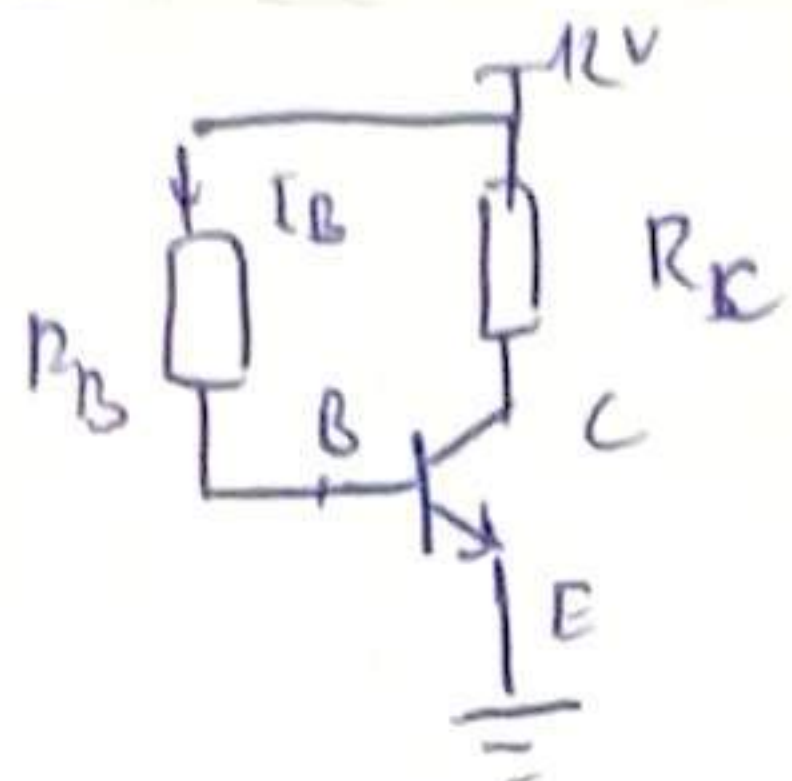
$$R_a^{NL} = \frac{V_o}{i_i}$$

$$\begin{aligned} R_a^T &= \frac{V_{RL}}{i_s} \\ &= \frac{V_o}{i_i} \cdot \frac{R_L}{R_L + Z_o} \cdot \frac{R_s}{R_s + Z_i} \\ &= R_a^{NL} \cdot \frac{R_L}{R_L + Z_o} \cdot \frac{R_s}{R_s + Z_i} \end{aligned}$$

$$R_a^T \approx R_a^{NL} \Rightarrow \begin{cases} \frac{R_L}{R_L + Z_o} \rightarrow 1 \\ \frac{R_s}{R_s + Z_i} \rightarrow 1 \\ Z_o \rightarrow 0 \\ Z_i \rightarrow 0 \end{cases}$$

a.

DC:



Ta có:

$$V_{CC} = V_{PB} + V_{BE}$$

$$V_{EC} = I_B R_B + V_{BE}$$

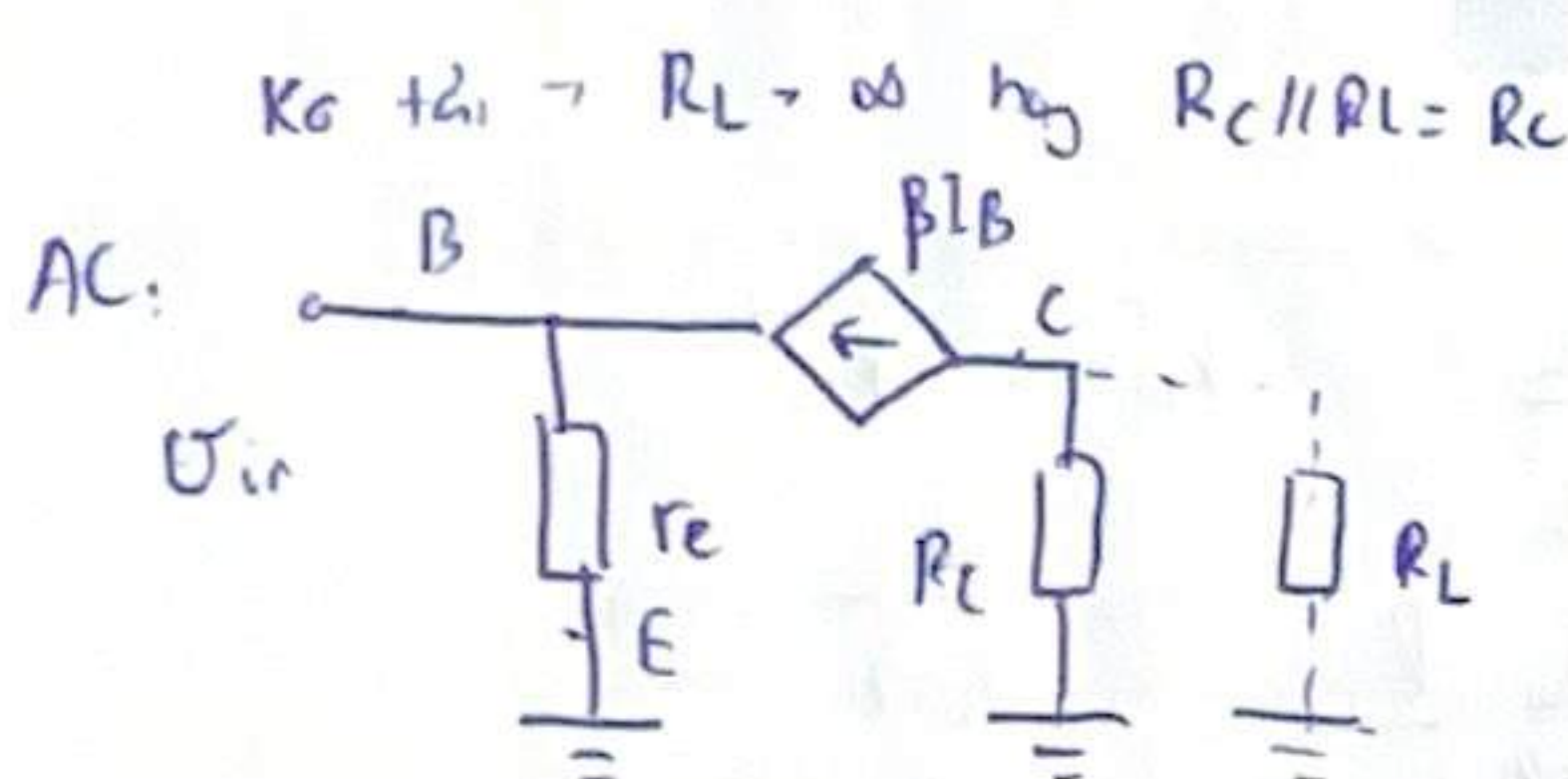
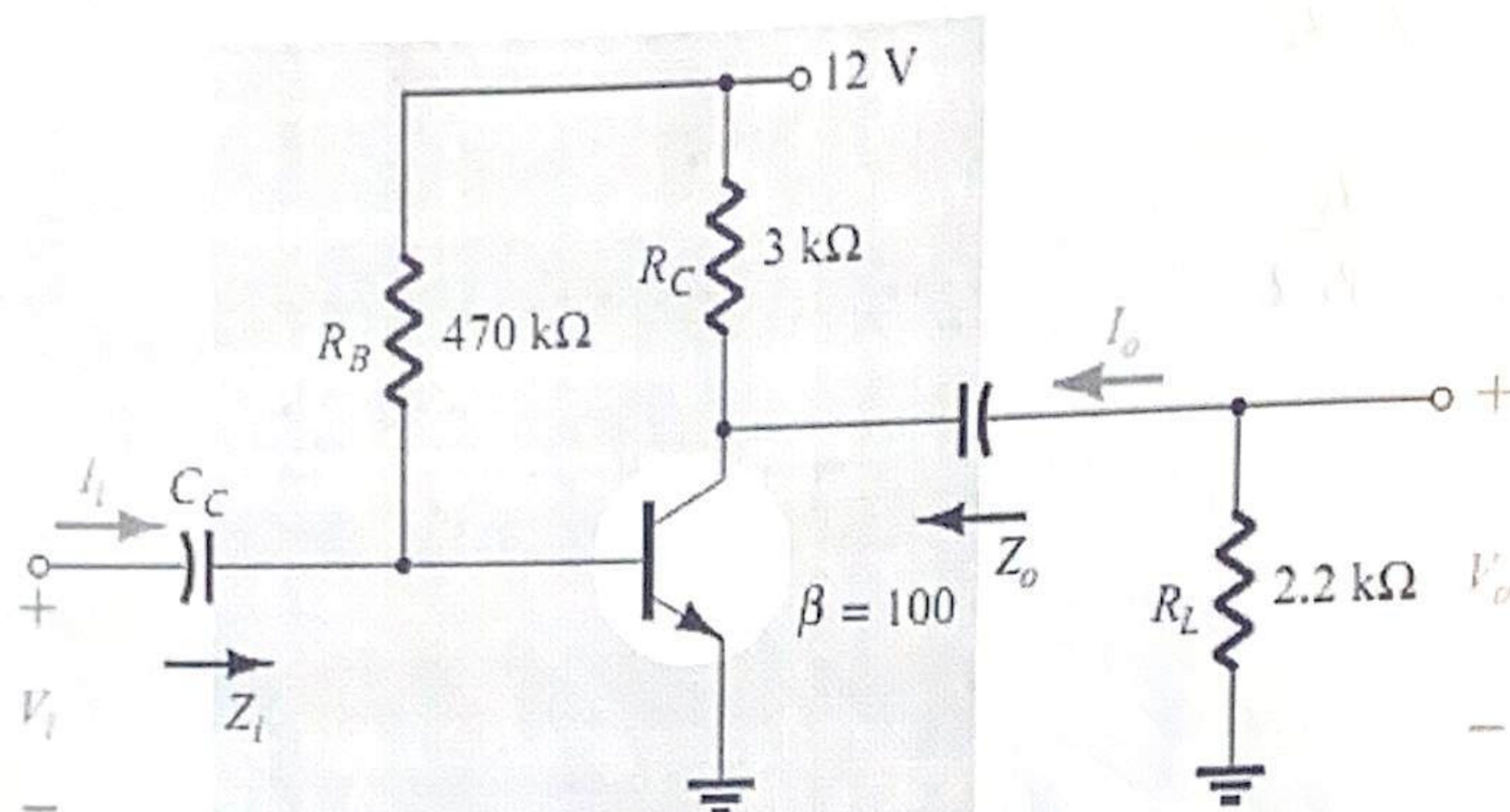
$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = 2,4 \cdot 10^{-5} \text{ (A)}$$

$$I_E = 2,43 \text{ mA} = (\beta + 1) I_B$$

$$r_e = \frac{26 \text{ mV}}{I_E} = 10,7 \Omega$$

C3.2

- Determine voltage gain, current gain, input and output impedances for the circuit without load, with load equal to 4 kΩ and for 4 Ω?
- Adding a darlington pair at the output and calculate again the gain with the same loads as in question (a).
- Comments



$$Z_i = \frac{V_i}{I_i} \Big|_{V_o=0} = \beta r_e$$

$$Z_o = \frac{V_o}{I_o} \Big|_{V_i=0} = R_C \parallel R_L$$

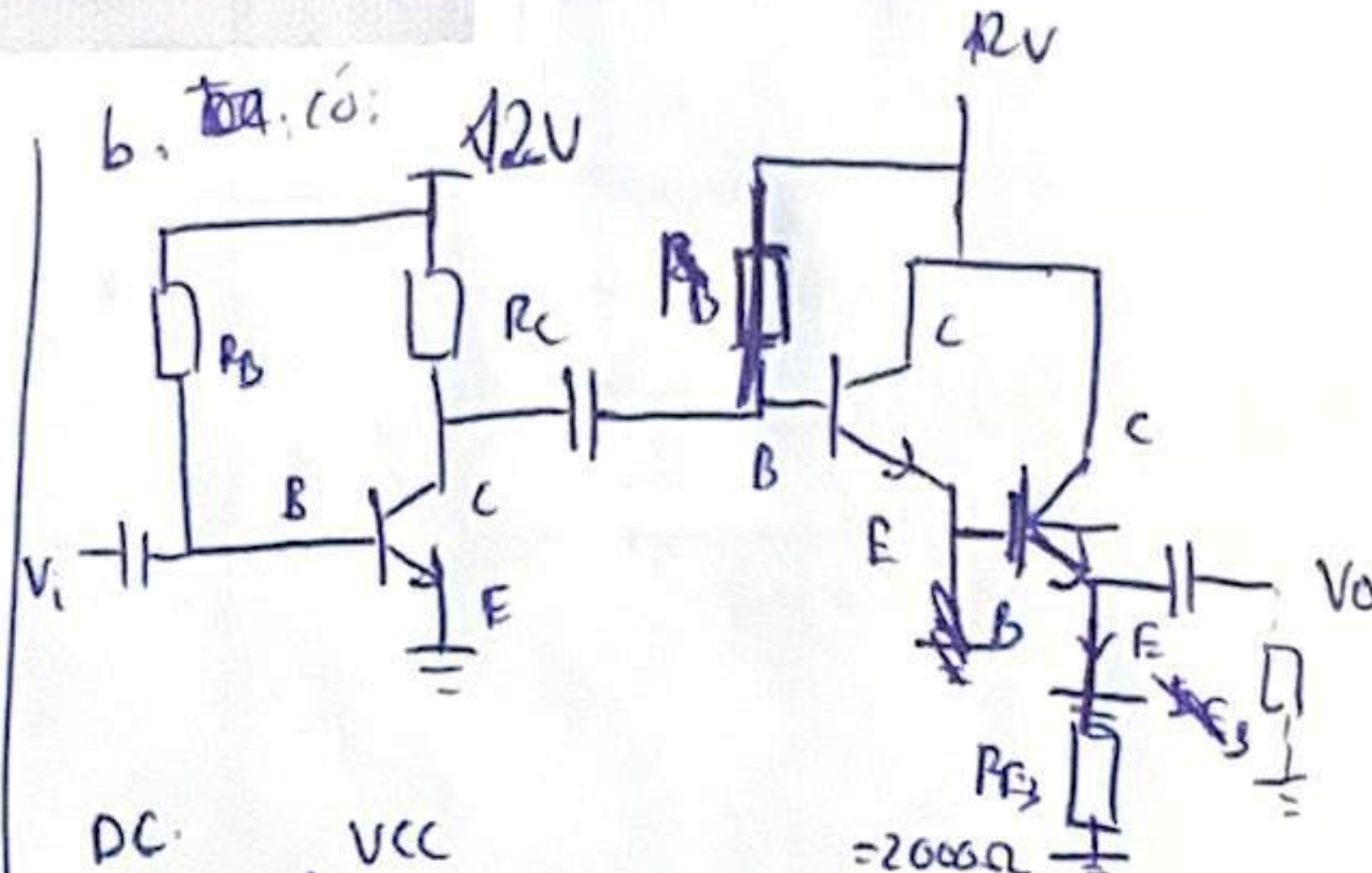
$$(K_o \text{ rđi} \rightarrow R_C \parallel R_L = R_C)$$

$$\left\{ \begin{array}{l} \text{no load} \\ \text{load} \end{array} \right. \left\{ \begin{array}{l} A_V^{NL} = \frac{V_o}{V_i} = -\frac{R_C \parallel R_L}{r_e} \text{ (chú ý BJT)} \\ A_i^{NL} \approx \beta \text{ (chú ý BJT)} \end{array} \right.$$

$$\rightarrow \text{có thể: } A_V^T = A_V^{NL} \cdot \frac{R_L}{R_L + R_C}; \quad A_i^T = A_i^{NL} \cdot \frac{-Z_i}{R_L}$$

$\Rightarrow$ Ta có bảng số liệu

	ω (Kđi)	4 kΩ	4 Ω
Z_i	1,031 kΩ	1,031 kΩ	1,031 kΩ
Z_o	3 kΩ	1714,3 Ω	4 Ω
A_V	-280	-160,21	-0,323
A_i	65	-12,4035	-12,404



DC:

VCC

$V_{CC} = V_{BE2} + V_{BE3} + I_{E3} R_{E3}$

$I_{E3} = \frac{V_{CC} - V_{BE2} - V_{BE3}}{R_{E3}} = 2,255 \text{ mA}$

$I_{E2} = I_{B3} = \frac{I_{E3}}{\beta} = 2,255 \text{ mA} = 53 \mu\text{A}$

$I_{E3} = \frac{I_{E2}}{\beta} = 0,2255 \text{ mA} = 225,5 \mu\text{A}$

$r_{E2} = 1,053 \Omega$

$r_{E3} = 0,1053 \Omega$

$r_{E2} = 4,906 \Omega$

$r_{E3} = 490,6 \Omega$

AC:

V_i

V_o

R_L

R_{E3}

R_{E2}

R_C

R_B

βI_{B1}

βI_{B2}

βI_{B3}

r_{E1}

r_{E2}

r_{E3}

r_{E4}

r_{E5}

$$A_v = \frac{I_{E3}(R_L \parallel R_{E3})}{I_{E1}(r_{E1} + R)} = \frac{(\beta+1) \cdot I_{B2} (R_{E3} \parallel R_L)}{-\frac{\beta+1}{\beta} I_{B2} \left(((\beta+1)r_{E2} + (\beta+1)^2(r_{E3} \parallel R_L)) \frac{1}{\beta} + 1 \right) (r_{E1})}$$

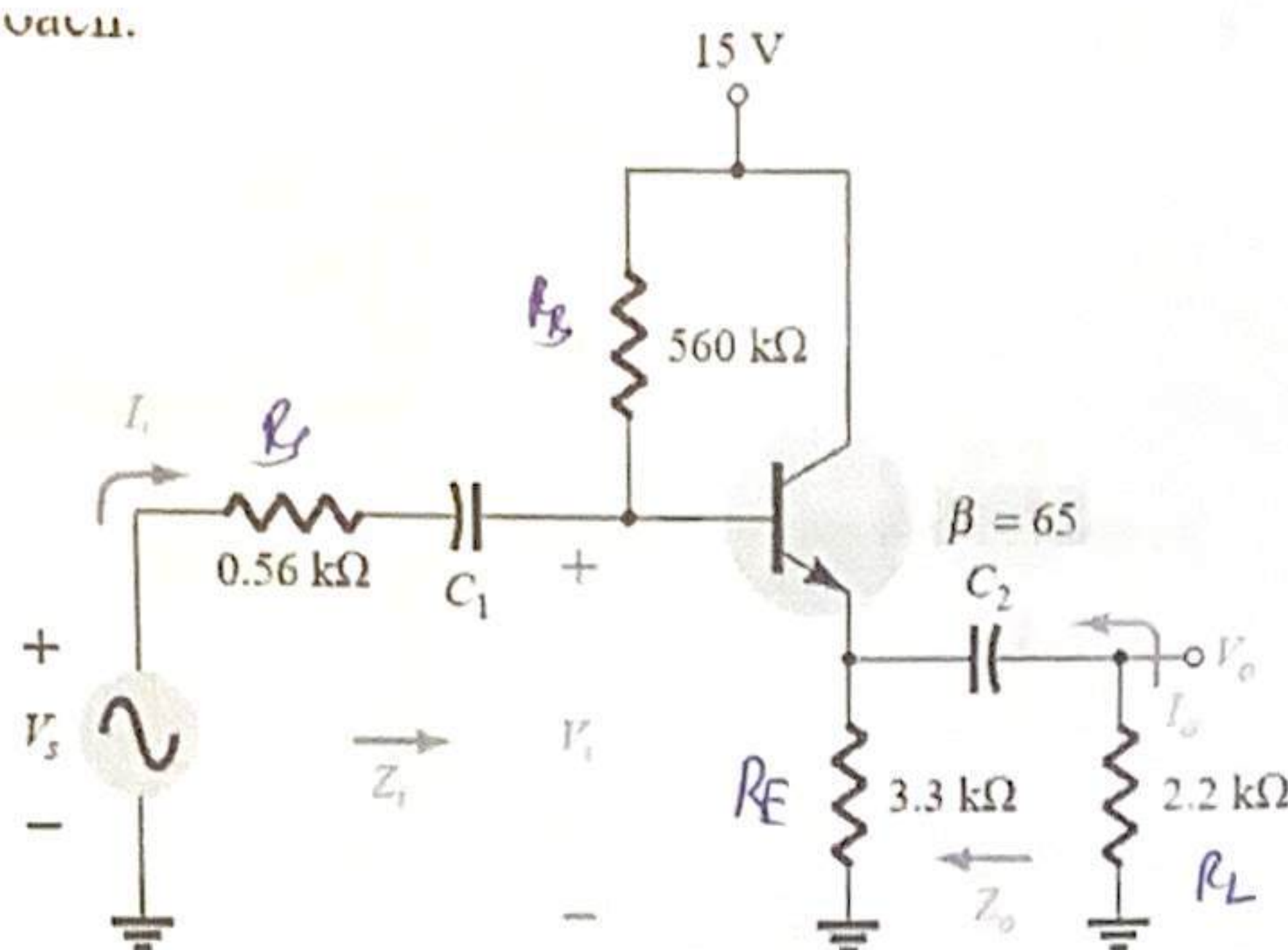
$$= \frac{-(\beta+1)^2 \cdot R_L (R_{E3} \parallel R_L)}{r_{E1} [R_L + (\beta+1)r_{E2} + (\beta+1)^2(r_{E3} \parallel R_L)]}$$

A_v	10 kΩ	4 kΩ	4 Ω
A_v	-225,11	-204,9	-2261

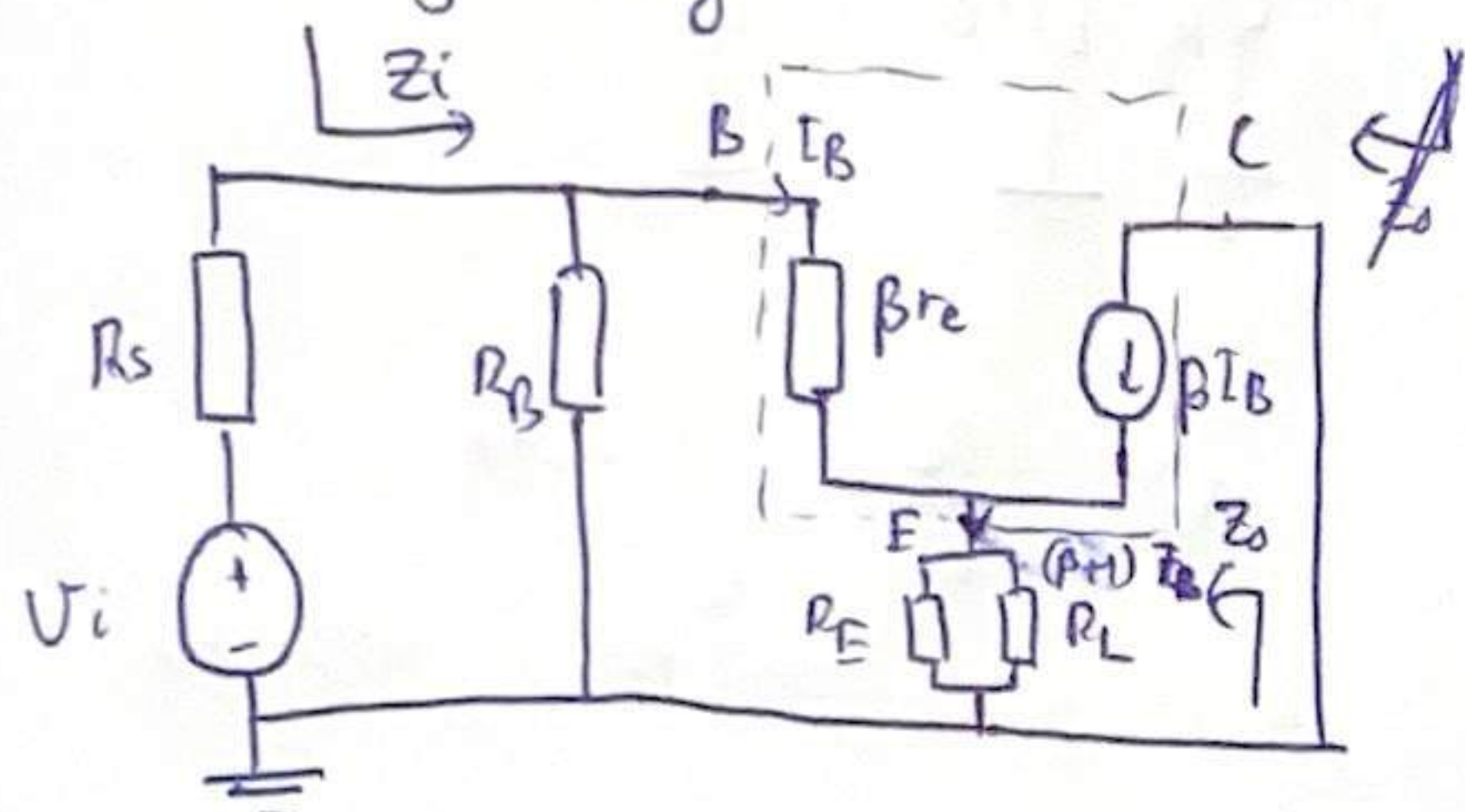
→ c. KL: R_L ảnh hưởng tới A_v . A_v càng nhỏ thì A_v càng nhỏ theo

C3.3 For the loaded emitter-follower configuration of fig with a source resistance and the no-load two-port parameters of $Z_i=157.54 \text{ k}\Omega$, $Z_o=21.6\Omega$, and $A_{vNL}=0.993$ with $r_e=21.74 \text{ V}$ and $\beta=65$, determine the new values of Z_i and Z_o as determined by the load and R_s .

100ac11.



Mô hình AC tương đương

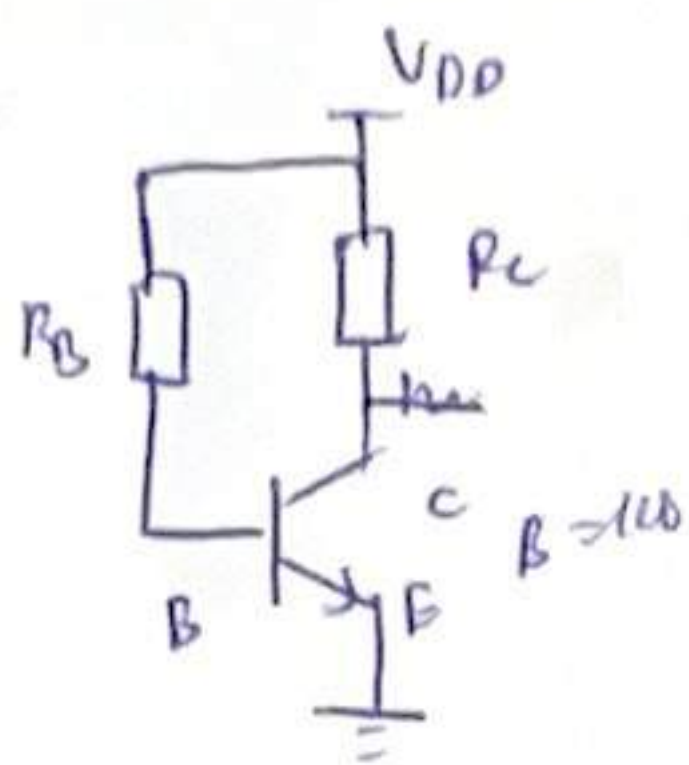


Nhìn vào đầu vào: $Z_i = R_B \parallel \beta(r_e + R_E \parallel R_L) = 754.6 \Omega$

Nhìn ra đầu ra: $Z_o = R_E \parallel \frac{R_s}{\beta+1} \parallel R_B = 6.44 \Omega$

Given $V_{DD} = 12V$

→ DC



$$V_{DD} = V_{R_B} + V_{BE}$$

$$\Rightarrow I_B = \frac{V_{DD} - V_{BE}}{R_B} = 2,4 \cdot 10^{-5} A$$

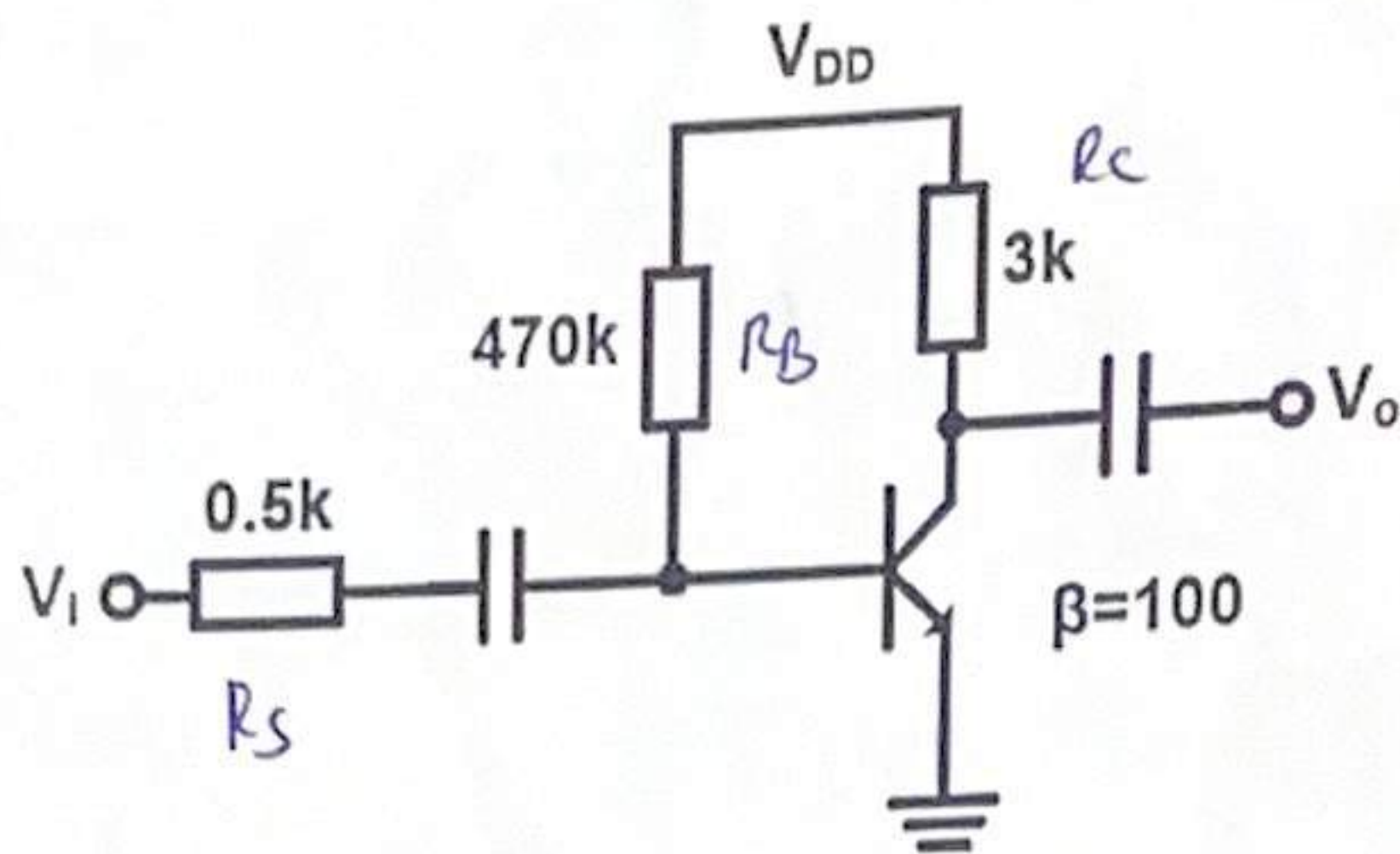
$$\Rightarrow I_E = 2,4 \text{ mA}$$

$$\Rightarrow r_e = \frac{26mV}{I_E} = 10,83 (\Omega)$$

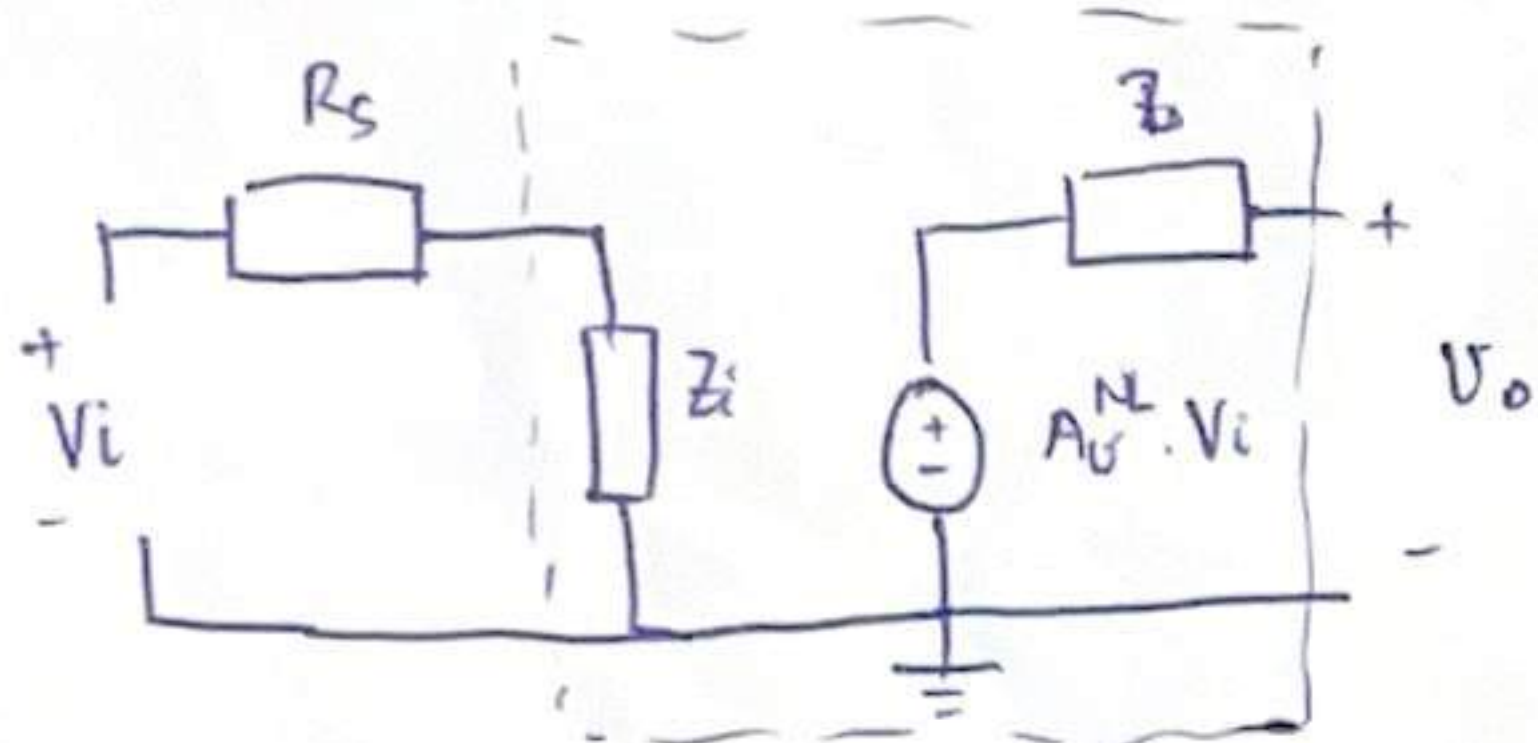
C3.4 In fig, a source with an internal resistance has been applied to the fixed-bias transistor amplifier

(a) Determine the voltage gain $A_v = V_o/V_s$. What percent of the applied signal appears at the input terminals of the amplifier?

(b) Determine the voltage gain $A_v = V_o/V_s$.



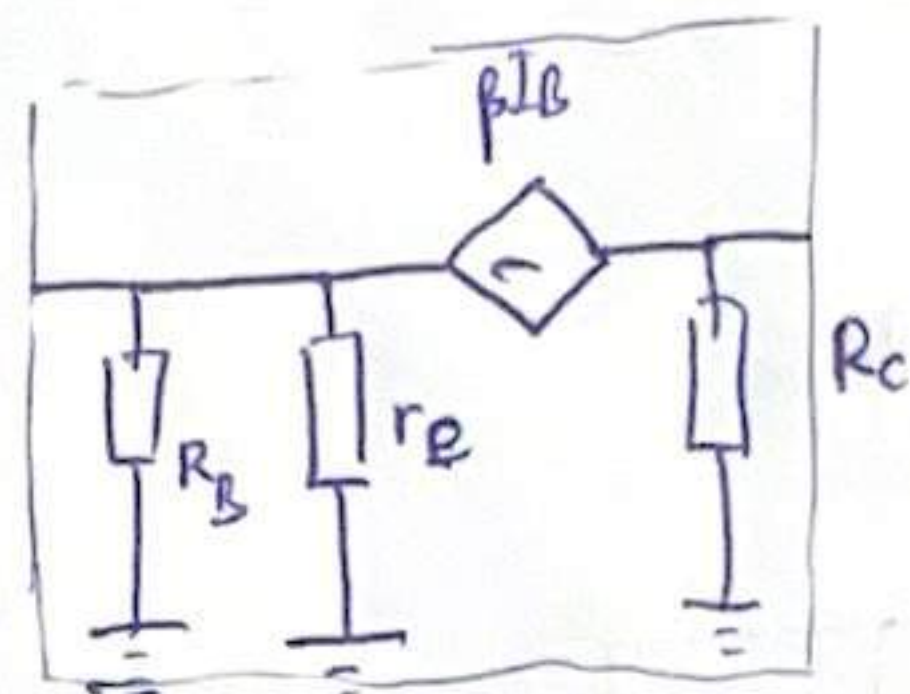
→ AC:



$$A_v^s = A_v^{NL} \cdot \frac{Z_i}{Z_i + R_S}$$

$$= -189,96$$

$$H = \frac{A_v^s}{A_v^{NL}} = \frac{-189,96}{-277} = 0,686 = 68,6\%$$



$$A_v^{NL} = \frac{-R_C}{r_e} = -277$$

$$Z_i = R_B \parallel (\beta r_e) = 1091,29$$

$$Z_o = R_C = 3k$$

(using BJT)